Computer Laboratory Manual

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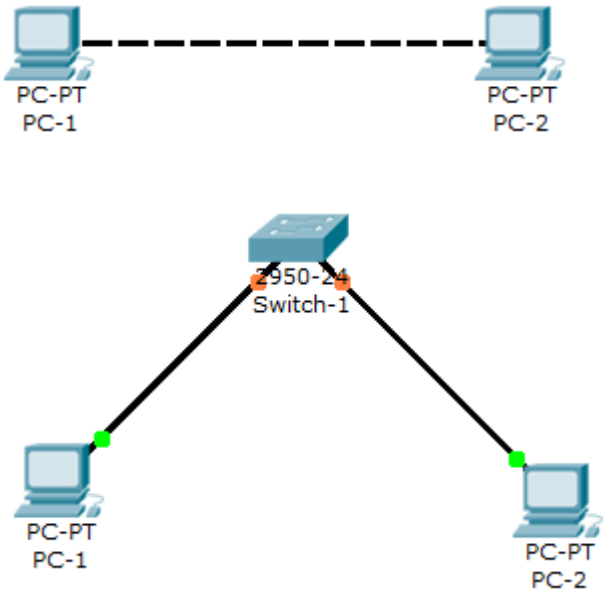
SECTION-I

Network Fundamentals

LAB-1

ETHERNET CABLES

Topology Diagram



Learning Objective

Upon completion of this lab, you will be able to:

- How to make the Ethernet Cables (Straight-Through & Crossover)
- Test Cables using LAN Cable Meter or LAN Cable Tester

Equipment Tools

- Connectors (RJ-45)
- Modular Plug Crimp Tool
- Cat-5, Cat-5e,Cat-6 or Cat-7 Cable
- LAN Cable Tester

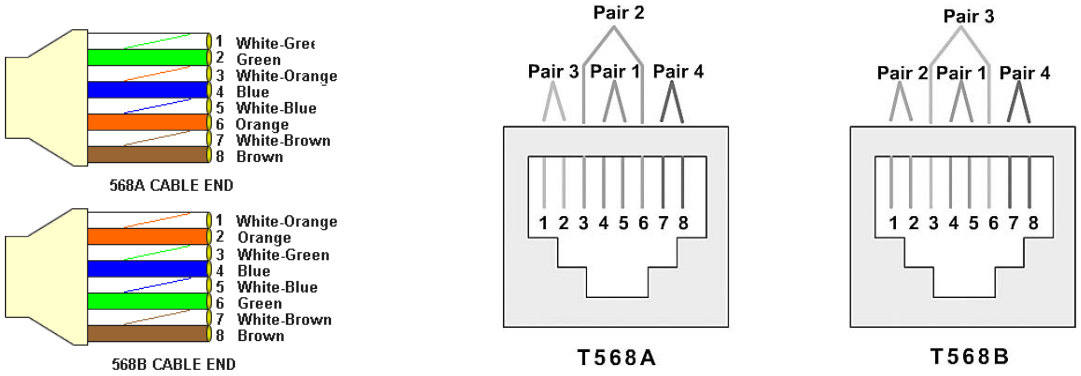
Background

Network devices can be connected via Ethernet Cables such as routers, switches, hubs and modems. They transmit data using the Ethernet protocol. The three most common Ethernet standards used today 10BASE-T (Ethernet), 100BASE-TX (Fast Ethernet, or FE), and 1000BASE-T (Gigabit Ethernet, or GE) use UTP cabling.

UTP Cabling Pin outs for 10BASE-T & 100BASE-TX

The wires in the UTP cable must be connected to the correct pin positions in the RJ-45 connectors in order for communication to work correctly. Interestingly, the IEEE does not actually define the official standards for cable manufacturing, as well as part of the details of the conventions used for the cabling pin outs. Two cooperate industry groups, the Telecommunication Industry Association (TIA) and Electronic Industry Alliance (EIA), define standards for UTP cabling, colour coding for wires, and standard pin outs on the cables. The following figure shows two pin out standards from the EIA/TIA, with the colour coding and pair numbers listed.

Figure: EIA/TIA Standard Ethernet Cabling Pin outs



To understand the acronyms listed in the figure, note that eight wires in a UTP cable have either a solid colour (green, orange, blue, or brown) or striped colour scheme using white and one of the other four colours. Also, a single-wire pair uses the same base colour. For example, the blue wire and the blue/white striped wire are paired and twisted.

A UTP cable needs two pairs of wires for 10BASE-T and 100BASE-T. All you need to remember, to properly configure the cables, are the diagrams for the two cable ends and the following rules:

- A straight-through cable has identical ends.
- A crossover cable has different ends.

Types of Ethernet Cables

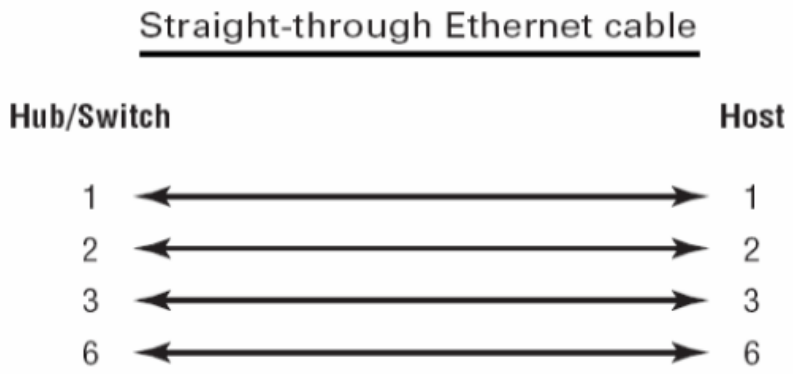
The following are three main types of Ethernet Cables

1- Straight-through Cable
2- Crossover Cable
3- Rollover Cable (Consol Cable)

Straight-Through Cable

A straight-through cable is a type of cable that's mainly used for sending data between from one end to the other end. An example usage of this is connecting a computer to a switch or hubs and router to switch and hubs. An Ethernet straight-through cable connects the wire at pin 1 on one end of the cable to pin 1 at the other end of the cable; the wire at pin 2 needs to connect to pin 2 on the other end of the cable; pin 3 on one end connects to pin 3 on the other; and so on. (To create a straight-through cable, both ends of the cable use the same EIA/TIA pin out standard on each end of the cable.) A Straight-through cable is used when the devices on the ends of the cable use opposite pins when they transmit data. In straight-through cable only pin 1, 2, 3 and 6 will be used as shown in the diagram.

Diagram



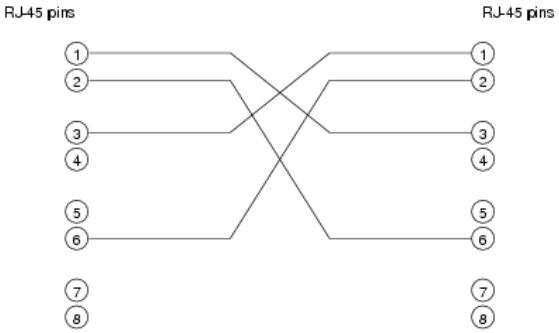
Crossover Cable

When connecting two devices that both use the same pins to transmit, the pin outs of the cable must be set up to swap the wire pair. A cable that swaps the wire pairs inside the cable is called a crossover cable. The network devices that can be connected via crossover cables are as follows:

- Switch-to-Switch
- Hub-to-hub
- Host-to-host
- Hub-to-Switch
- Router direct to Host

Diagram

The following figure shows the several conceptual views of a crossover cable.

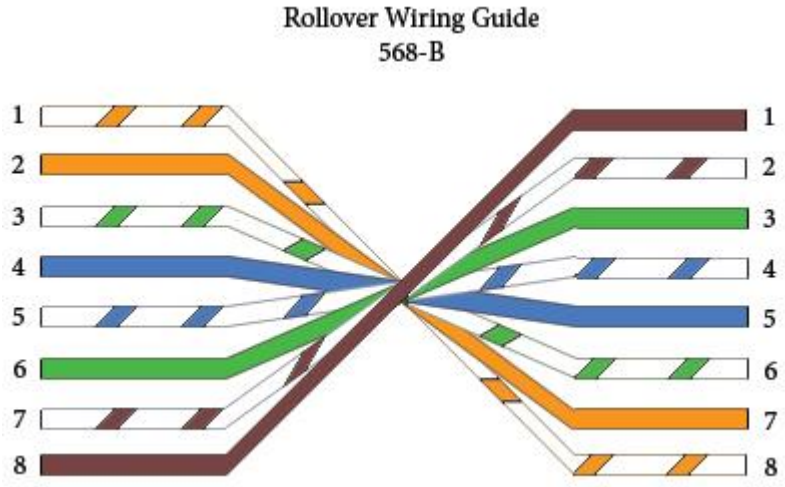


The figure shows the pins to which each wire is connected. Pin 1 on the left end connected to pin 3 on the right end, Pin 2 on the left to Pin 6 on the right, Pin 3 on the left to Pin 1 on the right, and Pin 6 on the left to Pin 2 on the right.

Rollover Cable

Rollover cable (also known as **Cisco console cable** or a **Yost cable**) is a type of null-modem cable that is often used to connect a computer terminal to a router's console port. This cable is typically flat (and has a light blue colour) to help distinguish it from other types of network cabling. It gets the name rollover because the pin outs on one end are reversed from the other, as if the wire had been rolled over and you were viewing it from the other side.

Diagram



To make, just cut the one side like a straight-through cable and reverse the other end.

Once you connect the cable from your PC to the Cisco router or Switch, you can start Hyper Terminal to connect a console connection and configure the device.

Unshielded Twisted Pair Connector

The standard connector for unshielded twisted pair cabling is a RJ-45 connector. This is a plastic connector that looks like a large telephone-style connector as shown in the below figure.



Figure: RJ-45 Connector

RJ stand for Registered Jack-*45*, an eight-wire connector used commonly to connect computers onto a local-area networks (LAN), especially Ethernets. RJ-45 connectors look similar to the ubiquitous RJ-11 connectors used for connecting telephone equipment, but they are somewhat wider.

Shielded Twisted Pair (STP) Cable

A disadvantage of UTP is that it may be susceptible to radio and electrical frequency interference. Shielded twisted pair (STP) is suitable for environments with electrical interference; however, the extra shielding can make the cable quite bulky. Shielded twisted pair is often used on networks using token ring.

Coaxial Cable

Coaxial cable is the kind of copper cable used by cable TV companies between the community antenna and user homes and businesses. Coaxial cable is sometimes used by telephone companies from their central office to the telephone poles near users. It is also widely installed for use in business and corporation Ethernet and other types of local area network. The main advantage of Coaxial cable is that it supports greater lengths between network devices than twisted pair cable.

This coaxial cable is also referred to as thinnet. 10Base2 refer to the specifications for thin cable carrying Ethernet signal. The 2 refer to the approximate maximum segment length being 200 meters. Thin Coaxial cable is famous in school networks, especially linear bus networks

Thick coaxial cable is also referred to as thicknet. 10Base5 refer to the specifications for thick coaxial cable carrying Ethernet signals. The 5 refer to the maximum segment length being 500 meter. It has extra proactive plastic cover helps keep moisture away from the centre conductor. The one disadvantage of thick coaxial cable is that it does not bend easily and is difficult to install.

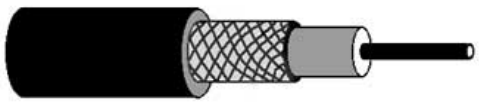


Figure: Coaxial Cable

Coaxial Cable Connector

The most common type of connector used in coaxial cable is Bayone-Neill-Concelman (BNC) connector. Different types of adapters are available for BNC connector, including a T-connector, barrel connector and terminator. Connectors on the cable are the weakest points in the network. To help avoid problems with your network, always use the BNC connector that crimp, rather than screw, onto the cable.



Figure: BNC Connector

Network Cable Tools

Modular Plug Crimp Tool. You will need a modular crimp tool. This one is very similar to the one I have been using for many years for all kinds of telephone cable work and it work just fine for Ethernet cables. You don't need a lot of bells and whistles, just a tool which will securely crimp RJ-45 connectors. Even though the crimper has cutters which can be used to cut the cable and individual wires, and possibly stripping the outer jacket.



Figure: Modular Plug Crimp Tool

LAN Cable Tester

Mis-wired and incomplete network cables could lead to headaches down the road. In addition, with power-over-Ethernet (PoE) making its way into the market place, crossed wire pairs could lead to physical damage of computers or phone system equipment, making it even more crucial that the pairs are in the correct order. A simple cable tester can quickly verify that information for you. Should you not have a network cable tester on hand, simply test connectivity pin to pin.



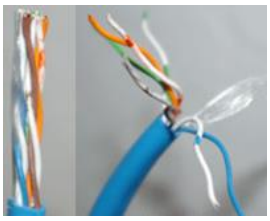
Figure: LAN Cable Tester

Procedures

To Make Ethernet Cables

STEP-1 Pull the cable off the reel to the desired length and cut. The total length of wire segments between a PC and a Hub or between two PC's cannot exceed 100 Meters for 100BASE-TX and 300 Meters for 10BASE-T.

STEP-2 Stripping at least a half of an inch off of the cable to expose the inner wires as shown in the below diagram and after the network cable jacket has been removed separate the wires within the cable so they can be put into the RJ-45 connector.



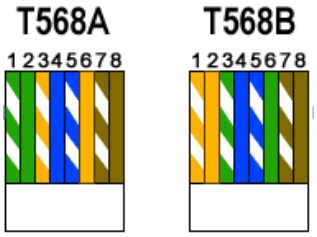
STEPS-3 Inspect the newly revealed wires for any cuts or scrapes that expose the copper wire inside. If you have breached the protective sheath of any wire, you will need to cut the entire segment of wires off and start over at step one. Exposed copper wire will lead to cross-talk, poor performance or no connectivity at all. It is important that the jacket for all network cables remains intact.



STEPS-4 Untwist the pairs so they will lay flat between your fingers. The white piece of thread can be cut off even with the jacket and disposed (see Warnings). For easier handling, cut the wires so that they are 3/4" (19 mm) long from the base of the jacket and even in length.



STEPS-5 According to IEEE there are two standards in Ethernet Cabling One is T568A and T568-B, each twisted-pair must be broken apart to create the layout as shown in the below figure.



STEPS-6 If you wish to create a straight-through cable both ends of the cable should be identical and should match the T568A IEEE standard as shown above figure. If you want to create a crossover cable one end of the cable should match T568A and the other should be matching T568B.

STEPS-7 Press all the wires flat and parallel between your thumb and forefinger. Verify the colors have remained in the correct order. Cut the top of the wires even with one another so

that they are 1/2" (12.5 mm) long from the base of the jacket, as the jacket needs to go into the 8P8C connector by about 1/8", meaning that you only have a 1/2" of room for the individual cables. Leaving more than 1/2" untwisted can jeopardize connectivity and quality. Ensure that the cut leaves the wires even and clean; failure to do so may cause the wire not to make contact inside the jack and could lead to wrongly guided cores inside the plug.

STEPS-8 Keep the wires flat and in order as you push them into the RJ-45 plug with the flat surface of the plug on top. The white/orange wire should be on the left if you're looking down at the jack. You can tell if all the wires made it into the jack and maintain their positions by looking head-on at the plug. You should be able to see a wire located in each hole, as seen at the bottom right. You may have to use a little effort to push the pairs firmly into the plug. The cabling jacket should also enter the rear of the jack about 1/4" (6 mm) to help secure the cable once the plug is crimped. You may need to stretch the sleeve to the proper length. Verify that the sequence is still correct before crimping.



STEPS-9 Place the wired plug into the crimping tool. Give the handle a firm squeeze. You should hear a ratcheting noise as you continue. Once you have completed the crimp, the handle will reset to the open position. To ensure all pins are set, some prefer to double-crimp by repeating this step.



STEPS-9 Repeat all of the above steps with the other end of the cable.

Assignment:

How many Computer LABs are in Sir Syed University name them? Write down the information about the Network Devices that are installed in each Computer Labs and which type of Ethernet cables category are implemented between each device in all the Computer Labs of SSUET.

LAB-2

Connectivity of Cisco Devices

Topology Diagram



Learning Objectives

Upon completion of this lab, you will be able to:

- Connect a router or switch and computer using a consol cable.
- Configure Host Terminal to establish a consol session with a Cisco IOS router

Equipment Tools

- PC (Personal Computer)
- Router/Switch
- Console Cable

Background

Cisco IOS Software

Cisco IOS (Internetwork Operating System) Software for network devices (Router/ Switches) implements the control logic and functions performed by a Cisco router and switches. Besides controlling the network devices performance and behaviour Cisco IOS also defines an interface for humans called the CLI. The Cisco IOS CLI (Command Line Interface) allows the user to use a terminal emulation program, which accepts text entered by the user. When the user presses Enter, the terminal emulator sends that text to the switch or router. The Router or Switch can be accessed through three methods the console, telnet and SSH (Secure Shell). Two of these methods (Telnet and SSH) use the IP (Internet Protocol) network and through these methods you can easily access the CLI mode of Router and Switch. The consol is a physical port built specifically to allow access to the CLI.

Console Cable (Rollover)

Rollover cables, like other cabling types, got their name from how they are wired. Rollover cables essentially have one end of the cable wired exactly opposite from the other. This essentially "rolls over" the wires- but why would we need to do such a thing? Rollover cables, also called Yost cables, usually connect a device to a router or switch's Console port. This allows a programmer to make a connection to the router or switch, and program it as needed. The following figures are showing the two different view of console cable.

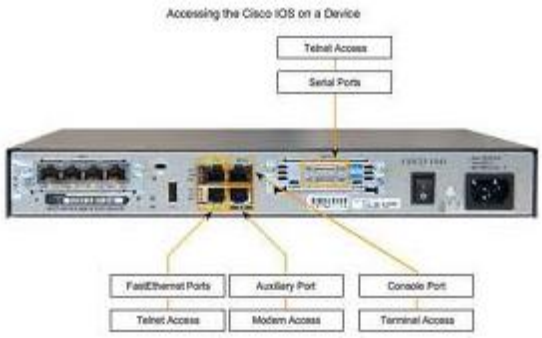




Router

A router is specialized computer connected to more than one network. A router runs software that allows the device to move data from one network to another. Routers operate at the network layer (OSI Model's layer 3). The primary function of a router is to connect networks together and keep certain kinds of broadcast traffic under control. There are several companies that make routers: Cisco (Linksys), Juniper, Nortel (Bay Networks), Redback, Lucent, 3Com, and HP just to name a few.

Physical Device View



The above diagram is showing the physical device view of router in which different ports are showing and the names and the purpose of these ports are as follows.

- **Fast Ethernet Ports:** Fast Ethernet ports are normally used for the creation of LAN (Local Area Network) and telnet access utility can be also used via these ports for accessing the Command Line Interface and through this CLI mode you can access the different modes of the router such as user mode, privilege mode and router configuration mode.
- **Auxiliary Port**: Auxiliary port normally used for Modem Access and Router can be configured remotely using a Modem.
- **Console Port:** Console port normally use for terminal access.
- **Serial Ports:** Serial Ports normally used for WAN (Wide Area Network) connectivity.

Switch

A network switch is a small hardware device that joins multiple computers together within one local area network (LAN). Technically, network switches operate at layer two (Data Link Layer) of the OSI model.

Network switches appear nearly identical to network hubs, but a switch generally contains more intelligence (and a slightly higher price tag) than a hub. Unlike hubs, network switches are capable of inspecting data packets as they are received, determining the source and destination device of each packet, and forwarding them appropriately. By delivering messages only to the connected device intended, a network switch conserves network bandwidth and offers generally better performance than a hub. As with hubs, Ethernet implementations of network switches are the most common. Mainstream Ethernet network switches support either 10/100Mbps Fast Ethernet or Gigabit Ethernet (10/100/1000) standards.

Physical Device View



Scenario

Set up a network similar to the one in the Topology Diagram. The following resources will be required:

- Connect a Computer with a serial interface RS-232 with Router or Switch console port
- Cisco router or Switch
- Consol (rollover) cable for connecting the workstation to the router

Task 1: Connect a Router or Switch and Computer with a Console Cable

Step 1: Set up basic physical connection.

1- Locate the consol port on the back of the router or Switch
2- Select the Console Cable for initially configure the Router or Switch
3- The console cable will need an RJ-45 to DB-9 or DB-25 adapter for Cisco router or Switch.
4- Use the correct adapter to connect the other end of the cable to your terminal or PC.

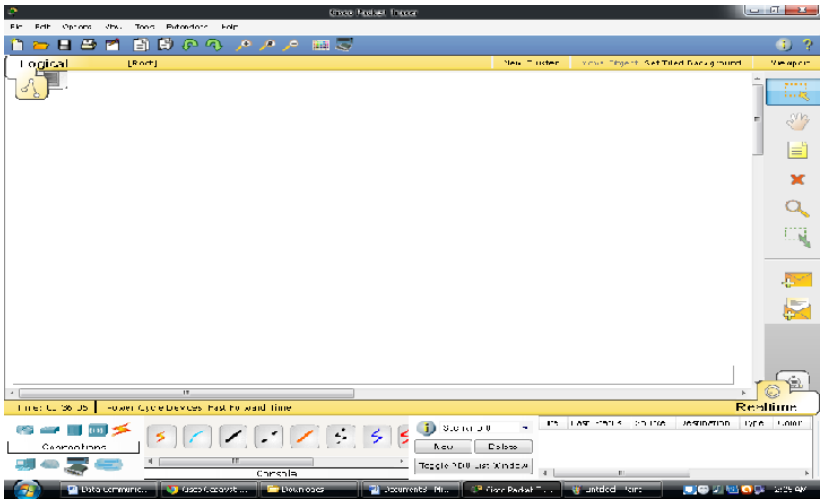
Step 2: Power on device

If not already powered on. Enable power to the computer and router.

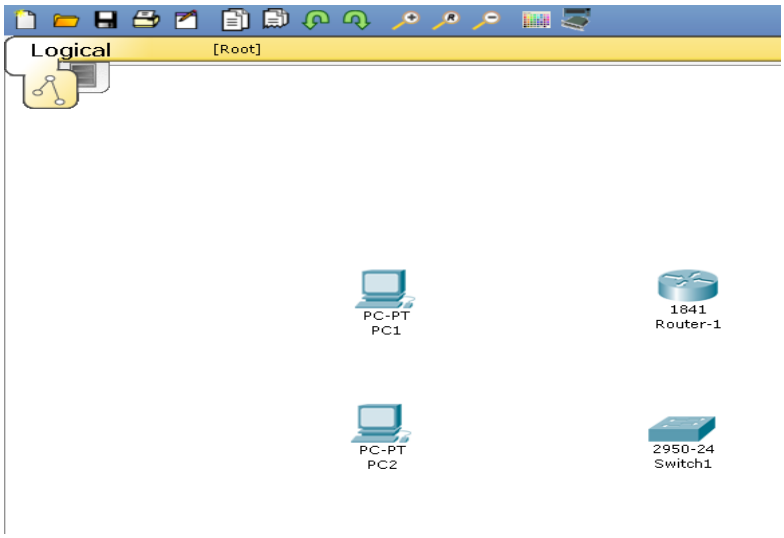
Task 2: Configure a Host Terminal to Establish a Console Session with a Cisco IOS Router or Switch.

The following steps are required for establishing a Consol Session with a Cisco IOS Router or Switch.

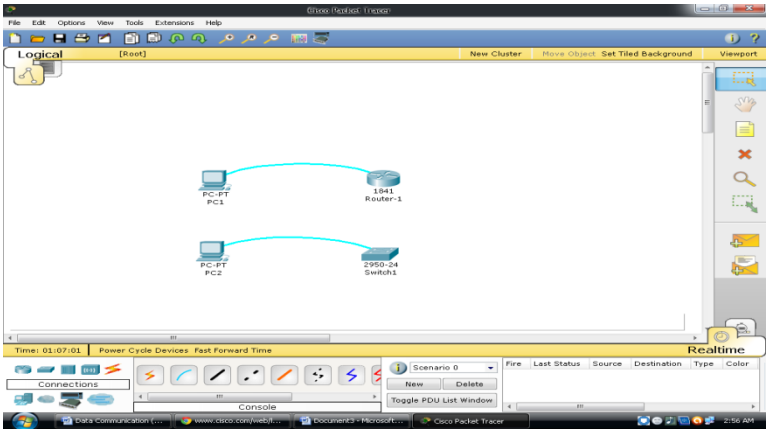
Step 1 Open the Packet Tracer



Step 2 Select the PC and Router or Switch and Place the Logical Window of Packet Tracer.



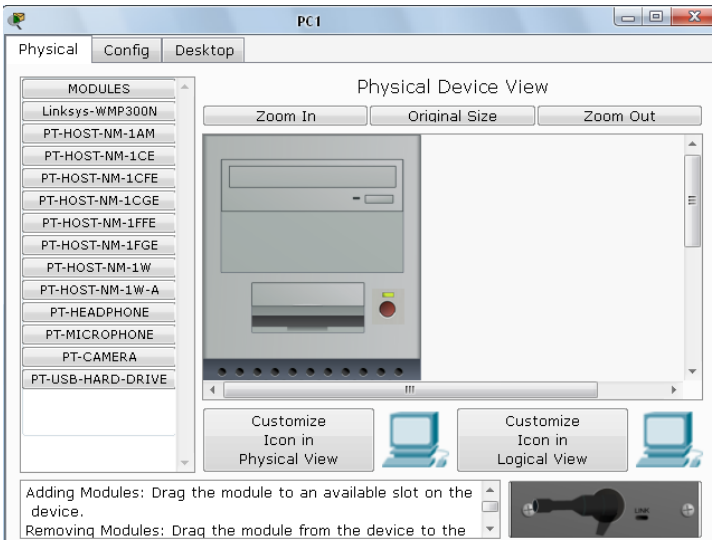
Step-3 Connect a PC RS-232 port to Router or Switch Console Port via Consol Cable.



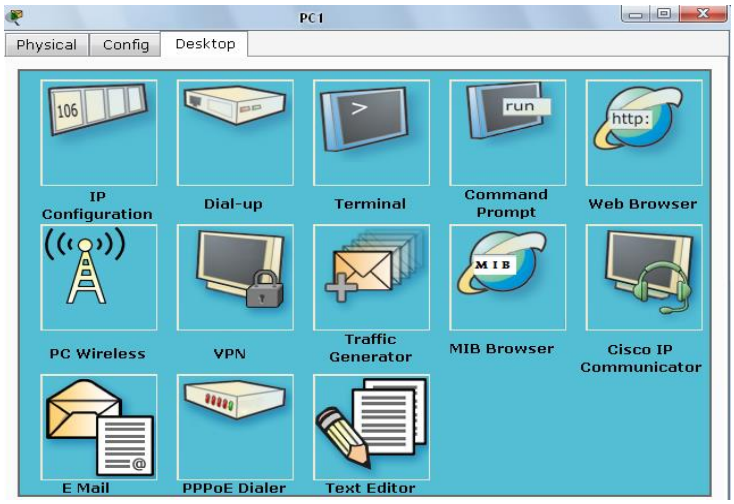
Step-4 Configure a Host Terminal for the establishment of consol session for Router or Switch.

The following steps are required for making a console session for Router or Switch.

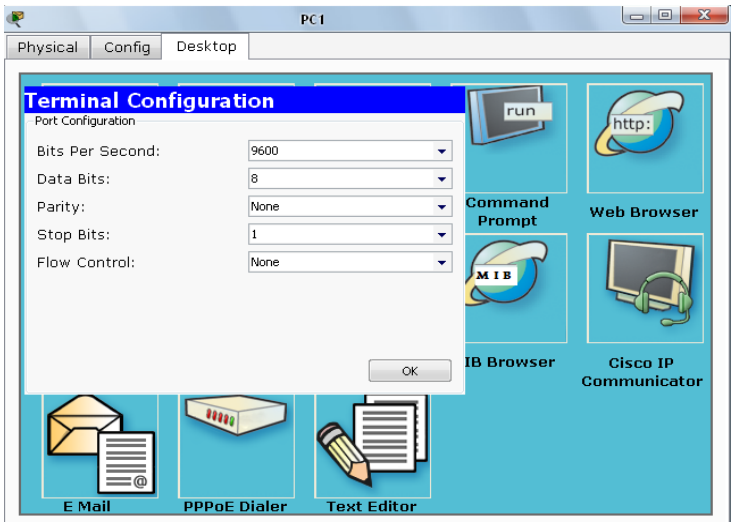
1- Double Click on PC



2- Select the Desktop tab as shown in the above window

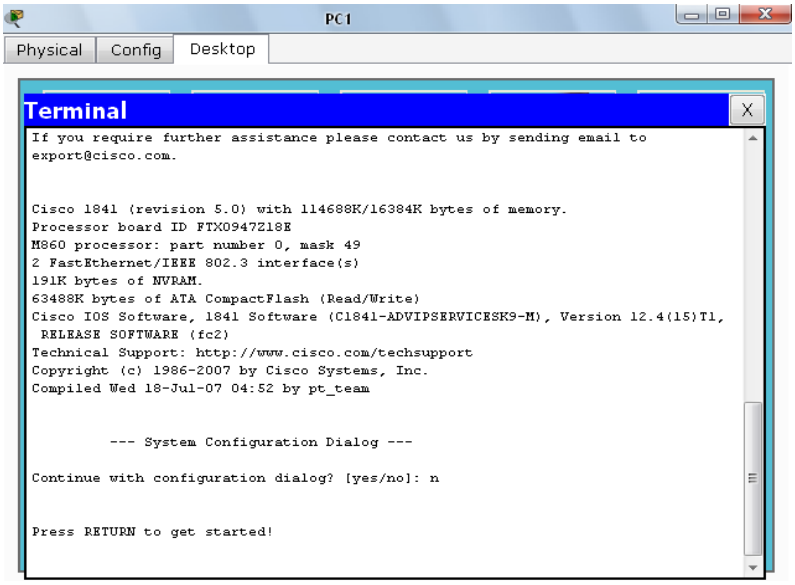


3- Open a Host Terminal as shown above window and press OK tab



The above window is showing the Terminal Configuration section which is referred to as default RS-232 COM-1 setting which is required for the initial configuration of router or switch.

4- Router Terminal or Switch Terminal



When the Host Terminal window comes up press the **OK** tab. There should be a response from the router. This indicates that connection has been successfully completed. If there is no connection troubleshoot as necessary.

SECTION-II

CISCO ROUTER CONFIGURATION

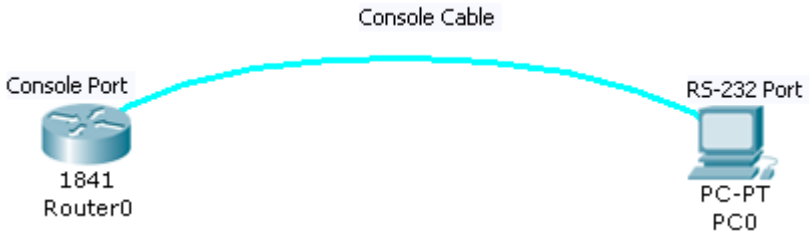
LAB-3

Basic Cisco Device Configuration

Learning Objective

- Study about the Internal Configuration Components of Router and Switch
- Configure a Cisco Router via Basic IOS command Modes
- Configure Cisco Router Password Access
- Save the router configuration file

Topology Diagram



Required Equipments

- Cisco Router 1841
- Computer Host (End Device)
- Consol (Rollover) Cable

Task-1 Study about the Internal Configuration Components of Router and Switch.

This section describes the internal configuration components of router and switch.

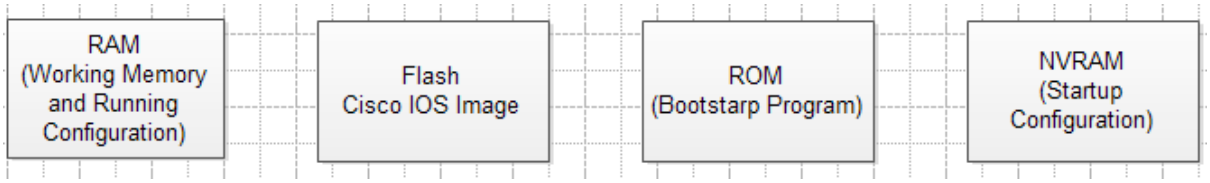
The internal configuration components of a router and Switch include:

1- **RAM/DRAM:** RAM/DRAM is the main storage component for the router and Switch. The running configuration of router and switch normally stored in RAM/DRAM.
2- **NVRAM:** Nonvolatile RAM (NVRAM) stores the initial or start-up configuration files that is used when router or switch is first powered on and when the router or switch is reloaded.
3- **ROM:** Read-Only-Memory (ROM) stores a bootstrap program that is loaded when the switch or router first powers on. The bootstrap program is responsible for finding the full Cisco IOS image and manages the process of loading Cisco IOS into RAM.
4- **Flash Memory:** The Cisco IOS image is stored in Flash Memory and it is a default location where the router and switch gets its Cisco IOS at boot time. Flash memory can be used to store any other files including backup copy of configuration file.

5- **Interfaces:** Interfaces are the network connections through which packets enter and exit the router.

Cisco IOS stores the collection of configuration commands in a configuration file. In fact, Router or Switch use multiple configuration file one for the initial configuration used when power on, and another configuration file for the active, currently used running configuration as stored in RAM.

Figure: Cisco Router/Switch Memory Types



Task-2 Configure a Cisco Router via Basic IOS command Modes.

Router Initialization and Command Modes

Each time the router is switched on, it goes through power-on self test diagnostics to verify basic operation of the CPU, memory and network interfaces. The system bootstrap software in ROM (boot image) executes and searches for valid router operating system software (Cisco IOS image).

There are three places to find the Cisco IOS image to load from flash: memory, a TFTP server on the network, or ROM. The source of the Cisco IOS image is determined from the boot field setting of the router's configuration register. The default setting for the configuration register indicates that the router should attempt to load a Cisco IOS image from flash memory. After finding a valid IOS image, it searches the valid configuration file. The configuration file, saved in NVRAM, is loaded into main memory and executed one line at a time. The purpose of these configuration commands start routing processes, supply address for interfaces, and set media characteristics.

The Cisco IOS use interface is divided into many different modes. The commands available at any given time depend on which mode you are currently in.

User Mode/ EXEC Mode: When you start a session on the router, you begin in use mode often called EXEC mode. Only limited subsets of the commands are available in EXEC Mode.

Syntax of User Mode/EXEC Mode: The user-level prompt consists of the host name followed by the angle bracket (>):

Router>

Privileged EXEC Mode: In order to have access to all commands, enter privileged EXEC mode. Normally, normally you must enter a password to enter privilege EXEC mode.

Syntax of Privilege EXEC Mode: The privilege EXEC mode prompt consists of the devices host name followed by the pound sign (#)

Router>en
Password:
Router#

Global Configuration Mode: Global configuration commands apply to feature that effect the system as a whole, rather than just one protocol or interface. Use the configure terminal privilege EXEC command to enter global configuration mode.

Syntax of Global Configuration Mode: The following example shows how to access global configuration mode.

Router#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#

Interface Configuration Mode: Interface configuration commands modify the operation of an interface such as Ethernet or Serial port. Interface configuration commands always follow an interface global command, which defines the interface type.

Syntax of Interface configuration Mode: In the following example, Fast Ethernet interface is about to be configured. The new prompt Router *(config-if) #* indicates interface configuration mode.

Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int f0/0
Router(config-if)#

Subinterface Configuration Mode: Multiple virtual interface (*called Subinterfaces*) can be configured on a single physical interface.

Syntax of Subinterface Configuration Mode:

Router (config)# interface serial 2
Router (config-if)# encapsulation frame-relay
Router (config-if)#interface serial 2.1
Router (config-subif)#

Router Basics Commands

This section describes how to log into a Cisco router and examine its configurable components using Cisco IOS commands. We will examine how to log into and out of a Cisco router, how Cisco IOS commands can be used to examine router status. Login into a Cisco router, access privileged mode, log out of a Cisco router, and use Cisco IOS commands to examine the configurable components of a Cisco router.

Login in to a Cisco Router

The login process is not complicated, but there are certain features to be aware of. The login screen requires you to hit the Return or Enter key, to continue to the next stage of

the login process. If no password is required you can easily access the Cisco IOS. A password is recommended first step in maintaining network security.

Press RETURN to get started.
User Access Verification
Password:

The router allows a finite period of 30 seconds to login and enter the password. If you fail to enter any portion of the password within 30 seconds of the initial prompt, it prompts again.

Press RETURN to get started.
User Access Verification
Password:
%Password: timeout expired!
Password:

User Mode Command List

When you first log in you see the user EXEC mode prompt. Type ? at the prompt to display a list of available commands.

```
Router> ?
Exec commands:
  <1-99>      Session number to resume
  connect     Open a terminal connection
  disable     Turn off privileged commands
  disconnect  Disconnect an existing network connection
  enable      Turn on privileged commands
  exit        Exit from the EXEC
  logout      Exit from the EXEC
  ping        Send echo messages
  resume      Resume an active network connection
  show        Show running system information
  ssh         Open a secure shell client connection
  telnet      Open a telnet connection
  terminal    Set terminal line parameters
  traceroute  Trace route to destination
```

Privileged Mode Command List

As in the user EXEC mode, display the available commands for the privileged EXEC mode by typing ? at the privileged prompt.

```
Router# ?
Exec commands:
  <1-99>      Session number to resume
  auto        Exec level Automation
  clear       Reset functions
  clock       Manage the system clock
  configure   Enter configuration mode
  connect     Open a terminal connection
```

```
copy        Copy from one file to another
debug       Debugging functions (see also 'undebug')
delete      Delete a file
dir         List files on a filesystem
disable     Turn off privileged commands
disconnect  Disconnect an existing network connection
enable      Turn on privileged commands
erase       Erase a file system
exit        Exit from the EXEC
logout      Exit from the EXEC
mkdir       Create new directory
more        Display the contents of a file
no          Disable debugging information's
ping        Send echo messages
reload      Halt and perform a cold restart
--More--
```

The *More* Prompt indicates that there are more commands to see. If you will press the *Return*, you will see the next line.

Router Examination

Several Cisco IOS commands can be used to examine the status of a router. Each command permits examination of a different configurable component of the router.

Router Status commands are as follows:

- ***show version:*** To display the configuration of the system hardware, the software version, the names and sources of configuration files and the boot images, use the show version EXEC command.
- ***Show processes:*** Use the show processes EXEC command to display information about the active process.
- ***show buffers:*** Use the show buffers EXEC command to display statistics for the buffer pools on the network server.
- ***show ip protocol:*** To display the parameters and current state of the active routing protocol process, use the show ip protocols EXEC command.
- ***show protocols:*** Use the show protocols EXEC command to display the configured protocols. This command shows the global and interface specific status of any configured level 3 protocol. For example, IP, DEC-net , IPX, AppleTalk etc.
- ***show memory:*** Use the show memory EXEC command to show statistics about memory.
- ***show ip route:*** Use show ip route EXEC command to display the current state of the routing table.
- ***show flash:*** List information about flash memory, including system image filenames and amounts of memory used and remaining.
- ***show running-config:*** To display the configuration information currently running on the terminal, use the show running-config EXEC command.
- ***show startup-config:*** To display contents of NVRAM. Use the show startup-config EXEC command.
- ***show interface:*** Use the show interface EXEC command to display statistics for all interfaces configured on the router or access server.
- ***show ip interface brief:*** Use for knowing the status of the Ethernet or Serial interface of the router

Task-3 Configure Cisco Router Password Access.

Access password are set for the privilege exec mode and user entry point such as console, aux and virtual lines. The privilege exec mode password is the most critical password, since it controls access to the configuration mode.

Step-1 Configure the privilege exec password.

Cisco IOS supports two commands that set access to the privileged exec mode. One command, **enable password,** contain weak cryptography and should never be used if the **enable secret** command is available. The **enable secret** command uses a very secure MD5 cryptographic hash algorithm. In production environment, strong passwords should be used at all times.

Set the privilege exec password to **cisco**.

Router(config)#enable secret Cisco
Router(config)#

Step-2 Configure the console password.

Set the console access password to **class.** The console password controls console access to the router.

Router(config)#line console 0
Router(config-line)#password class
Router(config-line)#login

Task-4 Save the router configuration file.

Cisco IOS refers to RAM configuration storage as running-configuration, and NVRAM configuration storage as startup-configuration. For configuration to survive rebooting or power restarts, the RAM configuration must be copied into non-volatile RAM (NVRAM). This doesn't occur automatically, NVRAM must be manually updated after any changes are made.

Step-1 Compare router RAM and NVRAM configuration.

Display the content of NVRAM. If the output of NVRAM is missing, it is because there is no saved configuration.

Router#sh startup-config
startup-config is not present
Router#
Display the content of RAM.

Router# show running-config

Use the output to answer the following question:

How large is the configuration file?.....

What is enable secret password?.....

Does your MOTD banner contain the information you entered earlier?.....

Do your interface descriptions contain the information you entered earlier?.....

Step 2 Save RAM configuration to NVRAM.

For a configuration to be used the next time the router is powered on or reloaded, it must be manually saved in NVRAM. Save the RAM configuration to NVRAM:

Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]

Assignment: Configure a Cisco Catalyst Switch 2950-24T with the Basic IOS commands.

Topology Diagram

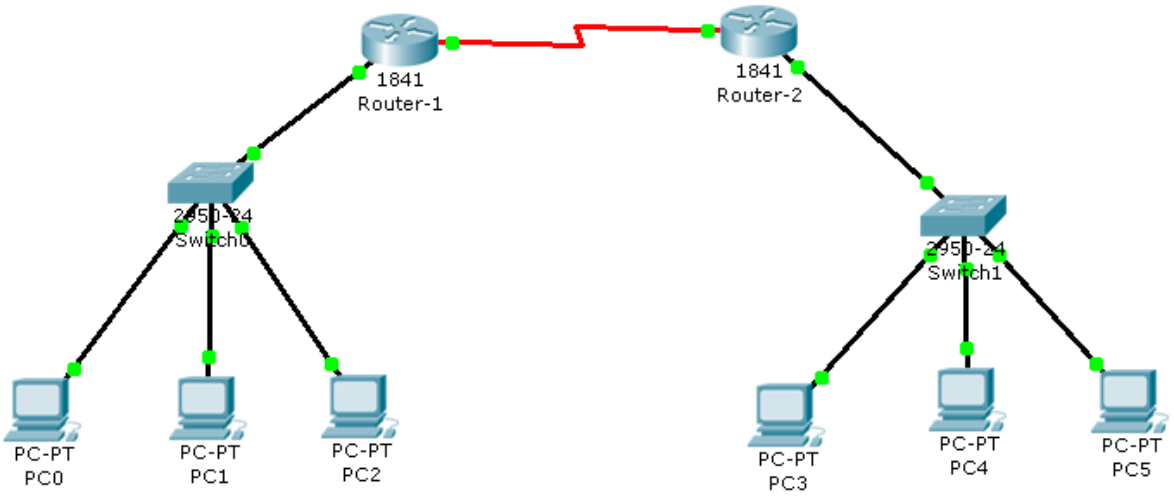


1- Connect the host to the Switch
2- Configure global configuration hostname setting
3- Configure the MOTD banner
4- Configure the Privilege EXEC password
5- Configure the Console password
6- Compare the Basic IOS commands of Router and Switch and write down your own words what the main difference between these commands.

LAB-4

Basic Router Configuration

Topology Diagram



Addressing Table

Device	Interface	IP Address	Subnet Mask	Def. Gateway
Router-1	Fa0/0	192.168.1.1	255.255.255.0	N/A
	S0/0/0	192.168.2.1	255.255.255.0	N/A
Router-2	Fa0/0	192.168.3.1	255.255.255.0	N/A
	S0/0/0	192.168.2.2	255.255.255.0	N/A
PC0	N/A	192.168.1.10	255.255.255.0	192.168.1.1
PC1	N/A	192.168.1.11	255.255.255.0	192.168.1.1
PC2	N/A	192.168.1.12	255.255.255.0	192.168.1.1
PC3	N/A	192.168.3.10	255.255.255.0	192.168.3.1
PC4	N/A	192.168.3.11	255.255.255.0	192.168.3.1
PC5	N/A	192.168.3.12	255.255.255.0	192.168.3.1

Learning Objectives

Upon completion of this lab, you will able to:

- Design and Cable a network according to Topology Diagram.
- Erase the startup configuration and reload a router to the default state.
- Perform basic configuration tasks on a router
- Configure and activate Ethernet interfaces & Serial Interfaces
- Test and verify configuration

Scenario

In this lab activity, you will create a network that is similar to the one shown in the Topology Diagram. Begin by cabling the network as shown in the Topology Diagram. You will then perform the initial router configuration required for connectivity. Use the IP addresses that are provided in the Topology Diagram to apply an addressing scheme to the network devices. When the network configuration is complete, examine the routing tables to verify that the network is operating properly.

Task 1: Design and Cable a Network.

Design and cable a network that is similar to the one in the Topology Diagram by using a Cisco Packet Tracer. In this network you will be required a router 1841. Be sure to use the appropriate type of Ethernet cable to connect from host to switch, switch to router and router to router. 1841 router don't have serial interface ports by default so you will be required to install by manually these ports. The following steps are required for installing a Serial Interface on 1841 router.

Step-1: Click on Router 1841

Step-2: Press the button Power Off

Step-3: You will see different Modules select the WIC-2T module which is called the 2-port asynchronous/synchronous serial network module.

Steps-4: Drag this module to empty slots of Router 1841

Step-5: Press the Power On button

When you will press the button power on you will see the router will boot again from the initial stage. To verify serial port attachment with the 1841 router write the following command on the CLI mode of the router.

Router# show ip interface brief

Router#sh ip int brief

Interface IP-Address OK? Method Status Protocol

FastEthernet0/0 unassigned YES unset administratively down down

FastEthernet0/1 unassigned YES unset administratively down down

Serial0/0/0 unassigned YES unset administratively down down

Serial0/0/1 unassigned YES unset administratively down down

Vlan1 unassigned YES unset administratively down down

The above output is showing that the serial interface ports have attached successfully on router 1841.

Task-2: Erase and Reload the Routers.

Step 1: Accessing a Router IOS (Internetwork Operating System)
Click on Router 1841 and Click on CLI (Command Line Interface) Tab. Cisco Packet Tracer software allow a directly access the Router CLI mode but in practical scenario it is not possible to direct access the router CLI mode.
Step 2: Enter privilege EXEC mode
Router > enable
Router#
Step-3: Clear the configuration
To clear the configuration, issue the **erase startup-config** command on privilege mode of router. Press Enter when prompted to conform that you really want to erase the configuration currently stored in NVRAM.
Router# erase startup-config
Erasing the nvram filesystem will remove all configuration files! Continue? [confirm]
[OK]
Erase of nvram: complete
Router#
Step-4: Reload Configuration
When the prompt returns, issue reload command
The result should look something like this:
Router>en
Router#reload
Proceed with reload? [confirm]
%SYS-5-RELOAD: Reload requested by console. Reload Reason: Reload Command.

System Bootstrap, Version 12.3(8r)T8, RELEASE SOFTWARE (fc1)
Cisco 1841 (revision 5.0) with 114688K/16384K bytes of memory.

Self decompressing the image :
##
[OK]

Step-5: Repeat steps 1 through 4 on router R2 to remove any startup configuration file that may be present.

Task-3: Perform Basic Configuration on Router-1

Step-1 Enter a Privileged EXEC mode.
Router> enable
Router#
Step-2 Enter global configuration mode.

Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Step-3: Configure the hostname as Router1
Enter the command hostname Router1
Router(config)#hostname Router1
Router1(config)#
Step-4: Configure EXEC Mode Password.
Step-5: Configure a message-of-the-day banner.
Router1(config)#ban
Router1(config)#banner motd &
Enter TEXT message. End with the character '&'.
!!!!!!!!!!!AUTHORIZED ACCESS ONLY !!!!!!!!!!
&
Router1(config)#
Step-6: Configure the console password on the router.
Use cisco as the password. When you are finished, exit from line configuration mode.
Step-7: Configure the Fast Ethernet0/0 interface.
Router1(config)#int
Router1(config)#interface fas
Router1(config)#interface fastEthernet 0/0
Router1(config-if)#ip address 192.168.1.1 255.255.255.0
Router1(config-if)#no shutdown
Step-8: Configure the Serial0/0/0 interface.
Router1(config)#int
Router1(config)#interface ser
Router1(config)#interface serial 0/0/0
Router1(config-if)#ip add
Router1(config-if)#ip address 192.168.2.1 255.255.255.0
Router1(config-if)#clock rate 64000
Router1(config-if)#no shutdown
Router1(config-if)#

Note: The interface will not be activated until the serial interface on Router2 is configured and activated.
Step-9: Return to privilege EXEC mode.
Use the end command to return to privilege EXEC mode.
Router1(config-if)#end
%SYS-5-CONFIG_I: Configured from console by console
Router1#
Step-10: Save the Router1 configuration
Save the Router1 Configuration using the copy running-config startup-config command.
Router1#copy run
Router1#copy running-config sta

Router1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router1#

Task-4: Perform Basic Configuration on Router-2

Step-1: *For Router2, repeat steps 1 through 6 from Task-2*
Step-2: Configure the Serial 0/0/0 interface.
Configure the Serial 0/0/0 interface with the IP address 192.168.2.2/24
Router2(config)#int
Router2(config)#interface ser
Router2(config)#interface serial 0/0/0
Router2(config-if)#ip add
Router2(config-if)#ip address 192.168.2.2 255.255.255.0
Router2(config-if)#no shutdown
Step-3: Configure the FastEthernet0/0 interface
Configure the FastEthernet0/0 interface with the ip address 192.168.3.1/24
Router2(config)#int
Router2(config)#interface fa
Router2(config)#interface fastEthernet 0/0
Router2(config-if)#ip adde4
Router2(config-if)#ip adde
Router2(config-if)#ip add
Router2(config-if)#ip address 192.168.3.1 255.255.255.0
Router2(config-if)#no shutdown
Step-4: Return to Privilege EXEC Mode.
Use the end command to return to privilege EXEC mode.
Router2(config-if)#end
Router2#
Step-5: Save the Router2 configuration
Router2#cop
Router2#copy run
Router2#copy running-config sta
Router2#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router2#

Task-5: Configure IP Addressing on the Host PCs

Step-1: Configure the host PC0.
Configure the host PC0 that is attached to Router1 via switch 0 with an ip address 192.168.1.10/24 and a default gateway of 192.168.1.1.
Step-2: Configure the host PC1.

Configure the host PC1 that is attached to Router1 via switch 0 with an ip address 192.168.1.11/24 and a default gateway of 192.168.1.1.

Step-3: Configure the host PC2.

Configure the host PC2 that is attached to Router1 via switch 0 with an ip address 192.168.1.12/24 and a default gateway of 192.168.1.1.

Step-4: Configure the host PC3.

Configure the host PC3 that is attached to Router2 via switch 1 with an ip address 192.168.3.10/24 and a default gateway of 192.168.3.1.

Step-5: Configure the host PC4.

Configure the host PC4 that is attached to Router2 via switch 1 with an ip address 192.168.3.11/24 and a default gateway of 192.168.3.1.

Step-6: Configure the host PC3.

Configure the host PC3 that is attached to Router2 via switch 1 with an ip address 192.168.3.12/24 and a default gateway of 192.168.3.1.

Task-6: Verify and Test the Configuration

Step-1: Verify the routing tables have the following routes using the *show ip route* command. The show ip route command shows the router's routing table. After implementing this command you will see the two routes between Router1 and Router2. Both routes are designated with a **C.** These are the directly connected networks that were activated when you configured the interfaces on each router.

Router1#sh ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP

i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area

** - candidate default, U - per-user static route, o - ODR*

P - periodic downloaded static route

Gateway of last resort is not set

C *192.168.1.0/24 is directly connected, FastEthernet0/0*

C *192.168.2.0/24 is directly connected, Serial0/0/0*

Router1#

Router2#sh ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP

i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area

** - candidate default, U - per-user static route, o - ODR*

P - periodic downloaded static route

Gateway of last resort is not set

C *192.168.2.0/24 is directly connected, Serial0/0/0*
C *192.168.3.0/24 is directly connected, FastEthernet0/0*
Router2#

Step-2: Verify interface configurations.
Another common problem is router interfaces that are not configured correctly or not activated. Use the *show ip interface brief* command to quickly verify the configuration of each router's interfaces. Your output should look similar to the following.
Router1#sh ip int brief
Interface IP-Address OK? Method Status Protocol

FastEthernet0/0 192.168.1.1 YES manual up ***up***

FastEthernet0/1 unassigned YES unset administratively down down

Serial0/0/0 192.168.2.1 YES manual up ***up***

Serial0/0/1 unassigned YES unset administratively down down

Vlan1 unassigned YES unset administratively down down

Router2#sh ip int br
Router2#sh ip int brief
Interface IP-Address OK? Method Status Protocol

FastEthernet0/0 192.168.3.1 YES manual up ***up***

FastEthernet0/1 unassigned YES unset administratively down down

Serial0/0/0 192.168.2.2 YES manual up ***up***

Serial0/0/1 unassigned YES unset administratively down down

Vlan1 unassigned YES unset administratively down down
Router2#

Step-3: Test connectivity by pinging from each host to the default gateway that has been configured for the host.
Step-4: Test connectivity between Router1 and Router2.

Assignment: Challenge Router Configuration

Topology Diagram



Addressing Table:

Device	Interface	IP Address	Subnet Mask	Def. Gateway
Router-1	**Fa0/0**	**192.168.1.1**	**255.255.255.0**	**N/A**
	S0/0/0	**192.168.2.1**	**255.255.255.0**	**N/A**
Router-2	**Fa0/0**	**192.168.3.1**	**255.255.255.0**	**N/A**
	S0/0/0	**192.168.2.2**	**255.255.255.0**	**N/A**
PC0	**N/A**	**192.168.1.10**	**255.255.255.0**	**192.168.1.1**
PC1	**N/A**	**192.168.3.10**	**255.255.255.0**	**192.168.3.1**

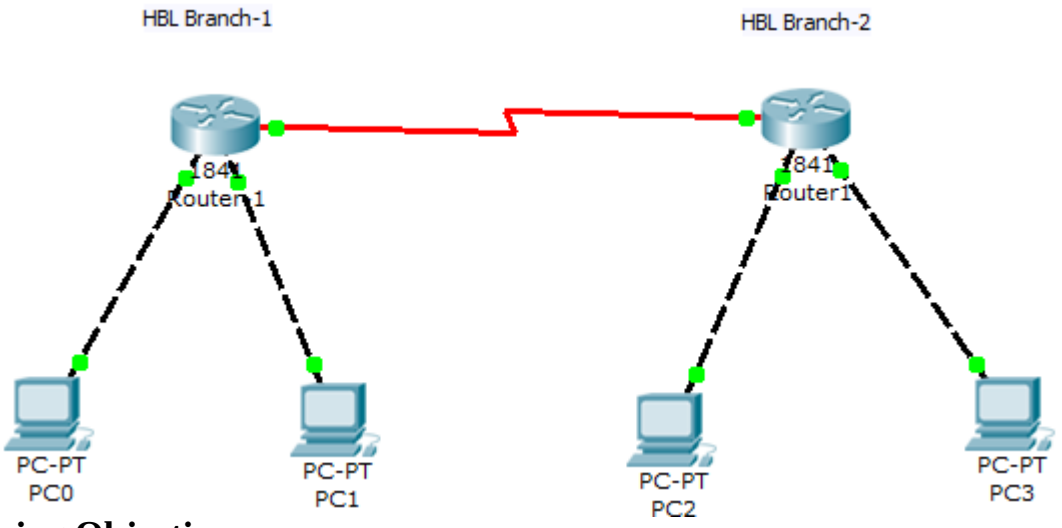
Required Tasks:

1. Cable a network according to the Topology Diagram
2. Erase the Start-up Configuration and reload a router to the default state.
3. Perform basic configuration task on a router
4. Configure and Activate Ethernet interface
5. Test and Verify configurations
6. Show the output of show running, show ip int brief and show ip route commands in task 3, task4 and task5.

LAB-5

Accessing Router through Telnet

Topology Diagram



Learning Objective

Upon completion of this lab, you will be able to:

- Design and cable a network according to topology Diagram
- Make the connectivity between two Routers
- Accessing a Routers via Telnet
- Accessing a PC2 or PC3 from PC0 or PC1 or Vice Versa
- Verify the connectivity of between Hosts

Background

Telnet is a protocol that allows you to connect to remote computers (called hosts) over a TCP/IP network (such as the Internet). You use software called a telnet client on your computer to make a connection to a telnet server (i.e., the remote host). Once your telnet client establishes a connection to the remote host, your client becomes a virtual terminal, allowing you to communicate with the remote host, which from your computer.

Telnet clients are available for all major operating systems.

Command-line telnet clients are built into most versions of Mac OS X, Windows, UNIX and Linux. To use them, go to their respective command lines (i.e., the

Terminal application in Mac OS X, the shell in UNIX or Linux, or the DOS prompt in Windows.

There are five passwords used to secure your Cisco router's consol, auxiliary, telnet (VTY), enable password, and enable secret. The first two passwords are used to set your enable password that's used to secure privileged mode. The other three are used to configure a password when user mode is accessed either through the consol port, through the auxiliary port, or via Telnet.

Scenario

Set up a network similar to the one in the Topology Diagram. Any router that meets the interface requirements may be used. Possible routers include 800, 1600, 1700, 1800, 2500 or a combination. The following resources will be required:

- Crossover cable is used for connecting a workstations to the router
- Serial cable is used between two routers.

Addressing Table:

Device	Interface	IP Address	Subnet Mask	Def. Gateway
Router-1	**Fa0/0** **Fa0/1**	**192.168.1.1** **192.168.2.1**	**255.255.255.0** **255.255.255.0**	**N/A** **N/A**
	S0/0/0	**10.0.0.1**	**255.0.0.0**	**N/A**
Router-2	**Fa0/0** **Fa0/1**	**192.168.3.1** **192.168.4.1**	**255.255.255.0** **255.255.255.0**	**N/A**
	S0/0/0	**10.0.0.2**	**255.0.0.0**	**N/A**
PC0	**N/A**	**192.168.1.2**	**255.255.255.0**	**192.168.1.1**
PC1	**N/A**	**192.168.2.2**	**255.255.255.0**	**192.168.2.1**
PC2	**N/A**	**192.168.3.2**	**255.255.255.0**	**192.168.3.1**
PC3	**N/A**	**192.168.4.2**	**255.255.255.0**	**192.168.4.1**

Task 1: Design and Cable a Network.

Design and cable a network that is similar to the one in the Topology Diagram by using a Cisco Packet Tracer. In this network you will be required a router 1841. Be sure to use the appropriate type of Ethernet cable to connect from host to router, and router to router. 1841 router don't have serial interface ports by default so you will be required to install by manually these ports. The following steps are required for installing a Serial Interface on 1841 router.

Step-1: Click on Router 1841

Step-2: Press the button Power Off

Step-3: You will see different Modules select the WIC-2T module which is called the 2-port asynchronous/synchronous serial network module.

Steps-4: Drag this module to empty slots of Router 1841

Step-5: Press the Power On button

When you will press the button power on you will see the router will boot again from the initial stage. To verify serial port attachment with the 1841 router write the following command on the CLI mode of the router.

Router# show ip interface brief

Router#sh ip int brief

Interface IP-Address OK? Method Status Protocol

FastEthernet0/0 unassigned YES unset administratively down down

FastEthernet0/1 unassigned YES unset administratively down down

Serial0/0/0 unassigned YES unset administratively down down

Serial0/0/1 unassigned YES unset administratively down down

Vlan1 unassigned YES unset administratively down down

The above output is showing that the serial interface ports have attached successfully on router 1841.

Task 2: Make the connectivity between two Routers

Perform Basic Configuration on Router-1

Step-1 Enter a Privileged EXEC mode.
Router1> enable
Router1#
Step-2 Enter global configuration mode.
Router1#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router1(config)#
Step-3: Configure the hostname as Router1
Enter the command hostname Router1
Router1(config)#hostname Router1
Router1(config)#
Step-4: Configure EXEC Mode Password.
Step-5: Configure a message-of-the-day banner.
Router1(config)#ban
Router1(config)#banner motd &
Enter TEXT message. End with the character '&'.
!!!!!!!!!!!AUTHORIZED ACCESS ONLY !!!!!!!!!!

&
Router1(config)#
Step-6: Configure the console password on the Router-1:
Use cisco as the password. When you are finished, exit from line configuration mode.
Router1(config)#line
Router1(config)#line cons
Router1(config)#line console 0
Router1(config-line)#pass
Router1(config-line)#password cisco
Router1(config-line)#logg
Router1(config-line)#logging synch
Router1(config-line)#logging synchronous
Router1(config-line)#exec
Router1(config-line)#exec-timeout 15 15
Router1(config-line)#exit
Router1(config)#
Step-7: Configure the telnet password on the Router-1
Router1(config)#line vty 0 4
Router1(config-line)#pass
Router1(config-line)#password cisco
Router1(config-line)#login
Router1(config-line)#logg
Router1(config-line)#logging synch
Router1(config-line)#logging synchronous
Router1(config-line)#exec
Router1(config-line)#exec-timeout 15 15
Router1(config-line)#exit
Router1(config)#
Step-8: Configure the Fast Ethernet0/0 interface.
Router1(config)#int
Router1(config)#interface fas
Router1(config)#interface fastEthernet 0/0
Router1(config-if)#ip address 192.168.1.1 255.255.255.0
Router1(config-if)#no shutdown
Router1(config-if)#exit
Router1(config)int fastEthernet 0/1
Router1(config-if)#ip address 192.168.2.1 255.255.255.0
Router1(config-if)#no shutdown
Step-8: Configure the Serial0/0/0 interface.
Router1(config)#int
Router1(config)#interface ser
Router1(config)#interface serial 0/0/0
Router1(config-if)#ip add
Router1(config-if)#ip address 10.0.0.1 255.0.0.0

Router1(config-if)#clock rate 64000
Router1(config-if)#no shutdown
Router1(config-if)#

Note: The interface will not be activated until the serial interface on Router2 is configured and activated.

Perform Basic Configuration on Router-2

Step-1 Enter a Privileged EXEC mode.
Router2> enable
Router2#
Step-2 Enter global configuration mode.
Router2#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router2(config)#
Step-3: Configure the hostname as Router2
Enter the command hostname Router2
Router2(config)#hostname Router1
Router2(config)#
Step-4: Configure EXEC Mode Password.
Step-5: Configure a message-of-the-day banner.
Router2(config)#ban
Router2(config)#banner motd &
Enter TEXT message. End with the character '&'.
!!!!!!!!!!!AUTHORIZED ACCESS ONLY !!!!!!!!!!
&
Router2(config)#
Step-6: Configure the console password on the Router-2:
Use cisco as the password. When you are finished, exit from line configuration mode.
Router2(config)#line
Router2(config)#line cons
Router2(config)#line console 0
Router2(config-line)#pass
Router2(config-line)#password cisco
Router2(config-line)#logg
Router2(config-line)#logging synch
Router2(config-line)#logging synchronous
Router2(config-line)#exec
Router2(config-line)#exec-timeout 15 15
Router2(config-line)#exit
Router2(config)#
Step-7: Configure the telnet password on the Router-2
Router2(config)#line vty 0 4
Router2(config-line)#pass

Router2(config-line)#password cisco
Router2(config-line)#login
Router2(config-line)#logg
Router2(config-line)#logging synch
Router2(config-line)#logging synchronous
Router2(config-line)#exec
Router2(config-line)#exec-timeout 15 15
Router2(config-line)#exit
Router2(config)#
Step-8: Configure the Fast Ethernet0/0 interface.
Router2(config)#int
Router2(config)#interface fas
Router2(config)#interface fastEthernet 0/0
Router2(config-if)#ip address 192.168.3.1 255.255.255.0
Router2(config-if)#no shutdown
Router2(config-if)#exit
Router2(config)int fastEthernet 0/1
Router2(config-if)#ip address 192.168.4.1 255.255.255.0
Router2(config-if)#no shutdown
Step-8: Configure the Serial0/0/0 interface.
Router2(config)#int
Router2(config)#interface ser
Router2(config)#interface serial 0/0/0
Router2(config-if)#ip add
Router2(config-if)#ip address 10.0.0.2 255.0.0.0
Router2(config-if)#no shutdown
Router2(config-if)#

Task-3 Accessing Routers via Telnet



Step-1: Telnet Router-2 from Router-1

Router-1#
Router-1#telnet 10.0.0.2
Trying 10.0.0.2 ...Open
User Access Verification
Password:
Router-2>en
Password:
Router-2#

Step-2: Telnet Router-1 from Router- 2

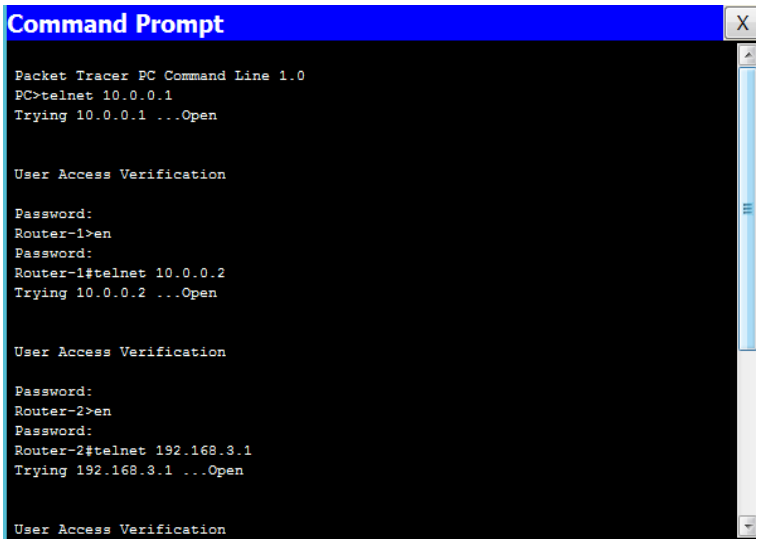
Router-2#
Router-2#telnet 10.0.0.1
Trying 10.0.0.1 ...Open
User Access Verification
Password:
Router-1>en
Password:
Router-1#

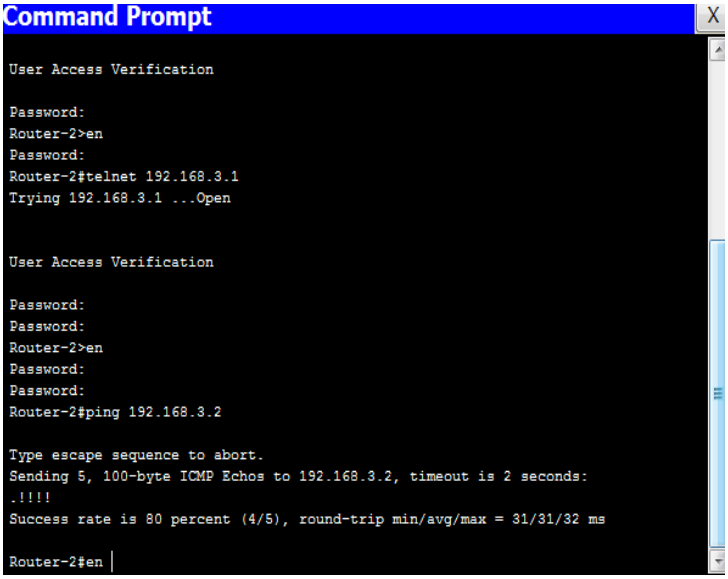
Task-4: Accessing a PC2 and PC3 from PC0 and PC1 or Vice Versa

Step-1 Click on PC0



Step-2 Click on command prompt





Task-5 Verify the Connectivity between Hosts

1) Ping PC1 from PC0

PC>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Reply from 192.168.2.1: bytes=32 time=32ms TTL=255

Reply from 192.168.2.1: bytes=32 time=31ms TTL=255

Reply from 192.168.2.1: bytes=32 time=31ms TTL=255

Reply from 192.168.2.1: bytes=32 time=31ms TTL=255

Ping statistics for 192.168.2.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 31ms, Maximum = 32ms, Average = 31ms

2) Ping PC0 from PC1

Packet Tracer PC Command Line 1.0

PC>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=63ms TTL=255

Reply from 192.168.1.1: bytes=32 time=31ms TTL=255

Reply from 192.168.1.1: bytes=32 time=31ms TTL=255

Reply from 192.168.1.1: bytes=32 time=31ms TTL=255

Ping statistics for 192.168.1.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 31ms, Maximum = 63ms, Average = 39ms

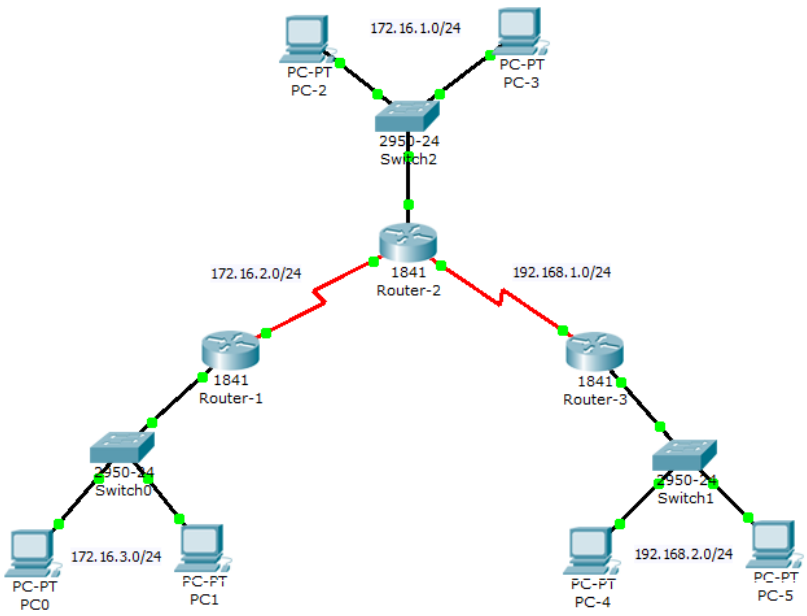
Assignment: Study about the Secure Shell (SSH)

SECTION-III

ROUTING

LAB-6 Static Route

Topology Diagram



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router-1	Fa0/0	172.16.3.1	255.255.255.0	N/A
	S0/0/0	172.16.2.1	255.255.255.0	N/A
Router-2	Fa0/0	172.16.1.1	255.255.255.0	N/A
	S0/0/0	172.16.2.2	255.255.255.0	N/A
	S0/0/1	192.168.1.2	255.255.255.0	N/A
Router-3	Fa0/0	192.168.2.1	255.255.255.0	N/A
	S0/0/1	192.168.1.1	255.255.255.0	N/A
PC0	NIC	172.16.3.3	255.255.255.0	172.16.3.1
PC1	NIC	172.16.3.4	255.255.255.0	172.16.3.1
PC2	NIC	172.16.1.2	255.255.255.0	172.16.1.1
PC3	NIC	172.16.1.3	255.255.255.0	172.16.1.1
PC4	NIC	192.168.2.3	255.255.255.0	192.168.2.1
PC5	NIC	192.168.2.4	255.255.255.0	192.168.2.1

Learning Objectives

Upon completion of this lab, you will be able to:

Cable a network according to the Topology Diagram.
Erase the start up configuration and reload a router to the default state.
Perform basic configuration tasks on a router.
Interpret debug ***ip routing*** output.
Configure and activate Serial and Ethernet interfaces.
Test Connectivity.
Gather information to discover causes for lack of connectivity between devices.
Configure a static route using an intermediate address.

Scenario

In this lab activity, you will create a network that is similar to the one shown in the Topology Diagram. Begin by cabling the network as shown in the Topology Diagram. You will then perform the initial router configurations required for connectivity. Use the IP addresses that are provided in the Addressing Table to apply an addressing scheme to the network devices. After completing the basic configuration, test connectivity between the devices on the network. First test the connections between directly connected devices, and then test connectivity between devices that are not directly connected. Static Routes must be configured on the routers for end-to-end communication to take place between the network hosts. You will configure the static routes that are needed to allow communication between the hosts. View the routing table after each static route is added to observe how the routing table has changed.

Task 1: Cable, Erase, and Reload the Routers.

Step 1: Cable a network that is similar to the one in the Topology Diagram.

Design and cable a network that is similar to the one in the Topology Diagram by using a Cisco Packet Tracer. In this network you will be required a router 1841. Be sure to use the appropriate type of Ethernet cable to connect from host to switch, switch to router and router to router. 1841 router don't have serial interface ports by default so you will be required to install by manually these ports. The following steps are required for installing a Serial Interface on 1841 router.

Step-1: Click on Router 1841

Step-2: Press the button Power Off

Step-3: You will see different Modules select the WIC-2T module which is called the 2-port asynchronous/synchronous serial network module.

Steps-4: Drag this module to empty slots of Router 1841

Step-5: Press the Power On button

When you will press the button power on you will see the router will boot again from the initial stage. To verify serial port attachment with the 1841 router write the following command on the CLI mode of the router.

Router# show ip interface brief

Router#sh ip int brief

Interface IP-Address OK? Method Status Protocol

FastEthernet0/0 unassigned YES unset administratively down down

FastEthernet0/1 unassigned YES unset administratively down down

Serial0/0/0 unassigned YES unset administratively down down

Serial0/0/1 unassigned YES unset administratively down down

Vlan1 unassigned YES unset administratively down down

Step 2: Clear the configuration on each router

Clear the configuration on each of the routers using the **erase startup-config** command and then **reload** the routers. Answer **no** if asked to save changes.

Task 2: Perform Basic Router Configuration.

Note: If you have difficulty with any of the commands in this task, see the Lab 5: Basic Router Configuration.

Step 1: Use global configuration commands.

On the routers, enter global configuration mode and configure the basic global configuration commands including:

- host name
- no ip domain-lookup
- enable secret

Step 2: Configure the consol and virtual terminal line passwords on each of the routers.

Enter first global configuration mode and then line configuration mode and use the following commands for implementing the consol and telnet password. Add the logging synchronous command to the consol and virtual terminal lines for preventing IOS messages delivered to the consol or Telnet lines from interrupting your keyboard input. Add the exec-timeout commands to set the interval that the EXEC command interpreter waits until user input is detected we can use the **exec-timeout** line configuration command.

- **Router-1(config) #line consol 0**
- **Router-1(config-line) #logging synchronous**
- **Router-1(config-line) #exec-timeout 15 15**
- **Router-1(config-line) #password cisco**
- **Router-1(config-line) #login**
- **Router-1(config) #line vty 0 4**
- **Router-1(config-line) #logging synchronous**
- **Router-1(config-line) #exec-timeout 15 15**
- **Router-1(config-line) #password cisco**
- **Router-1(config-line) #login**

Note: Implement the same commands on Router-2 and Router-3 for configuring the consol and telnet password.

Task-3: Interpreting Debug Output

Note: If you already configured IP addressing on R1, please remove all interface commands now before proceeding R1, R2 and R3 should be configured through the end of Task 2 without any interface configurations.

Step-1: On R1 from privileged EXEC mode, enter the debug ip routing command.

R1# debug ip routing

IP routing debugging is on

The **debug ip routing** command shows when routes are added, modified and deleted from the routing table. For example, every time you successfully configure and activate an interface. Cisco IOS adds a route to the routing table. We can verify this by observing output from the **debug ip routing** command.

Step-2: Enter interface configuration mode for R1's LAN interface.

R1# Router-1(config)#int f0/0
Router-1(config-if)#ip address 172.16.3.1 255.255.255.0
Router-1(config-if)#no shutdown
%LINK-5-CHANGED: Interface FastEthernet0/0, ***changed state to up***
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
RT: interface FastEthernet0/0 added to routing table
RT: SET_LAST_RDB for 172.16.3.0/24
NEW rdb: is directly connected
RT: add 172.16.3.0/24 via 0.0.0.0, connected metric [0/0]
RT: NET-RED 172.16.3.0/24
Router-1(config-if)#

The new network you configured on the LAN interface is now added to the routing table, as shown in the highlighted output.

If you don't see the route added to the routing table, the interface did not come up. Use the following systematic process to troubleshoot your connection.

1- Check your physical connections to the LAN interface. An interface will not come up unless it detects a carrier detect signal at the physical layer from another device. The interface must be connected with switch or Hub or PC.
2- Check the cabling, the correct cables connected to the devices.
3- The interface must be activated or enabled.

Step-3: Enter the command to verify that the new route is now in the routing table.

Your output should look similar to the following output. There should now be one route in the table for R1.

Router-1#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
** - candidate default, U - per-user static route, o - ODR*
P - periodic downloaded static route
Gateway of last resort is not set
172.16.0.0/24 is subnetted, 1 subnets
C 172.16.3.0 is directly connected, FastEthernet0/0

Step-4: Enter the interface configuration mode for R1's WAN interface connected to R2.
Router-1#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router-1(config)#inter
Router-1(config)#interface ser
Router-1(config)#interface serial 0/0/0
Router-1(config-if)#ip add
Router-1(config-if)#ip address 172.16.2.1 255.255.255.0
Is_up: 0 state: 0 sub state: 1 line: 0 has_route: False

As soon as you press the ENTER key, Cisco IOS debug output informs you that there is now a route, but its state is False. Because R1 is the DCE side of our lab environment, we must specify how fast the bits will be clocked between R1 and R2.

Step-5: Enter the clock rate command on R1.

You can specify and valid clocking speed. Use the ? to find the valid rates. Here, we used 64000 bps
Router-1(config-if)#clock rate 64000
Router-1(config-if)#no shutdown

Unlike configuring the LAN interface, fully configuring the WAN interface does not always guarantee that the route will be entered in the routing table, even if your cable connections are correct. The other side of the WAN link must also be configured.

Step-6: Configure the serial link of R2.
Enter the **debug ip routing** on privilege EXEC mode.

Router-2# debug ip routing

IP routing debugging is on

Enter interface configuration mode for R2's WAN interface connected to R1.

Router-2(config)#interface s0/0/0
Router-2(config-if)#ip address 172.16.2.2 255.255.255.0
Router-2(config-if)#no shutdown
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

After you enter the correct command, you should see debug output similar to the following example.

RT: interface Serial0/0/0 added to routing table

RT: SET_LAST_RDB for 172.16.2.0/24
NEW rdb: is directly connected

RT: add 172.16.2.0/24 via 0.0.0.0, connected metric [0/0]

RT: NET-RED 172.16.2.0/24

The new network that you configured on the WAN interface s now added to the routing table, as shown in the highlighted out put

Step-7: Enter the command to verify that the new route is now in the routing table for R1 and R2.
Your output should look similar to the following output. There should now be two routes in the routing table for R1 and one route in the table for R2. The highlighted area are showing that the 2 routes added to R1 routing table and one route added to R2 routing table.

Router-1#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
** - candidate default, U - per-user static route, o - ODR*

P - periodic downloaded static route

Gateway of last resort is not set

172.16.0.0/24 is subnetted, 2 subnets
C 172.16.2.0 is directly connected, Serial0/0/0
C 172.16.3.0 is directly connected, FastEthernet0/0

Router-2#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
** - candidate default, U - per-user static route, o - ODR*
P - periodic downloaded static route

Gateway of last resort is not set

172.16.0.0/24 is subnetted, 1 subnets
C 172.16.2.0 is directly connected, Serial0/0/0

Step-8: Turn off debugging on both routers using either *no debug ip routing* **or simply,** *undebug all***.**

Router-1#no debug ip routing
IP routing debugging is off

Router-2#no debug ip routing
IP routing debugging is off

Task-4: Finish Configuring Router Interfaces

Step-1: Configure Remaining R2 interfaces

Configure the Fast Ethernet Interface of R2 According to the Topology Diagram and Addressing Table.

Configure the remaining Serial Interface of R2 with R3 According to the Topology Diagram and Addressing Table.

Step-2: Configure R3 interfaces

Configure the Fast Ethernet Interface of R3 According to the Topology Diagram and Addressing Table.

Configure the Serial Interface of R3 with R2 According to the Topology Diagram and Addressing Table.

Task-5: Configure IP Addressing on the Host PCs.

Step-1: Configure the host PC0.

Configure the host PC0 with an IP address of 172.16.3.3/24 and default gateway of 172.16.3.1

Step-2: Configure the host PC1.

Configure the host PC1 with an IP address of 172.16.3.4/24 and default gateway of 172.16.3.1

Step-3: Configure the host PC2.

Configure the host PC2 with an IP address of 172.16.1.2/24 and default gateway of 172.16.1.1

Step-4: Configure the host PC3.

Configure the host PC3 with an IP address of 172.16.1.3/24 and default gateway of 172.16.1.1

Step-5: Configure the host PC4.

Configure the host PC4 with an IP address of 192.168.2.3/24 and default gateway of 192.168.2.1

Step-6: Configure the host PC5.

Configure the host PC5 with an IP address of 192.168.2.4/24 and default gateway of 192.168.2.1

Task-6: Test and Verify the Configurations.

Step-1: Test Connectivity

Test connectivity by pinging from each host to the default gateway that has been configured for that host.

Step-2: Use the ping command to test connectivity between directly connected routers.

From the router R2, it should be possible to ping R1 at 172.16.2.1

From the router R2, it should be possible to ping R3 at 192.168.1.2

Task-7: Gather Information

Step-1: Check Status of Interfaces

Check the status of the interfaces on each router with the command ***show ip interface brief***

The following output is for R1

Router-1#sh ip int brief

Interface IP-Address OK? Method Status Protocol

FastEthernet0/0 172.16.3.1 YES manual up up

FastEthernet0/1 unassigned YES unset administratively down down

Serial0/0/0 172.16.2.1 YES manual up up

Serial0/0/1 unassigned YES unset administratively down down

Vlan1 unassigned YES unset administratively down down

The following output for R2.

Router-2#sh ip int brief
Interface IP-Address OK? Method Status Protocol

FastEthernet0/0 172.16.1.1 YES manual up up

FastEthernet0/1 unassigned YES unset administratively down down

Serial0/0/0 172.16.2.2 YES manual up up

Serial0/0/1 192.168.1.2 YES manual up up

Vlan1 unassigned YES unset administratively down down

The following Output for R3.

Router-3#sh ip int brief
Interface IP-Address OK? Method Status Protocol

FastEthernet0/0 192.168.2.1 YES manual up up

FastEthernet0/1 unassigned YES unset administratively down down

Serial0/0/0 unassigned YES manual administratively down down

Serial0/0/1 192.168.1.1 YES manual up up

Vlan1 unassigned YES unset administratively down down

Step-2: View the routing table information for all three routers

Router-1#sh ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
** - candidate default, U - per-user static route, o - ODR*
P - periodic downloaded static route

Gateway of last resort is not set

172.16.0.0/24 is subnetted, 2 subnets
C 172.16.2.0 is directly connected, Serial0/0/0
C 172.16.3.0 is directly connected, FastEthernet0/0

Router-2#sh ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
** - candidate default, U - per-user static route, o - ODR*
P - periodic downloaded static route

Gateway of last resort is not set

172.16.0.0/24 is subnetted, 2 subnets
C 172.16.1.0 is directly connected, FastEthernet0/0
C 172.16.2.0 is directly connected, Serial0/0/0
C 192.168.1.0/24 is directly connected, serial 0/0/1

Router-3#sh ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

C 192.168.1.0/24 is directly connected, Serial0/0/1
C 192.168.2.0/24 is directly connected, Fast Ethernet 0/0

Task-8: Configure a Static Route Using a Next-Hop Address.

Step-1 To configure static routes with a next-hope specified, use the following syntax.

Router (config) # ip route *network-address subnet-mask ip-address*

Network-address: Destination network address of the remote network to be added to the routing table.

Subnet-Mask: Subnet mask of the remote network to be added to the routing table. The subnet mask can be modified o summarize a group of networks.

Ip-address: Commonly referred to as the next-hop router's IP address.

On the R3 router, configure a static route to the 172.16.1.0 network using the serial 0/0/1 interface of R2 as the next hope address.

Step-2 View the routing table to verify the new static route entry on each router.

Notice that the route is coded with an S, which means that the route is Static route.

Router-3#sh ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

172.16.0.0/24 is subnetted, 2 subnets
S 172.16.1.0 [1/0] via 192.168.1.2
S 172.16.3.0 [1/0] via 192.168.1.2
C 192.168.1.0/24 is directly connected, Serial0/0/1
C 192.168.2.0/24 is directly connected, FastEthernet0/0

The highlighted area is showing that Static Route has added in the R3 routing Table. With this route entered in the routing table, any packet that matches the first 24 left most bits of 172.16.1.0/24 will be forwarded to the next hope route at 192.168.1.2

Router-2#sh ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

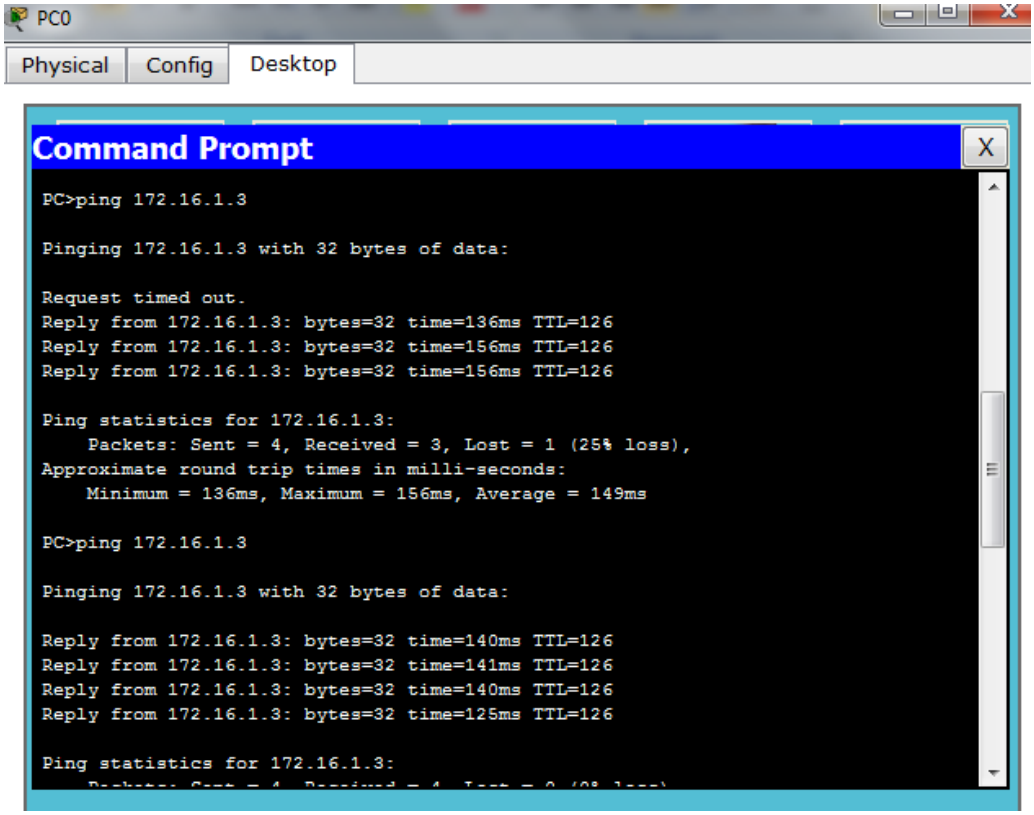
172.16.0.0/24 is subnetted, 3 subnets
C 172.16.1.0 is directly connected, FastEthernet0/0
C 172.16.2.0 is directly connected, Serial0/0/0
S 172.16.3.0 [1/0] via 172.16.2.1
C 192.168.1.0/24 is directly connected, Serial0/0/1
S 192.168.2.0/24 [1/0] via 192.168.1.1

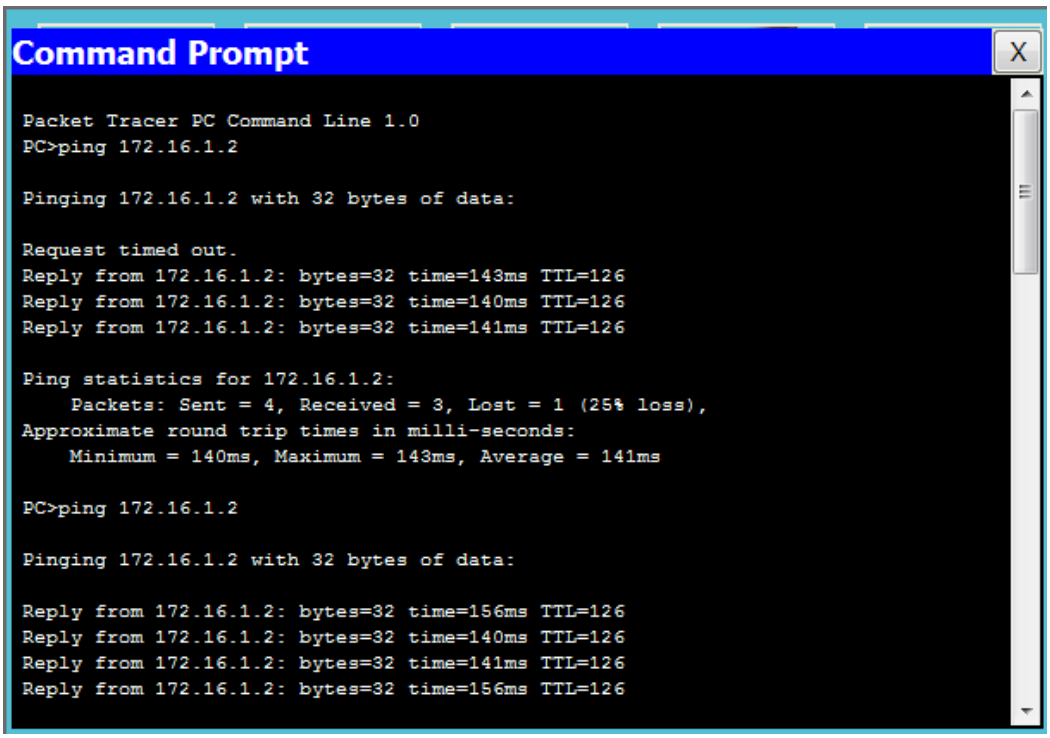
Router-1#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

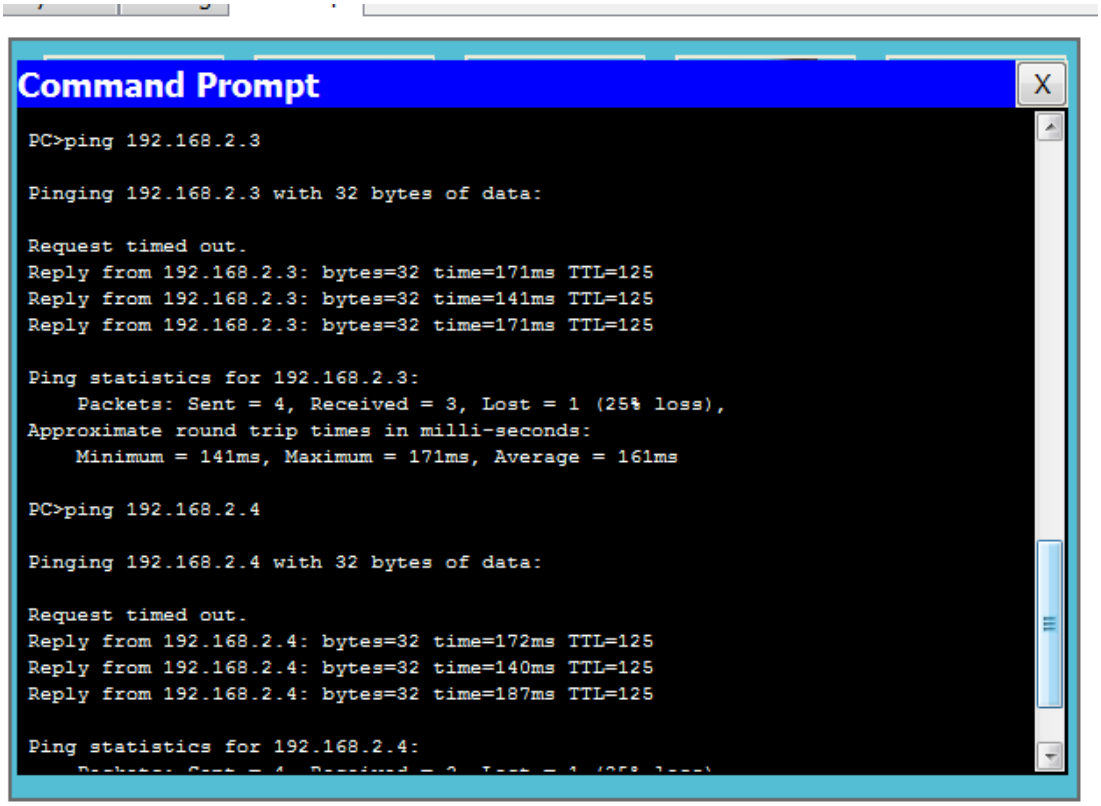
172.16.0.0/24 is subnetted, 3 subnets
S 172.16.1.0 [1/0] via 172.16.2.2
C 172.16.2.0 is directly connected, Serial0/0/0
C 172.16.3.0 is directly connected, FastEthernet0/0
S 192.168.2.0/24 [1/0] via 172.16.2.2

Step-3: Use the ping to check connectivity between the host PC0, PC1 and host PC2 and PC3.



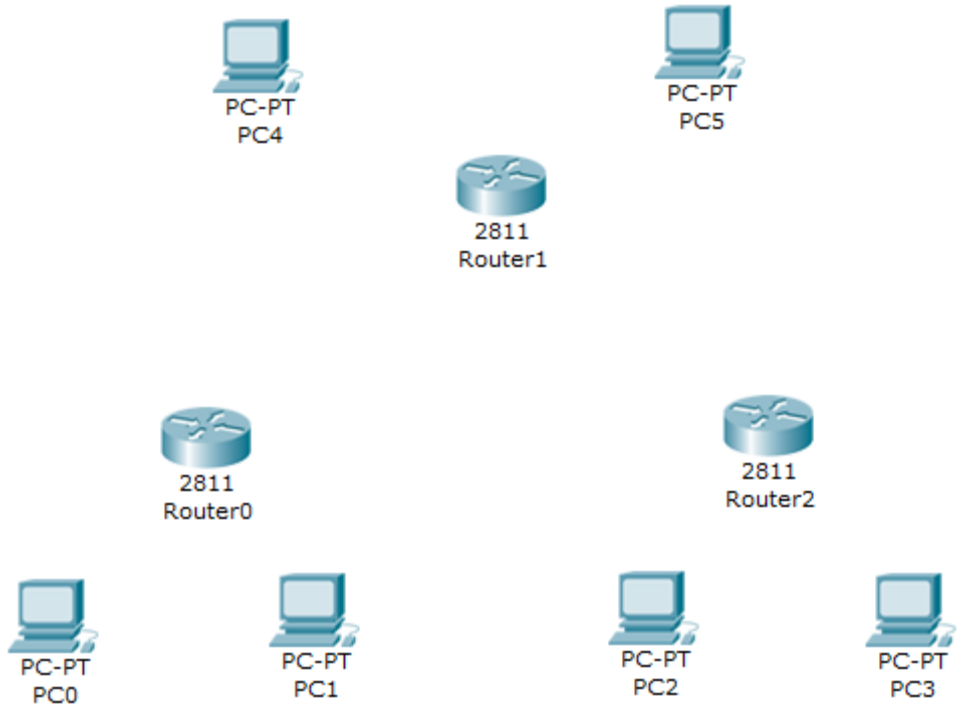


Step-4: Use the ping to check connectivity between the host PC0, PC1 and host PC4 and PC5.



Assignment: Configure the Static Route using an exit interface. What is the main difference between default static route and intermediate static route?

Topology Diagram:



Required Task:

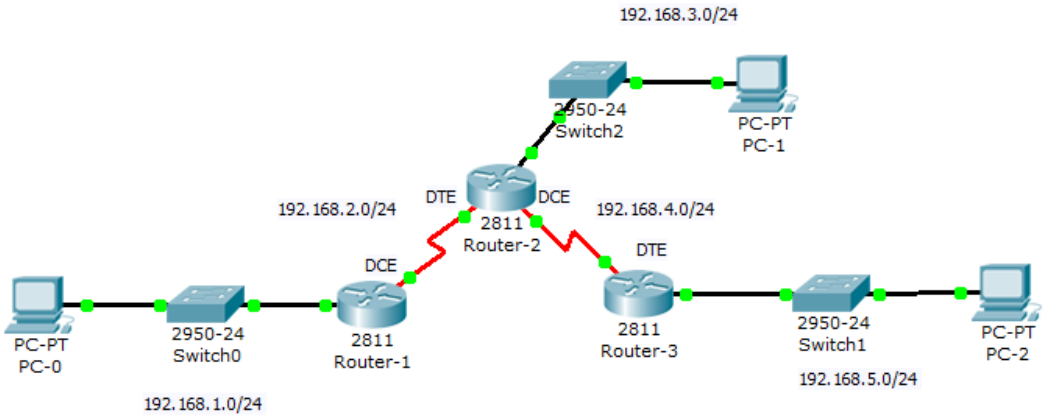
1. Cable a Network
2. Perform Basic Configuration Tasks on Router
3. Configure and Activate Serial and Ethernet Interface
4. Test connectivity result of each router
5. Configure a static route using an exit interface
6. Show the ping result between each connected hosts in the network.

Note: Pleas show the result of each required task in the Assignment.

LAB-7

RIP (Routing Information Protocol) Configuration

Topology Diagram



Learning Objective

Upon completion of this lab, you will be able to:

1. Cable a network according to the Topology Diagram.
2. Erase a startup configuration and reload a router to the default state.
3. Perform basic configuration tasks on a router.
4. Interpret debug ***ip routing*** output
5. Configure and active interfaces.
6. Configure RIP routing on all routers.
7. Verify RIP routing using **show** and **debug** commands.
8. Gather information about RIP processing using **debug ip rip** command.
9. Test Connectivity between the Hosts

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gatway
Router-1	Fa0/0	192.168.1.1	255.255.255.0	N/A
	S0/0/0	192.168.2.1	255.255.255.0	N/A
Router-2	Fa0/0	192.168.3.1	255.255.255.0	N/A
	S0/1/0	192.168.2.2	255.255.255.0	N/A
	S0/1/1	192.168.4.2	255.255.255.0	N/A
Router-3	Fa0/0 S0/1/1	192.168.5.1 192.168.4.1	255.255.255.0 255.255.255.0	N/A N/A
PC0	NIC	192.168.1.10	255.255.255.0	192.168.1.1
PC1	NIC	192.168.3.10	255.255.255.0	192.168.3.1
PC2	NIC	192.168.5.10	255.255.255.0	192.168.5.1

Scenario

In this lab activity, you will create a network that is similar to the one shown in the Topology Diagram. Begin by cabling the network as shown in the Topology Diagram. You will then perform the initial router configurations required for connectivity. Use the IP addresses that are provided in the Addressing Table to apply an addressing scheme to the network devices. After completing the basic configuration, test connectivity between the devices on the network. First test the connections between directly connected devices, and then test connectivity between devices that are not directly connected.

RIP (Routing Information Protocol)

RIP (Routing Information Protocol) is a widely-used protocol for managing router information within a self-contained network such as a corporate local area network (LAN) or an interconnected group of such LANs. RIP is classified by the Internet Engineering Task Force (IETF) as one of internal gateway protocols (Interior Gateway Protocol). RIP is widely used for routing traffic in the global Internet and is an interior gateway protocol (IGP), which means that it performs routing within a single autonomous system.

IGP (Interior Gateway Protocol): A routing protocol that was designed and intended for use inside a single autonomous system (AS).

Autonomous System (AS): An autonomous system is an internetwork under the administrative control of a single organization.

Task-1: Prepare the Network

Step-1: Cable the network that is similar to the one in the topology diagram.

Design and cable a network that is similar to the one in the Topology Diagram by using a Cisco Packet Tracer. In this network you will be required a router 2811. Be sure to use the appropriate type of Ethernet cable to connect from host to switch, switch to router. 2811 router don't have serial interface ports by default so you will be required to install by manually these ports. The following steps are required for installing a Serial Interface on 2811 router.

Step-1: Click on Router 2811

Step-2: Press the button Power Off

Step-3: You will see different Modules select the WIC-2T module which is called the 2-port asynchronous/synchronous serial network module.

Steps-4: Drag this module to empty slots of Router 2811

Step-5: Press the Power On button

When you will press the button power on you will see the router will boot again from the initial stage. To verify serial port attachment with the 2811 router write the following command on router's privilege mode in CLI.

Router#sh ip interface brief

Interface IP-Address OK? Method Status Protocol

FastEthernet0/0 unassigned YES unset administratively down down

FastEthernet0/1 unassigned YES unset administratively down down

Serial0/1/0 unassigned YES unset administratively down down

Serial0/1/1 unassigned YES unset administratively down down

Vlan1 unassigned YES unset administratively down down

Task-2: Erase, and Reload the Routers

Clear the configuration on each of the routers using the **erase startup-config** command and then **reload** the routers. Answer **no** if asked to save changes. The following commands will be use for erasing the NVRAM and reload the router for initial default state.

Router#erase startup-config

Erasing the nvram filesystem will remove all configuration files! Continue? [confirm]
[OK]
Erase of nvram: complete

Router#reload

Proceed with reload? [confirm]
%SYS-5-RELOAD: Reload requested by console. Reload Reason: Reload Command.

System Bootstrap, Version 12.1(3r)T2, RELEASE SOFTWARE (fc1)
Copyright (c) 2000 by cisco Systems, Inc.
cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory

Self decompressing the image :
##
[OK]

Task-3 Perform the Basic Router Configuration

Note: If you have difficulty with any of the commands in this task, see the Lab 5: Basic Router Configuration.

Step 1: Use global configuration commands.

On the routers, enter global configuration mode and configure the basic global configuration commands including:

- host name
- no ip domain-lookup
- enable secret

Step 2: Configure the consol and virtual terminal line passwords on each of the routers.

Enter first global configuration mode and then line configuration mode and use the following commands for implementing the consol and telnet password. Add the logging synchronous command to the consol and virtual terminal lines for preventing IOS messages delivered to the consol or Telnet lines from interrupting your keyboard input. Add the exec-timeout commands to set the interval that the EXEC command interpreter waits until user input is detected we can use the **exec-timeout** line configuration command.

- **Router-1(config) #line consol 0**
- **Router-1(config-line) #logging synchronous**
- **Router-1(config-line) #exec-timeout 15 15**
- **Router-1(config-line) #password cisco**
- **Router-1(config-line) #login**
- **Router-1(config) #line vty 0 4**
- **Router-1(config-line) #logging synchronous**
- **Router-1(config-line) #exec-timeout 15 15**
- **Router-1(config-line) #password cisco**
- **Router-1(config-line) #login**

Note: Implement the same commands on Router-2 and Router-3 for configuring the consol and telnet password.

Task-4: Interpreting Debug Output

Note: If you already configured IP addressing on R1, please remove all interface commands now before proceeding R1, R2 and R3 should be configured through the end of Task 3 without any interface configurations.

Step-1: On R1 from privileged EXEC mode, enter the debug ip routing command.

R1# debug ip routing

IP routing debugging is on

The **debug ip routing** command shows when routes are added, modified and deleted from the routing table. For example, every time you successfully configure and activate an interface. Cisco IOS adds a route to the routing table. We can verify this by observing output from the **debug ip routing** command. As shown in the following figure in which directly connected route is added to Router-1 routing table.

```
Router-1(config)#int f0/0
Router-1(config-if)#ip add
Router-1(config-if)#ip address 192.168.1.1 255.255.255.0
Router-1(config-if)#no shutdown

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state t
o up

RT: interface FastEthernet0/0 added to routing table
RT: SET_LAST_RDB for 192.168.1.0/24

    NEW rdb: is directly connected


RT: add 192.168.1.0/24 via 0.0.0.0, connected metric [0/0]

RT: NET-RED 192.168.1.0/24

Router-1(config-if)#
```

Note: Activate debug ip routing command on every router as shown in the topology diagram before going to Router's Interface Configuration.

Task-5: Interface Configurations

Step-1: Enter interface configuration mode for Router-1's LAN interface.

Router-1(config)#int f0/0
Router-1(config-if)#ip address 192.168.1.1 255.255.255.0
Router-1(config-if)#no shutdown

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Step-2: Enter the interface configuration mode for Router-1's WAN interface connected to Router-2.

Router-1(config)interface s0/1/0
Router-1(config-if)#ip address 192.168.2.1 255.255.255.0
Router-1(config-if)#clock rate 64000
Router-1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

Router-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

Step-3: Enter interface configuration mode for Router-2's LAN interface.

Router-2(config)#int f0/0
Router-2(config-if)#ip address 192.168.3.1 255.255.255.0
Router-2(config-if)#no shutdown
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Step-4: Enter the interface configuration mode for Router-2's WAN interface connected to Router-1 and Router-3.

Router-2(config)#int s0/1/0
Router-2(config-if)#ip address 192.168.2.2 255.255.255.0
Router-2(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

Router-2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

Router-2(config)#int s0/1/1
Router-2(config-if)#ip address 192.168.4.2 255.255.255.0
Router-2(config-if)#clock rate 64000
Router-2(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/1/1, changed state to up

Router-2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up

Step-5: Enter interface configuration mode for Router-3's LAN interface.

Router-3(config)#int f0/0
Router-3(config-if)#ip address 192.168.5.1 255.255.255.0
Router-3(config-if)#no shutdown
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Step-6: Enter the interface configuration mode for Router-3's WAN interface connected to Router-2.
Router-3(config)interface s0/1/1
Router-3(config-if)#ip address 192.168.4.1 255.255.255.0
Router-3(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

Router-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

Step-7: Enter the command to verify that the new route is now in the routing table of Router-1, Router-2 and Router-3.

Router-1#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
** - candidate default, U - per-user static route, o - ODR*
P - periodic downloaded static route

Gateway of last resort is not set

C 192.168.1.0/24 is directly connected, FastEthernet0/0
C 192.168.2.0/24 is directly connected, Serial0/1/0

According to the above routing table of Router-1 there are two directly connected routes that are added to routing table of Router-1.

Router-2#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
** - candidate default, U - per-user static route, o - ODR*
P - periodic downloaded static route

Gateway of last resort is not set

C 192.168.2.0/24 is directly connected, Serial0/1/0
C 192.168.3.0/24 is directly connected, FastEthernet0/0
C 192.168.4.0/24 is directly connected, Serial0/1/1

According to the above routing table of Router-2 there are three directly connected routes that are added to routing table of Router-2.

Router-3#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
** - candidate default, U - per-user static route, o - ODR*
P - periodic downloaded static route

Gateway of last resort is not set

C 192.168.4.0/24 is directly connected, Serial0/1/1
C 192.168.5.0/24 is directly connected, FastEthernet0/0

According to the above routing table of Router-3 there are two directly connected routes that are added to routing table of Router-3.

Step-8: Configure Ethernet interfaces of PC0, PC1 and PC2

Configure the Ethernet interfaces of PC0, PC1 and PC3 with the ip addresses and default gateways from the table under the Topology diagram.

Step-9: Test the PC configuration by pinging the default gateway from the PC.

Task-6: Configure RIP Routing on All Routers

Step-1 Enable a dynamic routing.

To enable a dynamic routing protocol, enter global configuration mode and use the **router** command. **Enter router?** at the global configuration prompt to see a list of available routing protocols on your router.

To enable a RIP, enter the command **router rip** in the global configuration mode.

Router-1#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router-1(config)#router rip
Router-1(config-router)#

Step-2 Enter Classful network Addresses.

Once you are in routing configuration mode, enter the classful network address for each directly connected network, using the **network** command.

Router-1(config)#router rip
*Router-1(config-router)#**network 192.168.1.0***
*Router-1(config-router)#**network 192.168.2.0***
Router-1(config-router)#

The **network** command:

- Enable RIP (Routing Information Protocol) on all interfaces that belong to this network. These interfaces will now both send and receive RIP updates.
- Advertises this network in RIP routing updates sent to other routers every 30 seconds

When you are finished with the RIP configuration, return to privileged EXEC mode and save the current configuration to NVRAM.

Router-1(config-router)#end
Router-1#copy run start

Step-3 Configure RIP on the Router-2 using the *router rip* and *network* command.

Router-2(config)#router rip
*Router-2(config-router)#**network 192.168.2.0***
*Router-2(config-router)#**network 192.168.3.0***
*Router-2(config-router)# **network 192.168.4.0***
Router-2(config-router)#end
Router-2# copy run start

When you are finished with the RIP configuration, return to privileged EXEC mode and save the current configuration to NVRAM.

Step-4 Configure RIP on the Router-3 using the *router rip* and *network* command.

Router-3(config)#router rip
*Router-3(config-router)#**network 192.168.4.0***
*Router-3(config-router)#**network 192.168.5.0***
Router-3(config-router)#end
Router-3# copy run start

When you are finished with the RIP configuration, return to privileged EXEC mode and save the current configuration to NVRAM.

Task-7: Verify RIP routing using *show* commands.

Step-1: Use the *show ip route* command to verify that each router has all of the networks in the topology entered in the routing table.

Routes learned through RIP are coded with an **R** in the routing table. If the tables are not converged as shown in the following output of Router-1, troubleshoot your configuration.

Router-1#sh ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
** - candidate default, U - per-user static route, o - ODR*
P - periodic downloaded static route

Gateway of last resort is not set

C 192.168.1.0/24 is directly connected, FastEthernet0/0
C 192.168.2.0/24 is directly connected, Serial0/1/0
R 192.168.3.0/24 [120/1] via 192.168.2.2, 00:00:06, Serial0/1/0
R 192.168.4.0/24 [120/1] via 192.168.2.2, 00:00:06, Serial0/1/0

Router-2#sh ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
** - candidate default, U - per-user static route, o - ODR*
P - periodic downloaded static route

Gateway of last resort is not set

R 192.168.1.0/24 [120/1] via 192.168.2.1, 00:00:04, Serial0/1/0
C 192.168.2.0/24 is directly connected, Serial0/1/0
C 192.168.3.0/24 is directly connected, FastEthernet0/0
C 192.168.4.0/24 is directly connected, Serial0/1/1
R 192.168.5.0/24 [120/1] via 192.168.4.1, 00:00:22, Serial0/1/1

Router-3#sh ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
** - candidate default, U - per-user static route, o - ODR*
P - periodic downloaded static route

Gateway of last resort is not set

R 192.168.1.0/24 [120/2] via 192.168.4.2, 00:00:23, Serial0/1/1
R 192.168.2.0/24 [120/1] via 192.168.4.2, 00:00:23, Serial0/1/1
R 192.168.3.0/24 [120/1] via 192.168.4.2, 00:00:23, Serial0/1/1
C 192.168.4.0/24 is directly connected, Serial0/1/1
C 192.168.5.0/24 is directly connected, FastEthernet0/0

Step-2: Use the *show ip protocols* command to view information about the routing processes.

The show ip protocols command can be used to view information about the routing processes that are occurring on the router. This output can be used to verify most RIP parameters to conform that.

- RIP routing is configured
- The correct interfaces send and receive RIP updates
- The router advertises the correct networks
- RIP neighbours are sending updates

Router-1#sh ip protocols

Routing Protocol is **"rip"**
Sending updates every 30 seconds, next due in 11 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 1, receive any version
Interface Send Recv Triggered RIP Key-chain
FastEthernet0/0 1 2 1
Serial0/1/0 1 2 1
Automatic network summarization is in effect
Maximum path: 4
Routing for Networks:
192.168.1.0
192.168.2.0
Passive Interface(s):
Routing Information Sources:
Gateway Distance Last Update
192.168.2.2 120 00:00:15
Distance: (default is **120**)
Router-1#

R1 is indeed configured with RIP. Router-1 is sending and receiving RIP updates on Fast Ethernet0/0 and Serial0/1/0. Router-1 is advertising networks 192.168.1.0 and 192.168.2.0. Router-1 has one routing information source. Router-2 is sending Router-1 updates.

Router-2#sh ip protocols

Routing Protocol is **"rip"**
Sending updates every 30 seconds, next due in 19 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 1, receive any version

Interface	Send	Recv	Triggered RIP	Key-chain
FastEthernet0/0	**1**	**2**	**1**	
Serial0/1/0	**1**	**2**	**1**	
Serial0/1/1	**1**	**2**	**1**	

Automatic network summarization is in effect
Maximum path: 4
Routing for Networks:
192.168.2.0
192.168.3.0
192.168.4.0
Passive Interface(s):
Routing Information Sources:

Gateway	Distance	Last Update
192.168.2.1	**120**	**00:00:10**
192.168.4.1	**120**	**00:00:06**

Distance: (default is **120**)

Router-3#sh ip protocols

Routing Protocol is **"rip"**
Sending updates every 30 seconds, next due in 27 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 1, receive any version

Interface	Send	Recv	Triggered RIP	Key-chain
FastEthernet0/0	**1**	**2**	**1**	
Serial0/1/1	**1**	**2**	**1**	

Automatic network summarization is in effect
Maximum path: 4
Routing for Networks:
192.168.4.0
192.168.5.0
Passive Interface(s):
Routing Information Sources:

Gateway	Distance	Last Update
192.168.4.2	**120**	**00:00:10**

Distance: (default is **120**)

Task-8: Gather information about RIP processing using debug ip rip command.

Step-1: Use the *debug ip rip* command to view the RIP messages being sent and received.

Router-1#debug ip rip
RIP protocol debugging is on
Router-1#RIP: received v1 update from 192.168.2.2 on **Serial0/1/0**

192.168.3.0 in 1 hops

192.168.4.0 in 1 hops

192.168.5.0 in 2 hops

RIP: sending v1 update to 255.255.255.255 via **FastEthernet0/0** (192.168.1.1)

RIP: build update entries

network 192.168.2.0 metric 1

network 192.168.3.0 metric 2

network 192.168.4.0 metric 2

network 192.168.5.0 metric 3

RIP: sending v1 update to 255.255.255.255 via **Serial0/1/0** (192.168.2.1)

RIP: build update entries

network 192.168.1.0 metric 1

RIP: received v1 update from 192.168.2.2 on **Serial0/1/0**

192.168.3.0 in 1 hops

192.168.4.0 in 1 hops

192.168.5.0 in 2 hops

The debug output shows that Router-1 receives an update from Router-2. Notice that how this update includes all the networks that Router-1 does not already have in its routing table. Because the Fast Ethernet0/0 interface belongs to the 192.168.1.0 network configured under

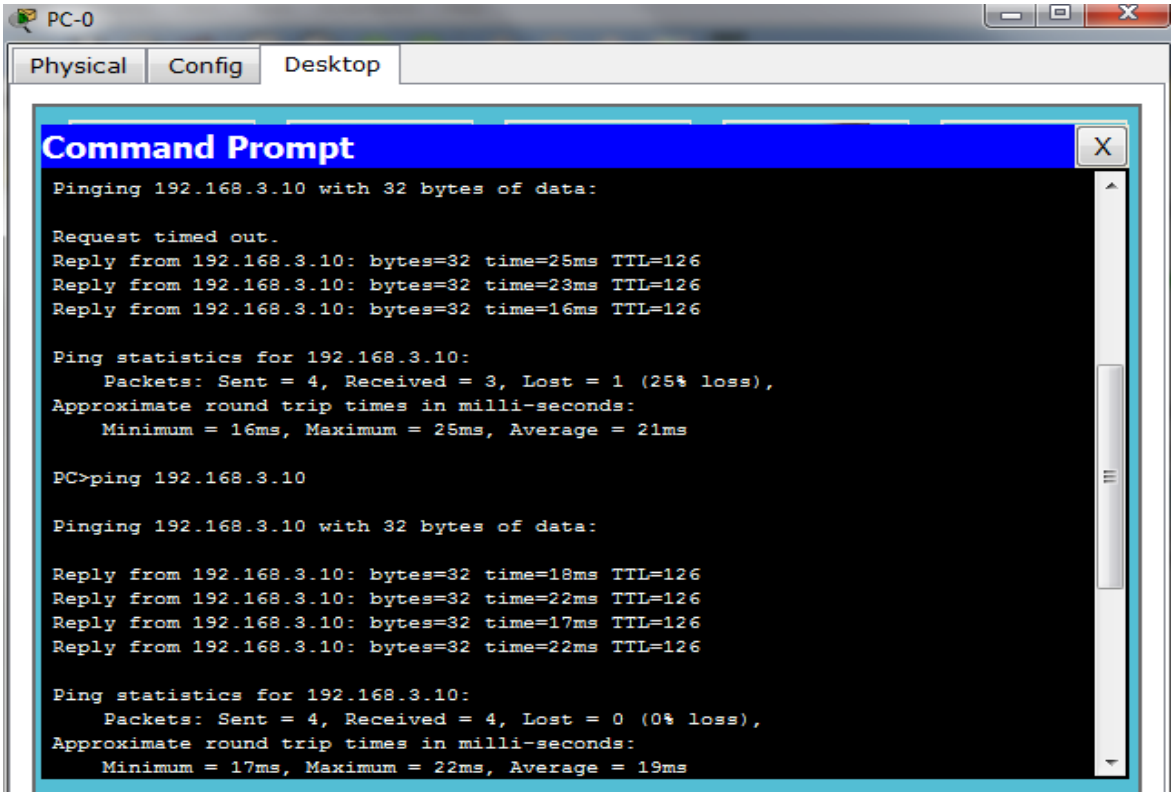
RIP. Router-1 builds an update to send out that interface. The update includes all networks known to Router-1 except the network of the interface. Finally, R1 builds an update to send to Router-2.

Note: Write down the command debug ip rip on Router-2 and Router-3 for knowing the information about RIP routing messages and their updates.

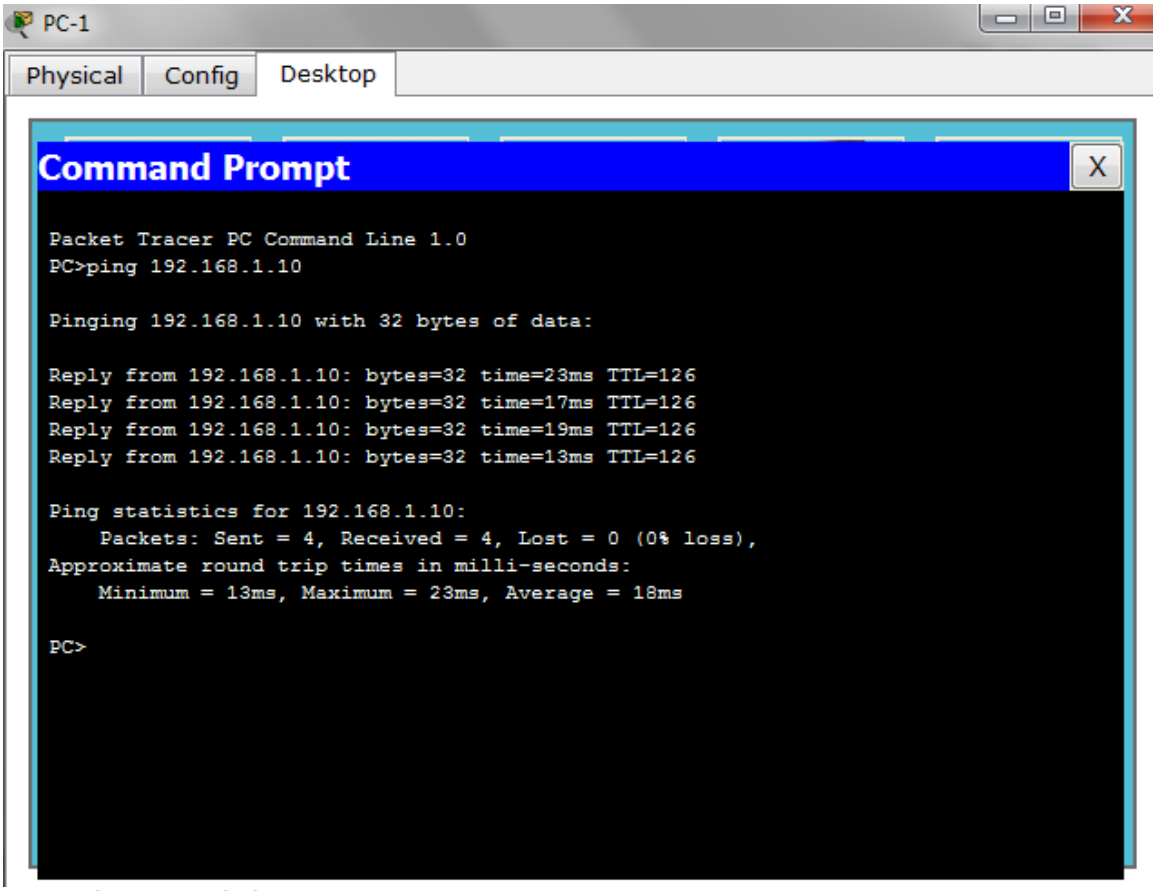
Task-9: Test Connectivity between the Hosts

Step-1: Use the ping command to check connectivity between the host PC0, PC1 and host PC2.

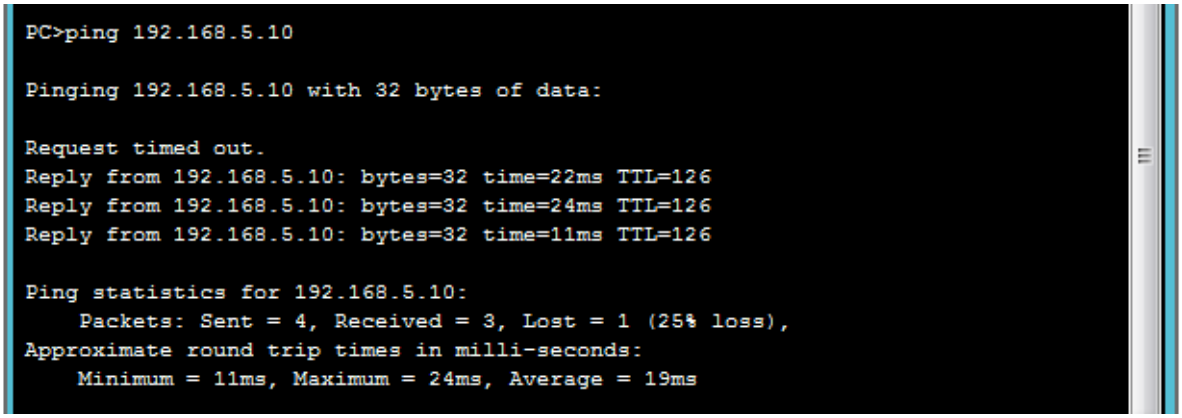
From PC0 to PC1



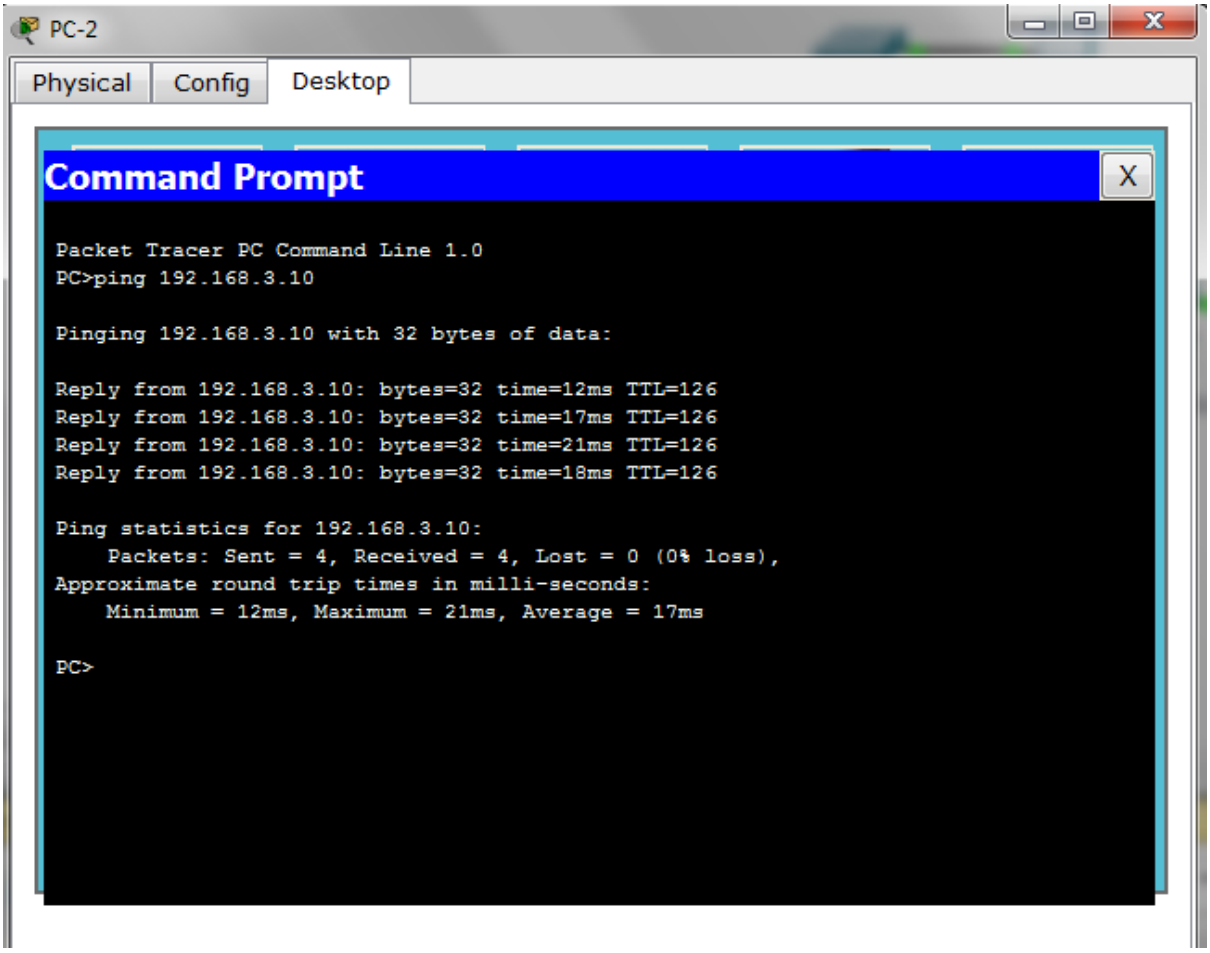
From PC1 to PC0



From PC 1 to PC 2



From PC2 to PC1



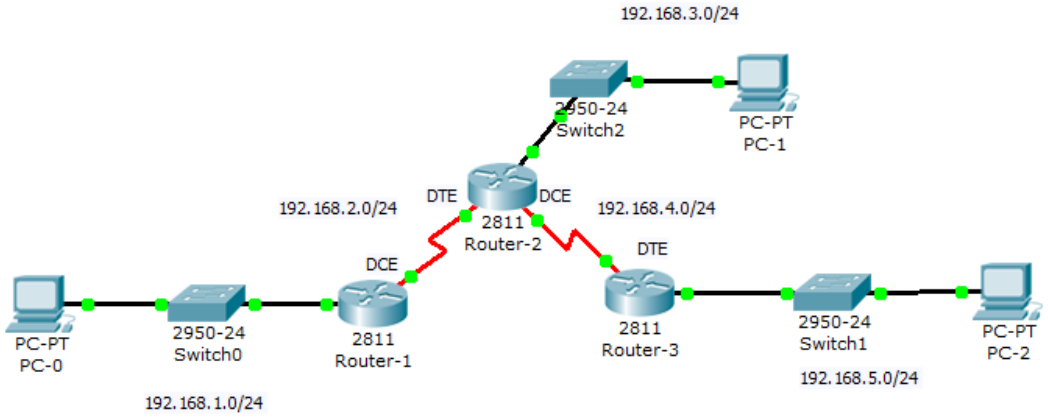
Assignment:

- What are the disadvantages in RIP V1 routing protocol?
- What is main difference between RIP V1 and RIP V2?
- Why we have required Passive Interface on Router's Fast Ethernet Interface during the Implementation of RIP routing protocols in the network?

LAB-8

EIGRP (Enhanced Integrated Routing Protocol) Configuration

Topology Diagram



Learning Objective

Upon completion of this lab, you will be able to:

1. Cable a network according to the Topology Diagram.
2. Erase a startup configuration and reload a router to the default state.
3. Perform basic configuration tasks on a router.
4. Interpret debug ***ip routing*** output
5. Configure and active interfaces.
6. Configure EIGRP routing protocol on all routers.
7. Verify EIGRP routing using **show** commands.
8. Test Connectivity between the Hosts

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gatway
Router-1	**Fa0/0**	**192.168.1.1**	**255.255.255.0**	**N/A**
	S0/0/0	**192.168.2.1**	**255.255.255.0**	**N/A**
Router-2	**Fa0/0**	**192.168.3.1**	**255.255.255.0**	**N/A**
	S0/1/0	**192.168.2.2**	**255.255.255.0**	**N/A**
	S0/1/1	**192.168.4.2**	**255.255.255.0**	**N/A**
Router-3	**Fa0/0**	**192.168.5.1**	**255.255.255.0**	**N/A**
	S0/1/1	**192.168.4.1**	**255.255.255.0**	**N/A**

PC0	**NIC**	**192.168.1.10**	**255.255.255.0**	**192.168.1.1**
PC1	**NIC**	**192.168.3.10**	**255.255.255.0**	**192.168.3.1**
PC2	**NIC**	**192.168.5.10**	**255.255.255.0**	**192.168.5.1**

Scenario

In this lab activity, you will create a network that is similar to the one shown in the Topology Diagram. Begin by cabling the network as shown in the Topology Diagram. You will then perform the initial router configurations required for connectivity. Use the IP addresses that are provided in the Addressing Table to apply an addressing scheme to the network devices. After completing the basic configuration, test connectivity between the devices on the network. First test the connections between directly connected devices, and then test connectivity between devices that are not directly connected.

EIGRP (Enhanced Integrated Routing Protocol)

EIGRP is a Cisco-proprietary Hybrid routing protocol, incorporating features of both Distance-Vector and Link-State routing protocols. Sometimes Cisco refers to EIGRP as simply an advanced distance vector protocol, but in other cases, Cisco refers to EIGRP as a new type: a balanced hybrid routing protocol.

EIGRP adheres to the following Hybrid characteristics:

- EIGRP uses **Diffusing Update Algorithm (DUAL)** to determine the best path among all "feasible" paths. DUAL also helps ensure a loop- free routing environment.
- EIGRP will form **neighbor** relationships with adjacent routers in the same **Autonomous System (AS**).
- **Reliable Transport Protocol (RTP)** is used to ensure delivery of most EIGRP packets.
- EIGRP routers **do not** send periodic, full-table routing updates. Updates are sent when a change occurs.
- EIGRP is a classless protocol, and thus supports VLSMs.

Task-1: Prepare the Network

Step-1: Cable the network that is similar to the one in the topology diagram.

Design and cable a network that is similar to the one in the Topology Diagram by using a Cisco Packet Tracer. In this network you will be required a router 2811. Be sure to use the appropriate type of Ethernet cable to connect from host to switch, switch to router. 2811 router don't have serial interface ports by default so you will be required to install by manually these ports. The following steps are required for installing a Serial Interface on 2811 router.

Step-1: Click on Router 2811

Step-2: Press the button Power Off

Step-3: You will see different Modules select the WIC-2T module which is called the 2-port asynchronous/synchronous serial network module.

Steps-4: Drag this module to empty slots of Router 2811

Step-5: Press the Power On button

When you will press the button power on you will see the router will boot again from the initial stage. To verify serial port attachment with the 2811 router write the following command on router's privilege mode in CLI.

Router#sh ip interface brief

Interface IP-Address OK? Method Status Protocol

FastEthernet0/0 unassigned YES unset administratively down down

FastEthernet0/1 unassigned YES unset administratively down down

Serial0/1/0 unassigned YES unset administratively down down

Serial0/1/1 unassigned YES unset administratively down down

Vlan1 unassigned YES unset administratively down down

Task-2: Erase, and Reload the Routers

Clear the configuration on each of the routers using the **erase startup-config** command and then **reload** the routers. Answer **no** if asked to save changes. The following commands will be use for erasing the NVRAM and reload the router for initial default state.

Router#erase startup-config

Erasing the nvram filesystem will remove all configuration files! Continue? [confirm]
[OK]
Erase of nvram: complete

Router#reload

Proceed with reload? [confirm]
%SYS-5-RELOAD: Reload requested by console. Reload Reason: Reload Command.

System Bootstrap, Version 12.1(3r)T2, RELEASE SOFTWARE (fc1)
Copyright (c) 2000 by cisco Systems, Inc.
cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory

Self decompressing the image :

##
[OK]

Task-3 Perform the Basic Router Configuration

Note: If you have difficulty with any of the commands in this task, see the Lab 5: Basic Router Configuration.

Step 1: Use global configuration commands.

On the routers, enter global configuration mode and configure the basic global configuration commands including:

- host name
- no ip domain-lookup
- enable secret

Step 2: Configure the consol and virtual terminal line passwords on each of the routers.

Enter first global configuration mode and then line configuration mode and use the following commands for implementing the consol and telnet password. Add the logging synchronous command to the consol and virtual terminal lines for preventing IOS messages delivered to the consol or Telnet lines from interrupting your keyboard input. Add the exec-timeout commands to set the interval that the EXEC command interpreter waits until user input is detected we can use the **exec-timeout** line configuration command.

- **Router-1(config) #line consol 0**
- **Router-1(config-line) #logging synchronous**
- **Router-1(config-line) #exec-timeout 15 15**
- **Router-1(config-line) #password cisco**
- **Router-1(config-line) #login**
- **Router-1(config) #line vty 0 4**
- **Router-1(config-line) #logging synchronous**
- **Router-1(config-line) #exec-timeout 15 15**
- **Router-1(config-line) #password cisco**
- **Router-1(config-line) #login**

Note: Implement the same commands on Router-2 and Router-3 for configuring the consol and telnet password.

Task-4: Interpreting Debug Output

Note: If you already configured IP addressing on R1, please remove all interface commands now before proceeding R1, R2 and R3 should be configured through the end of Task 3 without any interface configurations.

Step-1: On R1 from privileged EXEC mode, enter the debug ip routing command.

R1# debug ip routing

IP routing debugging is on

The **debug ip routing** command shows when routes are added, modified and deleted from the routing table. For example, every time you successfully configure and activate an interface. Cisco IOS adds a route to the routing table. We can verify this by observing output from the **debug ip routing** command. As shown in the following figure in which directly connected route is added to Router-1 routing table.

```
Router-1(config)#int f0/0
Router-1(config-if)#ip add
Router-1(config-if)#ip address 192.168.1.1 255.255.255.0
Router-1(config-if)#no shutdown

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state t
o up

RT: interface FastEthernet0/0 added to routing table
RT: SET_LAST_RDB for 192.168.1.0/24

    NEW rdb: is directly connected


RT: add 192.168.1.0/24 via 0.0.0.0, connected metric [0/0]

RT: NET-RED 192.168.1.0/24

Router-1(config-if)#
```

Note: Activate debug ip routing command on every router as shown in the topology diagram before going to Router's Interface Configuration.

Task-5: Interface Configurations

Step-1: Enter interface configuration mode for Router-1's LAN interface.

Router-1(config)#int f0/0
Router-1(config-if)#ip address 192.168.1.1 255.255.255.0
Router-1(config-if)#no shutdown

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Step-2: Enter the interface configuration mode for Router-1's WAN interface connected to Router-2.

Router-1(config)interface s0/1/0
Router-1(config-if)#ip address 192.168.2.1 255.255.255.0
Router-1(config-if)#clock rate 64000
Router-1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

Router-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

Step-3: Enter interface configuration mode for Router-2's LAN interface.

Router-2(config)#int f0/0
Router-2(config-if)#ip address 192.168.3.1 255.255.255.0
Router-2(config-if)#no shutdown
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Step-4: Enter the interface configuration mode for Router-2's WAN interface connected to Router-1 and Router-3.

Router-2(config)#int s0/1/0
Router-2(config-if)#ip address 192.168.2.2 255.255.255.0
Router-2(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

Router-2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

Router-2(config)#int s0/1/1
Router-2(config-if)#ip address 192.168.4.2 255.255.255.0
Router-2(config-if)#clock rate 64000
Router-2(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/1/1, changed state to up

Router-2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up

Step-5: Enter interface configuration mode for Router-3's LAN interface.

Router-3(config)#int f0/0
Router-3(config-if)#ip address 192.168.5.1 255.255.255.0

Router-3(config-if)#no shutdown
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Step-6: Enter the interface configuration mode for Router-3's WAN interface connected to Router-2.

Router-3(config)interface s0/1/1
Router-3(config-if)#ip address 192.168.4.1 255.255.255.0
Router-3(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

Router-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

Step-7: Enter the command to verify that the new route is now in the routing table of Router-1, Router-2 and Router-3.

Router-1#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
** - candidate default, U - per-user static route, o - ODR*
P - periodic downloaded static route

Gateway of last resort is not set

C 192.168.1.0/24 is directly connected, FastEthernet0/0
C 192.168.2.0/24 is directly connected, Serial0/1/0

According to the above routing table of Router-1 there are two directly connected routes that are added to routing table of Router-1.

Router-2#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
** - candidate default, U - per-user static route, o - ODR*
P - periodic downloaded static route

Gateway of last resort is not set

C 192.168.2.0/24 is directly connected, Serial0/1/0
C 192.168.3.0/24 is directly connected, FastEthernet0/0

C 192.168.4.0/24 is directly connected, Serial0/1/1

According to the above routing table of Router-2 there are three directly connected routes that are added to routing table of Router-2.

Router-3#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
** - candidate default, U - per-user static route, o - ODR*
P - periodic downloaded static route

Gateway of last resort is not set

C 192.168.4.0/24 is directly connected, Serial0/1/1
C 192.168.5.0/24 is directly connected, FastEthernet0/0

According to the above routing table of Router-3 there are two directly connected routes that are added to routing table of Router-3.

Step-8: Configure Ethernet interfaces of PC0, PC1 and PC2

Configure the Ethernet interfaces of PC0, PC1 and PC3 with the ip addresses and default gateways from the table under the Topology diagram.

Step-9: Test the PC configuration by pinging the default gateway from the PC.

Task-6: Configure EIGRP Routing on All Routers

Step-1 Enable a dynamic routing.

To enable a dynamic routing protocol, enter global configuration mode and use the **router** command. **Enter router?** at the global configuration prompt to see a list of available routing protocols on your router.

To enable an EIGRP, enter the command **router rip 1** in the global configuration mode.

Router-1#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router-1(config)#router eigrp 1
Router-1(config-router)#

Step-2 Enter Classful network Addresses.

Once you are in routing configuration mode, enter the classful network address for each directly connected network, using the **network** command.

*Router-1(config-router)#**network 192.168.1.0***
*Router-1(config-router)#**network 192.168.2.0***
Router-1(config-router)#

The **network** command:

- Enable EIGRP on all interfaces that belong to this network. These interfaces will now both send and receive EIGRP updates.

When you are finished with the EIGRP configuration, return to privileged EXEC mode and save the current configuration to NVRAM.

Router-1(config-router)#end
Router-1#copy run start

Step-3 Configure EIGRP on the Router-2 using the *router eigrp 1* and *network* command.

Router-2(config)#router eigrp 1
*Router-2(config-router)#**network 192.168.2.0***
*Router-2(config-router)#**network 192.168.3.0***
*Router-2(config-router)# **network 192.168.4.0***
Router-2(config-router)#end
Router-2# copy run start

When you are finished with the EIGRP configuration, return to privileged EXEC mode and save the current configuration to NVRAM.

Step-4 Configure EIGRP on the Router-3 using the *router eigrp 1* and *network* command.

Router-3(config)#router eigrp 1
*Router-3(config-router)#**network 192.168.4.0***
*Router-3(config-router)#**network 192.168.5.0***
Router-3(config-router)#end
Router-3# copy run start

When you are finished with the EIGRP configuration, return to privileged EXEC mode and save the current configuration to NVRAM.

Task-7: Verify EIGRP routing using *show* commands.

Step-1: Use the *show ip route* command to verify that each router has all of the networks in the topology entered in the routing table.

Routes learned through EIGRP are coded with an **D** in the routing table. If the tables are not converged as shown in the following output of Router-1, troubleshoot your configuration.

Router-1#sh ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
** - candidate default, U - per-user static route, o - ODR*
P - periodic downloaded static route

Gateway of last resort is not set

C 192.168.1.0/24 is directly connected, FastEthernet0/0
C 192.168.2.0/24 is directly connected, Serial0/1/0
D 192.168.3.0/24 [90/2172416] via 192.168.2.2, 00:11:35, Serial0/1/0
D 192.168.4.0/24 [90/2681856] via 192.168.2.2, 00:10:01, Serial0/1/0
D 192.168.5.0/24 [90/2684416] via 192.168.2.2, 00:10:01, Serial0/1/0

Router-2#sh ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
** - candidate default, U - per-user static route, o - ODR*
P - periodic downloaded static route

Gateway of last resort is not set

D 192.168.1.0/24 [90/2172416] via 192.168.2.1, 00:13:40, Serial0/1/0
C 192.168.2.0/24 is directly connected, Serial0/1/0
C 192.168.3.0/24 is directly connected, FastEthernet0/0
C 192.168.4.0/24 is directly connected, Serial0/1/1
D 192.168.5.0/24 [90/2172416] via 192.168.4.1, 00:11:32, Serial0/1/1

Router-3#sh ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
** - candidate default, U - per-user static route, o - ODR*
P - periodic downloaded static route

Gateway of last resort is not set
D 192.168.1.0/24 [90/2684416] via 192.168.4.2, 00:12:21, Serial0/1/1
D 192.168.2.0/24 [90/2681856] via 192.168.4.2, 00:12:21, Serial0/1/1
D 192.168.3.0/24 [90/2172416] via 192.168.4.2, 00:12:21, Serial0/1/1

C 192.168.4.0/24 is directly connected, Serial0/1/1
C 192.168.5.0/24 is directly connected, FastEthernet0/0

Step-2: Use the *show ip protocols* command to view information about the routing processes.

The show ip protocols command can be used to view information about the routing processes that are occurring on the router. This output can be used to verify most EIGRP parameters to conform that.

- EIGRP routing is configured
- The correct interfaces send and receive EIGRP updates
- The router advertises the correct networks
- EIGRP neighbours are sending updates

Router-1#sh ip protocols

Routing Protocol is "eigrp 1 "
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Default networks flagged in outgoing updates
Default networks accepted from incoming updates
EIGRP metric weight K1=1, K2=0, K3=1, K4=0, K5=0
EIGRP maximum hopcount 100
EIGRP maximum metric variance 1
Redistributing: eigrp 1
Automatic network summarization is in effect
Automatic address summarization:
Maximum path: 4
Routing for Networks:
192.168.1.0
192.168.2.0
Routing Information Sources:

Gateway	**Distance**	**Last Update**
192.168.2.2	**90**	**606203**

Distance: internal 90 external 170

Router-2#sh ip protocols

Routing Protocol is "eigrp 1 "
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Default networks flagged in outgoing updates
Default networks accepted from incoming updates
EIGRP metric weight K1=1, K2=0, K3=1, K4=0, K5=0
EIGRP maximum hopcount 100
EIGRP maximum metric variance 1
Redistributing: eigrp 1

Automatic network summarization is in effect
Automatic address summarization:
Maximum path: 4
Routing for Networks:
192.168.2.0
192.168.3.0
192.168.4.0
Routing Information Sources:

Gateway	Distance	Last Update
192.168.2.1	90	606218
192.168.4.1	90	734218

Distance: internal 90 external 170

Router-3#sh ip protocols

Routing Protocol is "eigrp 1 "
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Default networks flagged in outgoing updates
Default networks accepted from incoming updates
EIGRP metric weight K1=1, K2=0, K3=1, K4=0, K5=0
EIGRP maximum hopcount 100
EIGRP maximum metric variance 1
Redistributing: eigrp 1
Automatic network summarization is in effect
Automatic address summarization:
Maximum path: 4
Routing for Networks:
192.168.4.0
192.168.5.0
Routing Information Sources:

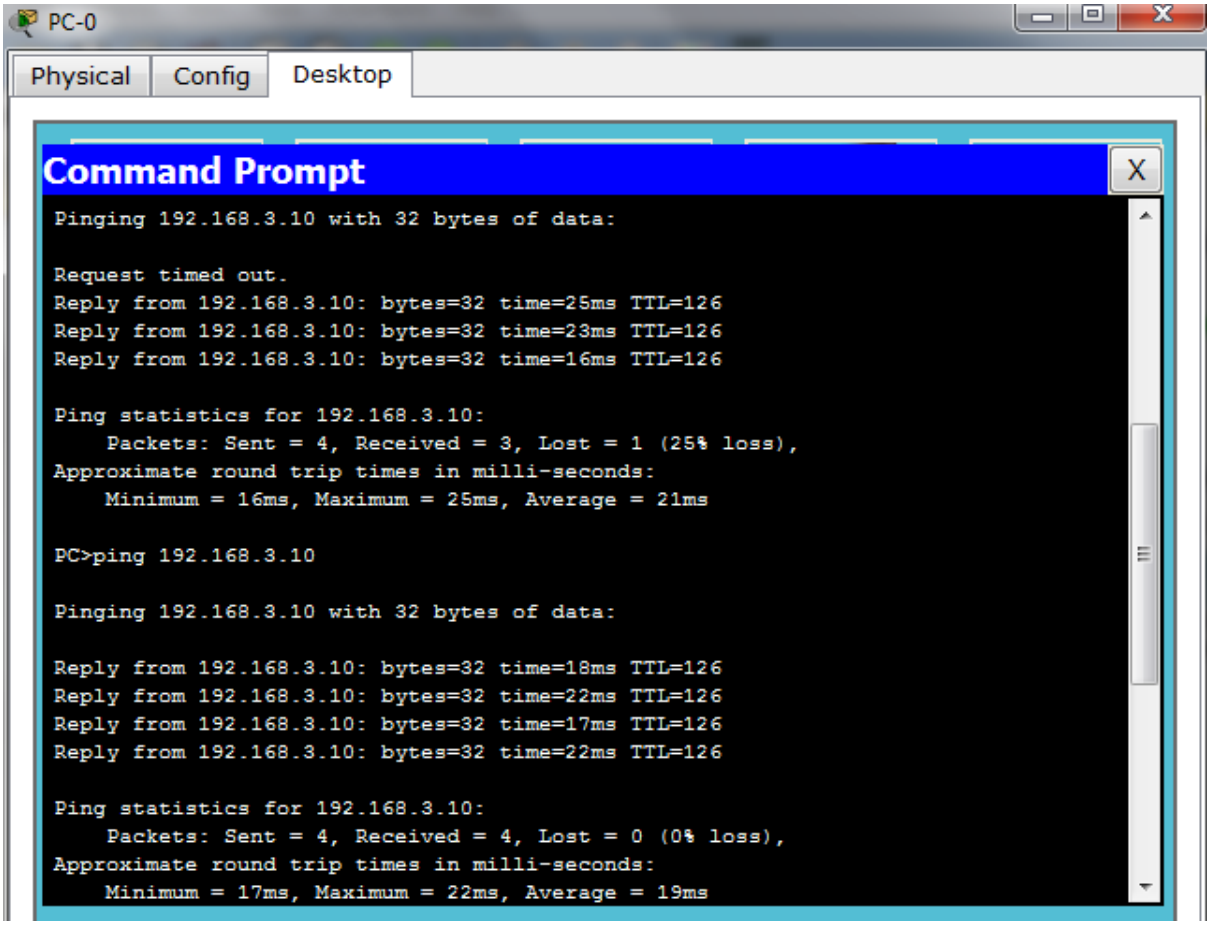
Gateway	Distance	Last Update
192.168.4.2	90	734203

Distance: internal 90 external 170

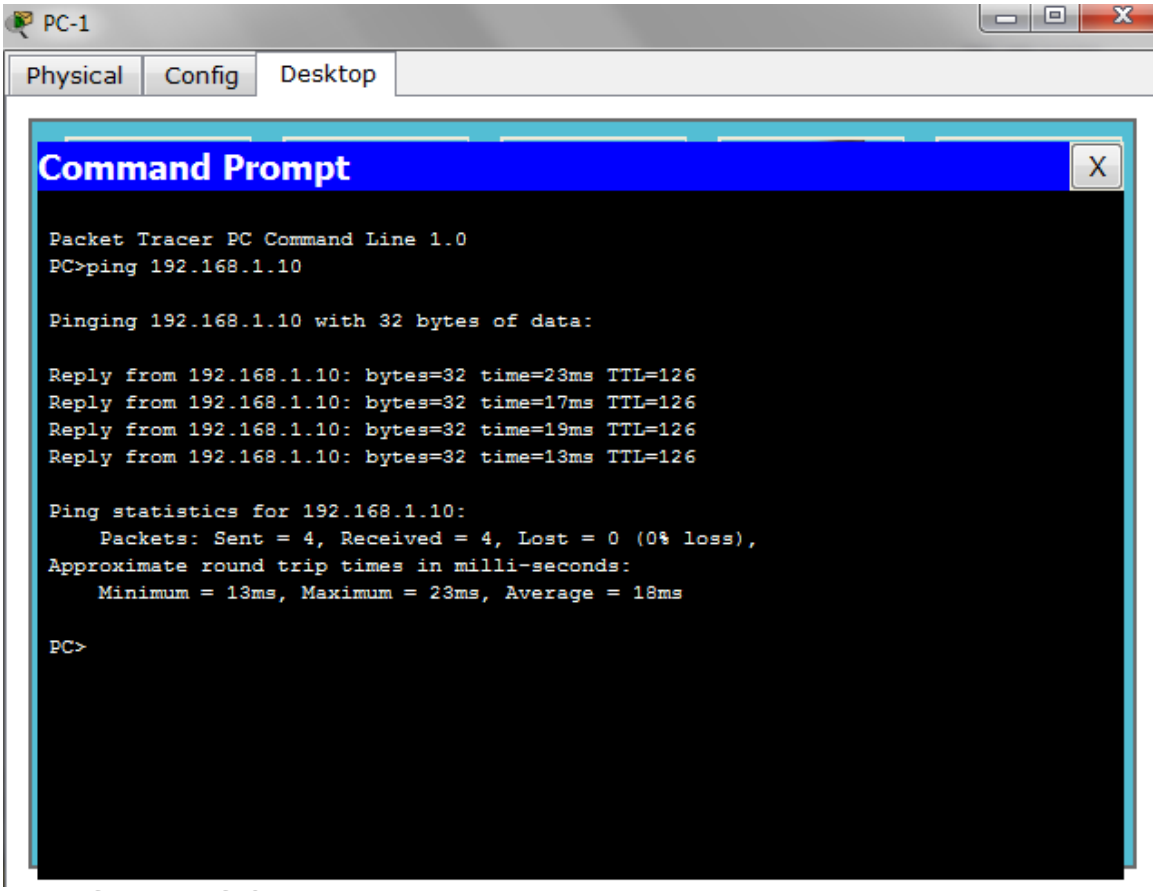
Task-8: Test Connectivity between the Hosts

Step-1: Use the ping command to check connectivity between the host PC0, PC1 and host PC2.

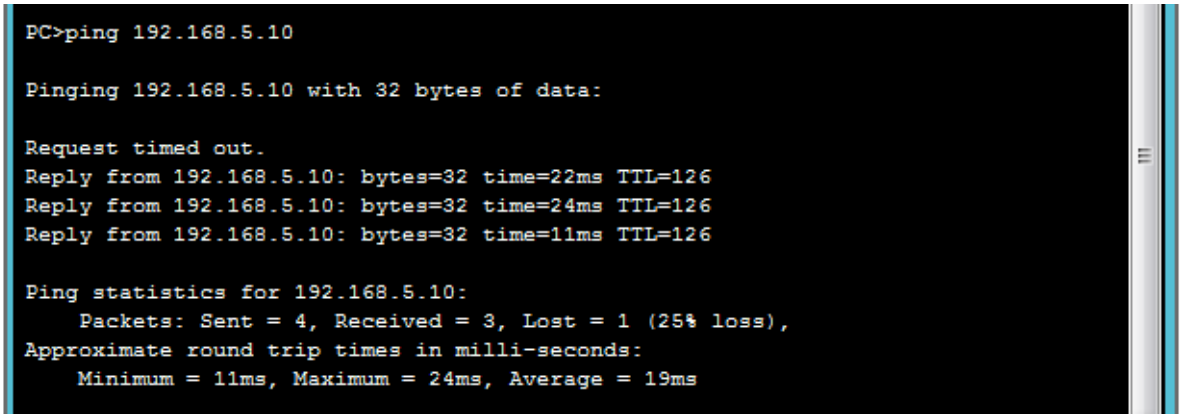
From PC0 to PC1


PC-0
Physical
Config
Desktop
Command Prompt
X
Pinging 192.168.3.10 with 32 bytes of data:
Request timed out.
Reply from 192.168.3.10: bytes=32 time=25ms TTL=126
Reply from 192.168.3.10: bytes=32 time=23ms TTL=126
Reply from 192.168.3.10: bytes=32 time=16ms TTL=126
Ping statistics for 192.168.3.10:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 16ms, Maximum = 25ms, Average = 21ms
PC>ping 192.168.3.10
Pinging 192.168.3.10 with 32 bytes of data:
Reply from 192.168.3.10: bytes=32 time=18ms TTL=126
Reply from 192.168.3.10: bytes=32 time=22ms TTL=126
Reply from 192.168.3.10: bytes=32 time=17ms TTL=126
Reply from 192.168.3.10: bytes=32 time=22ms TTL=126
Ping statistics for 192.168.3.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 17ms, Maximum = 22ms, Average = 19ms

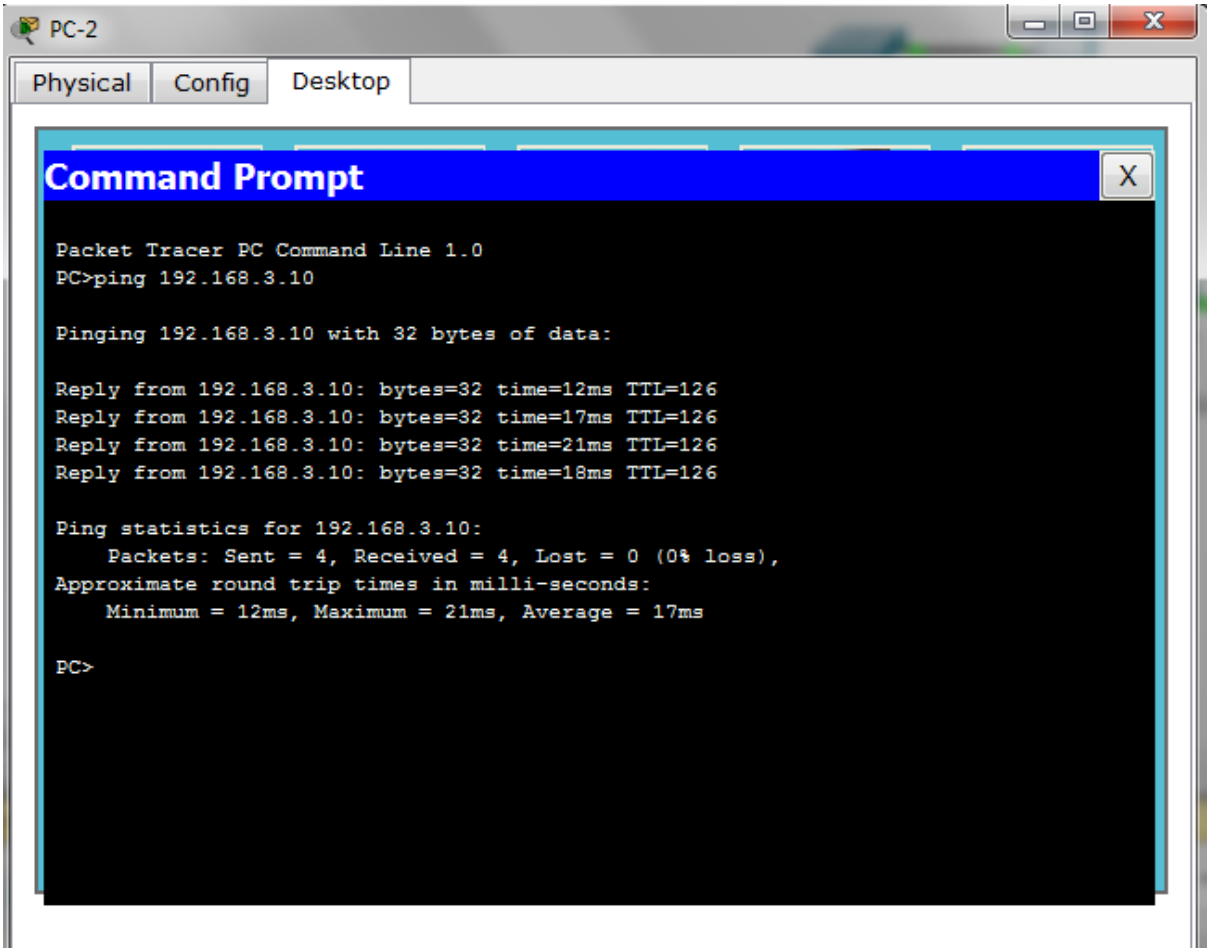
From PC1 to PC0



From PC 1 to PC 2



From PC2 to PC1



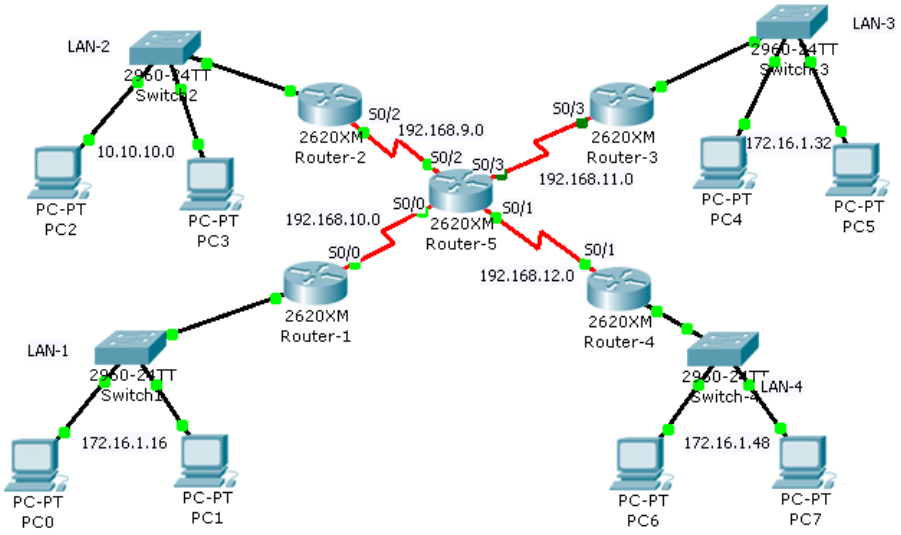
Assignment:

Write down about the Authentication Process of EIGRP. How we can implement Authentication Process in EIGPR routing Protocol.

LAB-9

OSPF (Open Shortest Path First Routing Protocol)

Topology Diagram



Learning Objective

Upon completion of this lab, you will be able to:

10. Cable a network according to the Topology Diagram.
11. Erase a start-up configuration and reload a router to the default state.
12. Perform basic configuration tasks on a router.
13. Interpret debug ***ip routing*** output
14. Configure and activate interfaces.
15. Examine the directly connected route on each router's routing table
16. Configure OSPF routing on all routers.
17. Configure OSPF Router IDs
18. Verify OSPF routing operation.
19. Test Connectivity between the Hosts

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router-1	**Fa0/0**	**172.16.1.17**	**255.255.255.240**	**N/A**
	S0/0	**192.168.10.1**	**255.255.255.252**	**N/A**
Router-2	**Fa0/0**	**10.10.10.1**	**255.255.255.0**	**N/A**

	S0/2	192.168.9.1	255.255.255.252	N/A
Router-3	Fa0/0	172.16.1.33	255.255.255.248	N/A
	S0/3	192.168.11.1	255.255.255.252	N/A
Router-4	F0/0	172.16.1.49	255.255.255.248	N/A
	S0/1	192.16.12.1	255.255.255.252	N/A
Router-5	S0/0	192.168.10.2	255.255.255.252	N/A
	S0/1	192.168.12.2	255.255.255.252	N/A
	S0/2	192.168.9.2	255.255.255.252	N/A
	S0/3	192.168.11.2	255.255.255.252	N/A
Router-6	F0/0 S0/1	172.16.1.56 192.168.13.2	255.255.255.240 255.255.255.252	N/A N/A
Router-7	F0/0 S0/1/1	172.16.1.40 192.16.14.2	255.255.255.240 255.255.255.252	N/A N/A
Router-8	F0/0 S0/0/1	10.10.11.1 192.168.15.2	255.255.255.0 255.	N/A
PC0	NIC	172.16.1.18	255.255.255.240	172.16.1.17
PC1	NIC	172.16.1.19	255.255.255.240	172.16.1.17
PC2	NIC	10.10.10.2	255.255.255.0	10.10.10.1
PC3	NIC	10.10.10.3	255.255.255.0	10.10.10.1
PC4	NIC	172.16.1.34	255.255.255.248	172.16.1.33
PC5	NIC	172.16.1.35	255.255.255.248	172.16.1.33
PC6	NIC	172.16.1.50	255.255.255.248	172.16.1.49
PC7	NIC	172.16.1.51	255.255.255.248	172.16.1.49

PC8	**NIC**	**172.16.1.57**	**255.255.255.240**	**172.16.1.56**
PC9	**NIC**	**172.16.1.58**	**255.255.255.240**	**172.16.1.56**
PC10	**NIC**	**172.16.1.41**	**255.255.255.240**	**172.16.1.40**
PC11	**NIC**	**172.16.1.42**	**255.255.255.240**	**172.16.1.40**
PC12	**NIC**	**172.16.1.43**	**255.255.255.240**	**172.16.1.40**
PC13	**NIC**	**10.10.11.2**	**255.255.255.0**	**10.10.11.1**
PC14	**NIC**	**10.10.11.3**	**255.255.255.0**	**10.10.11.1**

Scenario

In this lab activity, you will create a network that is similar to the one shown in the Topology Diagram. Begin by cabling the network as shown in the Topology Diagram. You will then perform the initial router configurations required for connectivity. Use the IP addresses that are provided in the Addressing Table to apply an addressing scheme to the network devices. After completing the basic configuration, test connectivity between the devices on the network. First test the connections between directly connected devices, and then test connectivity between devices that are not directly connected and after that you will implement the OSPF routing protocol on all the routers and will see the different operation of this routing protocol in this lab.

Task-1: Prepare the Network

Step-1: Cable the network that is similar to the one in the topology diagram.

Design and cable a network that is similar to the one in the Topology Diagram by using a Cisco Packet Tracer. In this network you will be required a router 2620. Be sure to use the appropriate type of Ethernet cable to connect from host to switch, switch to router. 2620 router don't have serial interface ports by default so you will be required to install by manually these ports. The following steps are required for installing a Serial Interface on 2620 router.

Step-1: Click on Router 2620

Step-2: Press the button Power Off

Step-3: You will see different Modules select the WIC-2T module which is called the 2-port asynchronous/synchronous serial network module.

Steps-4: Drag this module to empty slots of Router 2620

Step-5: Press the Power On button

When you will press the button power on you will see the router will boot again from the initial stage. To verify serial port attachment with the 2620 router write the following command on router's privilege mode in CLI.

Router#sh ip interface brief

Interface IP-Address OK? Method Status Protocol

FastEthernet0/0 unassigned YES unset administratively down down

Serial0/0 unassigned YES unset administratively down down

Serial0/1 unassigned YES unset administratively down down

Task-2: Erase, and Reload the Routers

Clear the configuration on each of the routers using the **erase startup-config** command and then **reload** the routers. Answer **no** if asked to save changes. The following commands will be use for erasing the NVRAM and reload the router for initial default state.

Router#erase startup-config

Erasing the nvram filesystem will remove all configuration files! Continue? [confirm]
[OK]
Erase of nvram: complete

Router#reload

Proceed with reload? [confirm]
%SYS-5-RELOAD: Reload requested by console. Reload Reason: Reload Command.

System Bootstrap, Version 12.1(3r)T2, RELEASE SOFTWARE (fc1)
Copyright (c) 2000 by cisco Systems, Inc.
cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory

Self decompressing the image :
##
[OK]

Task-3 Perform the Basic Router Configuration

Note: If you have difficulty with any of the commands in this task, see the Lab 5: Basic Router Configuration.

Step 1: Use global configuration commands.

On the routers, enter global configuration mode and configure the basic global configuration commands including:

- host name
- no ip domain-lookup
- enable secret

Note: no ip domain-lookup command is used to disable DNS ((Domain Name Server) host name-to-IP address translation on the routers.

Step 2: Configure the consol and virtual terminal line passwords on each of the routers.

Enter first global configuration mode and then line configuration mode and use the following commands for implementing the consol and telnet password. Add the logging synchronous command to the consol and virtual terminal lines for preventing IOS messages delivered to the consol or Telnet lines from interrupting your keyboard input. Add the exec-timeout commands to set the interval that the EXEC command interpreter waits until user input is detected we can use the **exec-timeout** line configuration command.

- **Router-1(config) #line consol 0**
- **Router-1(config-line) #logging synchronous**
- **Router-1(config-line) #exec-timeout 15 15**
- **Router-1(config-line) #password cisco**
- **Router-1(config-line) #login**
- **Router-1(config) #line vty 0 4**
- **Router-1(config-line) #logging synchronous**
- **Router-1(config-line) #exec-timeout 15 15**
- **Router-1(config-line) #password cisco**
- **Router-1(config-line) #login**

Note: Implement the same commands on Router-2, Router-3, Router-4 and Router5 for configuring the consol and telnet password.

Task-4: Interpreting Debug Output

Note: If you already configured IP addressing on Router-1, please remove all interface commands now before proceeding Router-1, Router-2 ,Router-3, Router-4 and Router-5 should be configured through the end of Task 3 without any interface configurations.

Step-1: On R1 from privileged EXEC mode, enter the debug ip routing command.

R1# debug ip routing

IP routing debugging is on

The **debug ip routing** command shows when routes are added, modified and deleted from the routing table. For example, every time you successfully configure and activate an interface. Cisco IOS adds a route to the routing table. We can verify this by observing output from the **debug ip routing** command. As shown in the following figure in which directly connected route is added to Router-1 routing table.

```
Router-1(config)#int f0/0
Router-1(config-if)#ip add
Router-1(config-if)#ip address 192.168.1.1 255.255.255.0
Router-1(config-if)#no shutdown

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state t
o up

RT: interface FastEthernet0/0 added to routing table
RT: SET_LAST_RDB for 192.168.1.0/24

    NEW rdb: is directly connected


RT: add 192.168.1.0/24 via 0.0.0.0, connected metric [0/0]

RT: NET-RED 192.168.1.0/24

Router-1(config-if)#
```

Note: Activate debug ip routing command on every router as shown in the topology diagram before going to Router's Interface Configuration.

Task-5: Interface Configurations

For Router-1

Step-1: Enter interface configuration mode for Router-1's LAN interface.

Router-1(config)#int f0/0
Router-1(config-if)#ip address 172.16.1.17 255.255.255.240
Router-1(config-if)#no shutdown

Step-2: Enter the interface configuration mode for Router-1's WAN interface connected to Router-5.
Router-1(config)interface s0/0
Router-1(config-if)#ip address 192.168.10.1 255.255.255.252
Router-1(config-if)#clock rate 64000
Router-1(config-if)#no shutdown

The Serial Interface0/0 of Router-1 will be DCE (Data Communication Equipment) and the other connected end will be DTE (Data Terminal Equipment).On DCE we normally give the clock rate.

Step-3: Enter interface configuration mode for Router-5's WAN interface connected to Router-1
Router-5(config)#int s0/0
Router-5(config-if)#ip address 192.168.10.2 255.255.255.252
Router-5(config-if)#no shutdown

For Router-2

Step-1: Enter the interface configuration mode for Router-2's LAN interface.

Router-2(config)#int f0/0
Router-2(config-if)#ip address 10.10.10.1 255.255.255.0
Router-2(config-if)#no shutdown

Step-2: Enter interface configuration mode for Router-2's WAN interface connected to Router-5

Router-2(config)#int s0/2
Router-2(config-if)#ip address 192.168.9.1 255.255.255.252
Router-2(config-if)#clock rate 64000
Router-2(config-if)#no shutdown

The Serial Interface0/2 of Router-2 will be DCE (Data Communication Equipment) and the other connected end will be DTE (Data Terminal Equipment).

Step-3: Enter interface configuration mode for Router-5's WAN interface connected to Router-2

Router-5(config)#int s0/0
Router-5(config-if)#ip address 192.168.9.2 255.255.255.252
Router-5(config-if)#no shutdown

For Router-3

Step-1: Enter interface configuration mode for Router-3's LAN interface.

Router-3(config)#int f0/0
Router-3(config-if)#ip address 172.16.1.33 255.255.255.248
Router-3(config-if)#no shutdown

Step-2: Enter the interface configuration mode for Router-3's WAN interface connected to Router-5.

Router-3(config)interface s0/3
Router-3(config-if)#ip address 192.168.11.1 255.255.255.252
Router-3(config-if)#no shutdown

Step-3: Enter interface configuration mode for Router-5's WAN interface connected to Router-3

Router-5(config)#int s0/3
Router-5(config-if)#ip address 192.168.11.2 255.255.255.252
Router-3(config-if)#clock rate 64000
Router-5(config-if)#no shutdown

The Serial Interface0/3 of Router-5 will be DCE (Data Communication Equipment) and the other connected end will be DTE (Data Terminal Equipment).

For Router-4

Step-1: Enter interface configuration mode for Router-4's LAN interface.

Router-4(config)#int f0/0
Router-4(config-if)#ip address 172.16.1.49 255.255.255.248
Router-4(config-if)#no shutdown

Step-2: Enter the interface configuration mode for Router-4's WAN interface connected to Router-5.
Router-4(config)interface s0/1
Router-4(config-if)#ip address 192.168.12.1 255.255.255.252
Router-4(config-if)#no shutdown
Step-3: Enter interface configuration mode for Router-5's WAN interface connected to Router-4
Router-5(config)#int s0/1
Router-5(config-if)#ip address 192.168.12.2 255.255.255.252
Router-4(config-if)#clock rate 64000
Router-5(config-if)#no shutdown

The Serial Interface0/1 of Router-5 will be DCE (Data Communication Equipment) and the other connected end will be DTE (Data Terminal Equipment).

Task-6: Examine the directly connected route on each router's routing table

Step-1: The ***show ip route*** command normally used to view the routing table.

For Router-1

Router-1#sh ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
** - candidate default, U - per-user static route, o - ODR*
P - periodic downloaded static route

Gateway of last resort is not set

172.16.0.0/28 is subnetted, 1 subnets
C 172.16.1.16 is directly connected, FastEthernet0/0
192.168.10.0/30 is subnetted, 1 subnets
C 192.168.10.0 is directly connected, Serial0/0

For Router-2

Router-2#sh ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
** - candidate default, U - per-user static route, o - ODR*
P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/24 is subnetted, 1 subnets
C 10.10.10.0 is directly connected, FastEthernet0/0
192.168.9.0/30 is subnetted, 1 subnets
C 192.168.9.0 is directly connected, Serial0/2

For Router-3

Router-3#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
** - candidate default, U - per-user static route, o - ODR*
P - periodic downloaded static route

Gateway of last resort is not set

172.16.0.0/29 is subnetted, 1 subnets
C 172.16.1.32 is directly connected, FastEthernet0/0
192.168.10.0/30 is subnetted, 1 subnets
C 192.168.11.0 is directly connected, Serial0/3

For Router-4

Router-4#sh ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
** - candidate default, U - per-user static route, o - ODR*
P - periodic downloaded static route

Gateway of last resort is not set

172.16.0.0/29 is subnetted, 1 subnets
C 172.16.1.48 is directly connected, FastEthernet0/0
192.168.10.0/30 is subnetted, 1 subnets
C 192.168.12.0 is directly connected, Serial0/1

For Router-5

Router-5#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
** - candidate default, U - per-user static route, o - ODR*
P - periodic downloaded static route

Gateway of last resort is not set

192.168.9.0/30 is subnetted, 1 subnets
C 192.168.9.0 is directly connected, Serial0/2
192.168.10.0/30 is subnetted, 1 subnets
C 192.168.10.0 is directly connected, Serial0/0
192.168.11.0/30 is subnetted, 1 subnets
C 192.168.11.0 is directly connected, Serial0/3
192.168.12.0/30 is subnetted, 1 subnets
C 192.168.12.0 is directly connected, Serial0/1

Step-2: Configure Ethernet interfaces of PC0, PC1, PC2, PC3, PC4, PC5, PC6 and PC7

Configure the Ethernet interfaces of PC0, PC1 and PC3 with the ip addresses and default gateways from the table under the Topology diagram.

Step-7: Test the host's connectivity by pinging the default gateway of Routers from the PC (host) command prompt.

Task-7: OSPF Configuration on All Routers

Step-1 Enable a dynamic routing.

To enable a dynamic routing protocol, enter global configuration mode and use the **router** command. **Enter router?** at the global configuration prompt to see a list of available routing protocols on your router.

For Router -1

Use the router ospf command in global configuration mode to enable OSPF on the Router-1. Enter the process ID of 1 for *process-ID* parameter.

Router-1#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router-1(config)#router ospf 1
Router-1(config-router)#

Step-2 Configure the network statement for the LAN network

Once you are in the Router OSPF configuration sub-mode, configure the LAN network 172.16.1.16/28 to be included in the OSPF updates that are sent out of Router-1.

The OSPF **network** command uses a combination of ***network-address*** and ***wildcard mask*** similar to that which can be used by EIGRP. Unlike EIGRP, the wildcard mask in OSPF is required.

Router-1(config-router)# network 172.16.1.16 0.0.0.15 area 0
Router-1(config-router)#

Step-3 Configure the router to advertise the 192.168.10.0/30 network attached to the Serial0/0 interface.

Router-1(config-router)# network 192.168.10.0 0.0.0.3 area 0

When you are finished with the OSPF configuration, return to privileged EXEC mode and save the current configuration to NVRAM.

Router-1(config-router)#end
Router-1#copy run start

For Router-2

Step-1 Configure OSPF on the Router-2 using the *router ospf 1*.

Router-2#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router-2(config)#router ospf 1
Router-2(config-router)#

Step-2 Configure the network statement for the LAN network

Once you are in the Router OSPF configuration sub-mode, configure the LAN network 10.10.10.0/24 to be included in the OSPF updates that are sent out of Router-2.

The OSPF **network** command uses a combination of ***network-address*** and ***wildcard mask*** similar to that which can be used by EIGRP. Unlike EIGRP, the wildcard mask in OSPF is required.

Router-2(config-router)# network 10.10.10.0 0.0.0.255 area 0
Router-2(config-router)#

Step-3 Configure the router to advertise the 192.168.9.0/30 network attached to the Serial0/2 interface.

Router-2(config-router)# network 192.168.9.0 0.0.0.3 area 0

When you are finished with the OSPF configuration, return to privileged EXEC mode and save the current configuration to NVRAM.

Router-2(config-router)#end
Router-2#copy run start
For Router-3

Step-1 Configure OSPF on the Router-3 using the *router ospf 1*.

Router-3#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router-3(config)#router ospf 1
Router-3(config-router)#

Step-2 Configure the network statement for the LAN network

Once you are in the Router OSPF configuration sub-mode, configure the LAN network 172.16.1.32/28 to be included in the OSPF updates that are sent out of Router-3.

The OSPF **network** command uses a combination of ***network-address*** and ***wildcard mask*** similar to that which can be used by EIGRP. Unlike EIGRP, the wildcard mask in OSPF is required.

Router-3(config-router)# network 172.16.1.32 0.0.0.7 area 0
Router-3(config-router)#

Step-3 Configure the router to advertise the 192.168.11.0/30 network attached to the Serial0/3 interface.

Router-3(config-router)# network 192.168.11.0 0.0.0.3 area 0

When you are finished with the OSPF configuration, return to privileged EXEC mode and save the current configuration to NVRAM.

Router-3(config-router)#end

Router-3#copy run start

For Router-4

Step-1 Configure OSPF on the Router-4 using the *router ospf 1*.

Router-4#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router-4(config)#router ospf 1
Router-4(config-router)#

Step-2 Configure the network statement for the LAN network

Once you are in the Router OSPF configuration sub-mode, configure the LAN network 172.16.1.48/29 to be included in the OSPF updates that are sent out of Router-4.

The OSPF **network** command uses a combination of ***network-address*** and ***wildcard mask*** similar to that which can be used by EIGRP. Unlike EIGRP, the wildcard mask in OSPF is required.

Router-4(config-router)# network 172.16.1.48 0.0.0.7 area 0
Router-4(config-router)#

Step-3 Configure the router to advertise the 192.168.12.0/30 network attached to the Serial0/1 interface.

Router-4(config-router)# network 192.168.12.0 0.0.0.7 area 0

When you are finished with the OSPF configuration, return to privileged EXEC mode and save the current configuration to NVRAM.

Router-4(config-router)#end
Router-4#copy run start

For Router-5

Step-1 Configure OSPF on the Router-5 using the *router ospf 1*.

Router-5#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router-5(config)#router ospf 1
Router-5(config-router)#

Step-2 Configure the router to advertise the following networks attached to the different serial ports of Router-5

Router-5(config-router)# network 192.168.10.0 0.0.0.3 area 0
Router-5(config-router)# network 192.168.9.0 0.0.0.3 area 0
Router-5(config-router)# network 192.168.11.0 0.0.0.3 area 0

Router-5(config-router)# network 192.168.12.0 0.0.0.3 area 0

Task-8: Configure OSPF Router IDs

The OSPF router ID is used to uniquely identify the router in the OSPF routing domain and Router ID is the tie-breaker for OSPF path selection. A router ID is an IP address. Cisco routers derive the Router ID in one of the three ways and with the following precedence.

1. IP address configured with the OSPF router-id command
2. Highest IP address of any of the router's loopback address
3. Highest active IP address on any of the router's physical interface

Step-1 Examine the current router IDs in the topology.

Since no router IDs or loopback interface have been configured on the five routers, the router ID for each router is determined by the highest IP address of any active interface.

The router ID can also be seen in the output of the ***show ip protocols***, ***show ip ospf***, and ***show ip ospf interfaces*** commands. Here we will use one command ***show ip protocols*** on each routers for finding Routers IDs in OSPF routing domain**.** You can also use other two commands for finding the router ID in the OSPF routing domain as well.

For Router-1

Router-1#sh ip protocols

Routing Protocol is "ospf 1"
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Router ID 192.168.10.1
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Maximum path: 4

For Router-2

Router-2#sh ip protocols

Routing Protocol is "ospf 1"
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Router ID 192.168.9.1
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Maximum path: 4

For Router-3

Router-3#sh ip protocols

Routing Protocol is "ospf 1"
Outgoing update filter list for all interfaces is not set

Incoming update filter list for all interfaces is not set
Router ID 192.168.11.2
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Maximum path: 4

For Router-4

Router-4#sh ip protocols

Routing Protocol is "ospf 1"
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Router ID 192.168.12.2
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Maximum path: 4

For Router-5

Routing Protocol is "ospf 1"
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Router ID 192.168.12.1
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Maximum path: 4

Step-2 Use loopback addresses to change the router IDs of the routers in the topology.

For Router-1

Router-1(config)#interface loopback 0

Router-1(config-if)#ip address 10.1.1.1 255.255.255.255

For Router-2

Router-2(config)#interface loopback 0

Router-2(config-if)#ip address 10.2.2.2 255.255.255.255

For Router-3

Router-3(config)#interface loopback 0

Router-3(config-if)#ip address 10.3.3.3 255.255.255.255

For Router-4

Router-4(config)#interface loopback 0

Router-4(config-if)#ip address 10.4.4.4 255.255.255.255

For Router-5

Router-5(config)#interface loopback 0

Router-5(config-if)#ip address 10.5.5.5 255.255.255.255

Step-3 Reload the routers to force the new Router IDs to be used.

When a new Router ID is configured, it will not be used until the OSPF process is required. Make sure that the current configuration is saved to NVRAM , and then use the **reload command to restart each router**.

Step-4 Use the *show ip ospf interfaces* command o verify that the router IDs have changed.

For Router-1

Router-1#sh ip ospf interface
FastEthernet0/0 is up, line protocol is up
Internet address is 172.16.1.17/28, Area 0
Process ID 1, **Router ID 10.1.1.1**, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DR, Priority 1
Designated Router (ID) 10.1.1.1, Interface address 172.16.1.17
No backup designated router on this network
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5

Note that Router-1 ID has been changed after configuring the loopback address on Router-1 before that in Task-7 ***Router ID was 192.168.10.1.*** The loopback address having the highest priority for selection of the Router ID.

For Router-2

Router-2#sh ip ospf interface
FastEthernet0/0 is up, line protocol is up
Internet address is 10.10.10.1/24, Area 0
Process ID 1, **Router ID 10.2.2.2**, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DR, Priority 1
Designated Router (ID) 10.2.2.2, Interface address 10.10.10.1
No backup designated router on this network
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5

Note that Router-2 ID has been changed after configuring the loopback address on Router-2 before that in Task-7 ***Router ID was 192.168.9.1.***

For Router-3

Router-3#sh ip ospf interface
FastEthernet0/0 is up, line protocol is up
Internet address is 172.16.1.33/29, Area 0

Process ID 1, Router ID 10.3.3.3, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DR, Priority 1
Designated **Router (ID) 10.3.3.3**, Interface address 172.16.1.33
No backup designated router on this network
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5

Note that Router-3 ID has been changed after configuring the loopback address on Router-3 before that in Task-7 ***Router ID was 192.168.11.2.***

For Router-4

Router-4#sh ip ospf interface
FastEthernet0/0 is up, line protocol is up
Internet address is 172.16.1.49/29, Area 0
Process ID 1, **Router ID 10.4.4.4**, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DR, Priority 1
Designated Router (ID) 10.4.4.4, Interface address 172.16.1.49
No backup designated router on this network
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
Note that Router-4 ID has been changed after configuring the loopback address on Router-4 before that in Task-7 ***Router ID was 192.168.12.2.***

For Router-5

Router-5#sh ip ospf interface
Serial0/0 is up, line protocol is up
Internet address is 192.168.10.2/30, Area 0
Process ID 1, **Router ID 10.5.5.5**, Network Type POINT-TO-POINT, Cost: 64
Transmit Delay is 1 sec, State POINT-TO-POINT, Priority 0
No designated router on this network
No backup designated router on this network
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5

Note that Router-5 ID has been changed after configuring the loopback address on Router-5 before that in Task-7 ***Router ID was 192.168.12.1.***

Task-9: Verify OPSF Operation

Step-1

On the Router-1, use the show ip ospf neighbor command to view the information about the OSPF neighbor **router.** You should be able to see the neighbor ID and IP address of each adjacent router, and the interface that R1 uses to reach that OSPF neighbor.

For Router-1

Router-1#sh ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.5.5.5	0	FULL/ -	00:00:35	192.168.10.2	Serial0/0

For Router-2

Router-2#sh ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.5.5.5	0	FULL/ -	00:00:31	192.168.9.2	Serial0/2

For Router-3

Router-3#sh ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.5.5.5	0	FULL/ -	00:00:33	192.168.11.1	Serial0/3

For Router-4

Router-4#sh ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.5.5.5	0	FULL/ -	00:00:35	192.168.12.1	Serial0/1

For Router-5

Router-5#sh ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.4.4.4	0	FULL/ -	00:00:32	192.168.12.2	Serial0/1
10.3.3.3	0	FULL/ -	00:00:38	192.168.11.2	Serial0/3
10.2.2.2	0	FULL/ -	00:00:33	192.168.9.1	Serial0/2
10.1.1.1	0	FULL/ -	00:00:33	192.168.10.1	Serial0/0

Step-2: Use the *show ip protocols* command to view the information about the routing protocol operation on different routers.

Notice that the information that was configured in the previous Tasks, such as protocols, process ID, neighbour ID, and networks, is shown in the output. The IP address of the adjacent neighbours are also shown.

For Router-1

Router-1#sh ip protocols

Routing Protocol is "ospf 1"
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Router ID 10.1.1.1
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Maximum path: 4
Routing for Networks:
172.16.1.16 0.0.0.15 area 0
192.168.10.0 0.0.0.3 area 0
Routing Information Sources:

Gateway	*Distance*	*Last Update*
10.2.2.2	***110***	***00:02:12***
10.3.3.3	***110***	***00:02:06***
10.4.4.4	***110***	***00:02:04***
10.5.5.5	***110***	***00:02:41***
10.1.1.1	***110***	***00:02:40***
192.168.10.1	***110***	***00:13:10***

Distance: (default is 110)

For Router-2

Router-2#sh ip protocols

Routing Protocol is "ospf 1"
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Router ID 10.2.2.2
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Maximum path: 4
Routing for Networks:
10.10.10.0 0.0.0.255 area 0
192.168.9.0 0.0.0.3 area 0
Routing Information Sources:

Gateway	Distance	Last Update
10.1.1.1	**110**	**00:00:13**
10.2.2.2	**110**	**00:07:10**
10.3.3.3	**110**	**00:07:05**
10.4.4.4	**110**	**00:07:03**
10.5.5.5	**110**	**00:00:13**
192.168.10.1	**110**	**00:07:39**

Distance: (default is 110)

For Router-3

Router-3#sh ip protocols

Routing Protocol is "ospf 1"
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Router ID 10.3.3.3
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Maximum path: 4
Routing for Networks:
172.16.1.32 0.0.0.7 area 0
192.168.11.0 0.0.0.3 area 0
Routing Information Sources:

Gateway	Distance	Last Update
10.1.1.1	**110**	**00:08:57**
10.2.2.2	**110**	**00:15:54**
10.3.3.3	**110**	**00:15:48**
10.4.4.4	**110**	**00:15:47**
10.5.5.5	**110**	**00:08:57**
192.168.10.1	**110**	**00:16:23**

Distance: (default is 110)

For Router-4
Router-4#sh ip protocols

Routing Protocol is "ospf 1"
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Router ID 10.4.4.4
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Maximum path: 4
Routing for Networks:
172.16.1.48 0.0.0.7 area 0
192.168.12.0 0.0.0.3 area 0
Routing Information Sources:

Gateway	Distance	Last Update
10.1.1.1	**110**	**00:04:25**
10.2.2.2	**110**	**00:11:22**
10.3.3.3	**110**	**00:11:16**
10.4.4.4	**110**	**00:11:14**
10.5.5.5	**110**	**00:04:25**
192.168.10.1	**110**	**00:11:51**

Distance: (default is 110)

For Router-5

Router-5#sh ip protocols

Routing Protocol is "ospf 1"
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Router ID 10.5.5.5
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Maximum path: 4
Routing for Networks:
192.168.10.0 0.0.0.3 area 0
192.168.9.0 0.0.0.3 area 0
192.168.11.0 0.0.0.3 area 0
192.168.12.0 0.0.0.3 area 0
Routing Information Sources:

Gateway	Distance	Last Update
10.1.1.1	**110**	**00:11:33**
10.2.2.2	**110**	**00:18:29**
10.3.3.3	**110**	**00:18:24**
10.4.4.4	**110**	**00:18:22**
10.5.5.5	**110**	**00:11:33**
192.168.10.1	**110**	**00:18:58**

Distance: (default is 110)

Step-3: Use the *show ip route* command to verify that each router has all of the networks in the topology entered in the routing table.

Routes learned through OSPF are coded with an **O** in the routing table. If the tables are not converged as shown in the following output of any Router, troubleshoot your configuration.

For Router-1

Router-1#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 10.1.1.1/32 is directly connected, Loopback0
O 10.10.10.0/24 [110/129] via 192.168.10.2, 00:12:56, Serial0/0
172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
C 172.16.1.16/28 is directly connected, FastEthernet0/0
O 172.16.1.32/29 [110/129] via 192.168.10.2, 00:12:56, Serial0/0

O 172.16.1.48/29 [110/129] via 192.168.10.2, 00:12:56, Serial0/0
192.168.9.0/30 is subnetted, 1 subnets
O 192.168.9.0 [110/128] via 192.168.10.2, 00:12:56, Serial0/0
192.168.10.0/30 is subnetted, 1 subnets
C 192.168.10.0 is directly connected, Serial0/0
192.168.11.0/30 is subnetted, 1 subnets
O 192.168.11.0 [110/128] via 192.168.10.2, 00:12:56, Serial0/0
192.168.12.0/30 is subnetted, 1 subnets
O 192.168.12.0 [110/128] via 192.168.10.2, 00:12:56, Serial0/0

For Router-2

Router-2#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 10.2.2.2/32 is directly connected, Loopback0
C 10.10.10.0/24 is directly connected, FastEthernet0/0
172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
O 172.16.1.16/28 [110/129] via 192.168.9.2, 00:18:58, Serial0/2
O 172.16.1.32/29 [110/129] via 192.168.9.2, 01:26:42, Serial0/2
O 172.16.1.48/29 [110/129] via 192.168.9.2, 01:26:32, Serial0/2
192.168.9.0/30 is subnetted, 1 subnets
C 192.168.9.0 is directly connected, Serial0/2
192.168.10.0/30 is subnetted, 1 subnets
O 192.168.10.0 [110/128] via 192.168.9.2, 00:19:08, Serial0/2
192.168.11.0/30 is subnetted, 1 subnets
O 192.168.11.0 [110/128] via 192.168.9.2, 01:26:42, Serial0/2
192.168.12.0/30 is subnetted, 1 subnets
O 192.168.12.0 [110/128] via 192.168.9.2, 01:26:42, Serial0/2

For Router-3

Router-3#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 10.3.3.3/32 is directly connected, Loopback0
O 10.10.10.0/24 [110/129] via 192.168.11.1, 01:28:07, Serial0/3
172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
O 172.16.1.16/28 [110/129] via 192.168.11.1, 00:20:28, Serial0/3
C 172.16.1.32/29 is directly connected, FastEthernet0/0
O 172.16.1.48/29 [110/129] via 192.168.11.1, 01:28:07, Serial0/3
192.168.9.0/30 is subnetted, 1 subnets
O 192.168.9.0 [110/128] via 192.168.11.1, 01:28:07, Serial0/3
192.168.10.0/30 is subnetted, 1 subnets
O 192.168.10.0 [110/128] via 192.168.11.1, 00:20:42, Serial0/3
192.168.11.0/30 is subnetted, 1 subnets
C 192.168.11.0 is directly connected, Serial0/3
192.168.12.0/30 is subnetted, 1 subnets
O 192.168.12.0 [110/128] via 192.168.11.1, 01:28:07, Serial0/3

For Route-4

Router-4#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 10.4.4.4/32 is directly connected, Loopback0
O 10.10.10.0/24 [110/129] via 192.168.12.1, 01:29:23, Serial0/1
172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
O 172.16.1.16/28 [110/129] via 192.168.12.1, 00:21:42, Serial0/1
O 172.16.1.32/29 [110/129] via 192.168.12.1, 01:29:23, Serial0/1
C 172.16.1.48/29 is directly connected, FastEthernet0/0
192.168.9.0/30 is subnetted, 1 subnets
O 192.168.9.0 [110/128] via 192.168.12.1, 01:29:23, Serial0/1
192.168.10.0/30 is subnetted, 1 subnets
O 192.168.10.0 [110/128] via 192.168.12.1, 00:21:55, Serial0/1
192.168.11.0/30 is subnetted, 1 subnets
O 192.168.11.0 [110/128] via 192.168.12.1, 01:29:23, Serial0/1
192.168.12.0/30 is subnetted, 1 subnets
C 192.168.12.0 is directly connected, Serial0/1

For Router-5

Router-5#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 10.5.5.5/32 is directly connected, Loopback0
O 10.10.10.0/24 [110/65] via 192.168.9.1, 01:30:41, Serial0/2
172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
O 172.16.1.16/28 [110/65] via 192.168.10.1, 00:23:03, Serial0/0
O 172.16.1.32/29 [110/65] via 192.168.11.2, 01:30:41, Serial0/3
O 172.16.1.48/29 [110/65] via 192.168.12.2, 01:30:41, Serial0/1
192.168.9.0/30 is subnetted, 1 subnets
C 192.168.9.0 is directly connected, Serial0/2
192.168.10.0/30 is subnetted, 1 subnets
C 192.168.10.0 is directly connected, Serial0/0
192.168.11.0/30 is subnetted, 1 subnets
C 192.168.11.0 is directly connected, Serial0/3
192.168.12.0/30 is subnetted, 1 subnets
C 192.168.12.0 is directly connected, Serial0/1

Task-10: Test Connectivity between the Hosts

Step-1: Use the ping command to check connectivity between the host PC0, PC2, PC3, PC4, PC5, PC6 and PC7

From PC0 to PC2

```
PC>ping 10.10.10.2

Pinging 10.10.10.2 with 32 bytes of data:

Reply from 10.10.10.2: bytes=32 time=187ms TTL=125
Reply from 10.10.10.2: bytes=32 time=187ms TTL=125
Reply from 10.10.10.2: bytes=32 time=188ms TTL=125
Reply from 10.10.10.2: bytes=32 time=141ms TTL=125

Ping statistics for 10.10.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 141ms, Maximum = 188ms, Average = 175ms
```

From PC1 to PC3

```
PC>ping 10.10.10.3

Pinging 10.10.10.3 with 32 bytes of data:

Reply from 10.10.10.3: bytes=32 time=156ms TTL=125
Reply from 10.10.10.3: bytes=32 time=142ms TTL=125
Reply from 10.10.10.3: bytes=32 time=188ms TTL=125
Reply from 10.10.10.3: bytes=32 time=171ms TTL=125

Ping statistics for 10.10.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 142ms, Maximum = 188ms, Average = 164ms
```

From PC 2 to PC4

```
PC>ping 172.16.1.34

Pinging 172.16.1.34 with 32 bytes of data:

Reply from 172.16.1.34: bytes=32 time=156ms TTL=125
Reply from 172.16.1.34: bytes=32 time=188ms TTL=125
Reply from 172.16.1.34: bytes=32 time=172ms TTL=125
Reply from 172.16.1.34: bytes=32 time=158ms TTL=125

Ping statistics for 172.16.1.34:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 156ms, Maximum = 188ms, Average = 168ms
```

From PC3 to PC5

```
PC>ping 172.16.1.35

Pinging 172.16.1.35 with 32 bytes of data:

Reply from 172.16.1.35: bytes=32 time=156ms TTL=125
Reply from 172.16.1.35: bytes=32 time=187ms TTL=125
Reply from 172.16.1.35: bytes=32 time=158ms TTL=125
Reply from 172.16.1.35: bytes=32 time=141ms TTL=125

Ping statistics for 172.16.1.35:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 141ms, Maximum = 187ms, Average = 160ms
```

From PC4 to PC6

```
PC>ping 172.16.1.50

Pinging 172.16.1.50 with 32 bytes of data:

Reply from 172.16.1.50: bytes=32 time=187ms TTL=125
Reply from 172.16.1.50: bytes=32 time=187ms TTL=125
Reply from 172.16.1.50: bytes=32 time=158ms TTL=125
Reply from 172.16.1.50: bytes=32 time=125ms TTL=125

Ping statistics for 172.16.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 125ms, Maximum = 187ms, Average = 164ms
```

From PC5 to PC7

```
PC>ping 172.16.1.51

Pinging 172.16.1.51 with 32 bytes of data:

Reply from 172.16.1.51: bytes=32 time=172ms TTL=125
Reply from 172.16.1.51: bytes=32 time=127ms TTL=125
Reply from 172.16.1.51: bytes=32 time=125ms TTL=125
Reply from 172.16.1.51: bytes=32 time=156ms TTL=125

Ping statistics for 172.16.1.51:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 125ms, Maximum = 172ms, Average = 145ms
```

Assignment:

What is the main difference between EIGRP and OSPF describe briefly. How we can implement the authentication process in OSPF Routing Protocol shows the command syntax for the implementation of OPSF Routing Authentication Process.

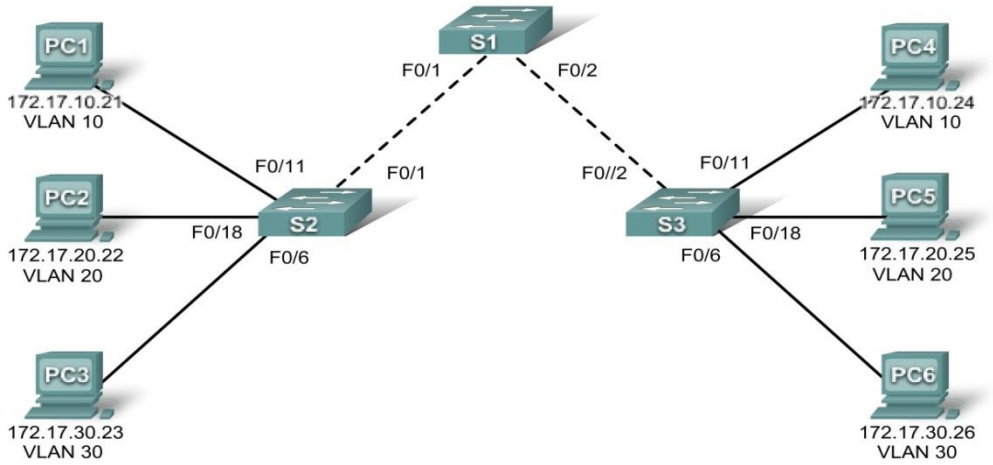
SECTION IV

SWITCHING

Lab -10

Basic VLAN Configuration

Topology Diagram



Addressing Table

Device (Hostname)	Interface	IP Address	Subnet Mask	Default Gateway
S1	**VLAN 99**	172.17.99.11	255.255.255.0	N/A
S2	**VLAN 99**	172.17.99.12	255.255.255.0	N/A
S3	**VLAN 99**	172.17.99.13	255.255.255.0	N/A
PC1	**NIC**	172.17.10.21	255.255.255.0	172.17.10.1
PC2	**NIC**	172.17.20.22	255.255.255.0	172.17.20.1
PC3	**NIC**	172.17.30.23	255.255.255.0	172.17.30.1
PC4	**NIC**	172.17.10.24	255.255.255.0	172.17.10.1
PC5	**NIC**	172.17.20.25	255.255.255.0	172.17.20.1
PC6	**NIC**	172.17.30.26	255.255.255.0	172.17.30.1

Initial Port Assignments (Switches 2 and 3)

Ports	Assignment	Network
Fa0/1 – 0/5	802.1q Trunks (Native VLAN 99)	172.17.99.0 /24
Fa0/6 – 0/10	VLAN 30 – Guest (Default)	172.17.30.0 /24
Fa0/11 – 0/17	VLAN 10 – Faculty/Staff	172.17.10.0 /24
Fa0/18 – 0/24	VLAN 20 – Students	172.17.20.0 /24

Learning Objectives

Upon completion of this lab, you will be able to:

- Cable a network according to the topology diagram
- Erase the startup configuration and reload a switch to the default state
- Perform basic configuration tasks on a switch
- Create VLANs
- Assign switch ports to a VLAN
- Add, move, and change ports
- Verify VLAN configuration
- Enable trunking on inter-switch connections
- Verify trunk configuration
- Save the VLAN configuration

Task 1: Prepare the Network

Step 1: Cable a network that is similar to the one in the topology diagram.

You can use any current switch in your lab as long as it has the required interfaces shown in the topology.

Note: If you use 2900 or 2950 switches, the outputs may appear different. Also, certain commands may be different or unavailable.

Step 2: Clear any existing configurations on the switches, and initialize all ports in the shutdown state.

If necessary, refer to Lab 2.5.1, Appendix 1, for the procedure to clear switch configurations.

It is a good practice to disable any unused ports on the switches by putting them in shutdown. Disable all ports on the switches:

Switch#**config term**
Switch(config)#**interface range fa0/1-24**
Switch(config-if-range)#**shutdown**

Task 2: Perform Basic Switch Configurations

Step 1: Configure the switches according to the following guidelines.

- Configure the switch hostname.
- Disable DNS lookup.
- Configure an EXEC mode password of **class**.
- Configure a password of **cisco** for console connections.
- Configure a password of **cisco** for vty connections.

Step 2: Re-enable the user ports on S2 and S3.

S2(config)#**interface range fa0/6, fa0/11, fa0/18**

S2(config-if-range)#**switchport mode access**

S2(config-if-range)#**no shutdown**

S3(config)#**interface range fa0/6, fa0/11, fa0/18**

S3(config-if-range)#**switchport mode access**

S3(config-if-range)#**no shutdown**

Task 3: Configure and Activate Ethernet Interfaces

Step 1: Configure the PCs.

You can complete this lab using only two PCs by simply changing the IP addressing for the two PCs specific to a test you want to conduct. For example, if you want to test connectivity between PC1 and PC2, then configure the IP addresses for those PCs by referring to the addressing table at the beginning of the lab. Alternatively, you can configure all six PCs with the IP addresses and default gateways.

Task 4: Configure VLANs on the Switch

Step 1: Create VLANs on switch S1.

Use the **vlan***vlan-id* command in global configuration mode to add a VLAN to switch S1. There are four VLANS configured for this lab: VLAN 10 (faculty/staff); VLAN 20 (students); VLAN 30 (guest); and VLAN 99 (management). After you create the VLAN, you will be in vlan configuration mode, where you can assign a name to the VLAN with the **name***vlan name* command.

```
S1(config)#vlan 10
S1(config-vlan)#name faculty/staff
S1(config-vlan)#vlan 20
S1(config-vlan)#name students
S1(config-vlan)#vlan 30
S1(config-vlan)#name guest
S1(config-vlan)#vlan 99
S1(config-vlan)#name management
S1(config-vlan)#end
S1#
```

Step 2: Verify that the VLANs have been created on S1.

Use the **show vlan brief** command to verify that the VLANs have been created.

```
S1#show vlan brief

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
-
1    default                           active    Fa0/1, Fa0/2, Fa0/4, Fa0/5
                                                 Fa0/6, Fa0/7, Fa0/8, Fa0/9
```

```
                                                Fa0/10, Fa0/11, Fa0/12,
Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16,
Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20,
Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24,
Gi0/1
                                                Gi0/2
10   faculty/staff                  active
20   students                       active
30   guest                          active
99   management                     active
```

Step 3: Configure and name VLANs on switches S2 and S3.

Create and name VLANs 10, 20, 30, and 99 on S2 and S3 using the commands from Step 1. Verify the correct configuration with the **show vlan brief** command.

What ports are currently assigned to the four VLANs you have created?

_____

Step 4: Assign switch ports to VLANs on S2 and S3.

Refer to the port assignment table on page 1. Ports are assigned to VLANs in interface configuration mode, using the **switchport access vlan***vlan-id* command. You can assign each port individually or you can use the **interface range** command to simplify this task, as shown here. The commands are shown for S3 only, but you should configure both S2 and S3 similarly. Save your configuration when done.

```
S3(config)#interface range fa0/6-10
S3(config-if-range)#switchport access vlan 30
S3(config-if-range)#interface range fa0/11-17
S3(config-if-range)#switchport access vlan 10
S3(config-if-range)#interface range fa0/18-24
S3(config-if-range)#switchport access vlan 20
S3(config-if-range)#end
S3#copy running-config startup-config
Destination filename [startup-config]? [enter]
Building configuration...
[OK]
```

Step 4: Determine which ports have been added.

Use the **show vlan id** *vlan-number* command on S2 to see which ports are assigned to VLAN 10.

Which ports are assigned to VLAN 10?

_____

Note: The **show vlan id** *vlan-name* displays the same output.

You can also view VLAN assignment information using the **show interfaces** *interface***switchport** command.

Step 5: Assign the management VLAN.

A management VLAN is any VLAN that you configure to access the management capabilities of a switch. VLAN 1 serves as the management VLAN if you did not specifically define another VLAN. You assign the management VLAN an IP address and subnet mask. A switch can be managed via HTTP, Telnet, SSH, or SNMP. Because the out-of-the-box configuration of a Cisco switch has VLAN 1 as the default VLAN, VLAN 1 is a bad choice as the

management VLAN. You do not want an arbitrary user who is connecting to a switch to default to the management VLAN. Recall that you configured the management VLAN as VLAN 99 earlier in this lab.

From interface configuration mode, use the **ip address** command to assign the management IP address to the switches.

```
S1(config)#interface vlan 99
S1(config-if)#ip address 172.17.99.11 255.255.255.0
S1(config-if)#no shutdown

S2(config)#interface vlan 99
S2(config-if)#ip address 172.17.99.12 255.255.255.0
S2(config-if)#no shutdown

S3(config)#interface vlan 99
S3(config-if)#ip address 172.17.99.13 255.255.255.0
S3(config-if)#no shutdown
```

Assigning a management address allows IP communication between the switches, and also allows any host connected to a port assigned to VLAN 99 to connect to the switches. Because VLAN 99 is configured as the management VLAN, any ports assigned to this VLAN are considered management ports and should be secured to control which devices can connect to these ports.

Step 6: **Configure trunking and the native VLAN for the trunking ports on all switches.**

Trunks are connections between the switches that allow the switches to exchange information for all VLANS. By default, a trunk port belongs to all VLANs, as opposed to an access port, which can only belong to a single VLAN. If the switch supports both ISL and 802.1Q VLAN encapsulation, the trunks must specify which method is being used. Because the 2960 switch only supports 802.1Q trunking, it is not specified in this lab.

A native VLAN is assigned to an 802.1Q trunk port. In the topology, the native VLAN is VLAN 99. An 802.1Q trunk port supports traffic coming from many VLANs (tagged traffic) as well as traffic that does not come from a VLAN (untagged traffic). The 802.1Q trunk port places untagged traffic on the native VLAN. Untagged traffic is generated by a computer attached to a switch port that is configured with the native VLAN. One of the IEEE 802.1Q specifications for Native VLANs is to maintain backward compatibility with untagged traffic common to legacy LAN scenarios. For the purposes of this lab, a native VLAN serves as a common identifier on opposing ends of a trunk link. It is a best practice to use a VLAN other than VLAN 1 as the native VLAN.

Use the **interfacerange** command in global configuration mode to simplify configuring trunking.

```
S1(config)#interface range fa0/1-5

S1(config-if-range)#switchport mode trunk

S1(config-if-range)#switchport trunk native vlan 99

S1(config-if-range)#no shutdown

S1(config-if-range)#end

S2(config)# interface range fa0/1-5

S2(config-if-range)#switchport mode trunk
```

```
S2(config-if-range)#switchport trunk native vlan 99

S2(config-if-range)#no shutdown

S2(config-if-range)#end
S3(config)# interface range fa0/1-5

S3(config-if-range)#switchport mode trunk

S3(config-if-range)#switchport trunk native vlan 99

S3(config-if-range)#no shutdown

S3(config-if-range)#end
```

Verify that the trunks have been configured with the **show interface trunk** command.

```
S1#show interface trunk

Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      99
Fa0/2       on           802.1q         trunking      99

Port      Vlans allowed on trunk
Fa0/1       1-4094
Fa0/2       1-4094

Port        Vlans allowed and active in management domain
Fa0/1       1,10,20,30,99
Fa0/2       1,10,20,30,99

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,20,30,99
Fa0/2       1,10,20,30,99
```

Step 7: Verify that the switches can communicate.

From S1, ping the management address on both S2 and S3.

```
S1#ping 172.17.99.12
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.17.99.12, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/9 ms
S1#ping 172.17.99.13
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.17.99.13, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/1 ms
```

Step 8: Ping several hosts from PC2.

Ping from host PC2 to host PC1 (172.17.10.21). Is the ping attempt successful? _____

Ping from host PC2 to the switch VLAN 99 IP address 172.17.99.12. Is the ping attempt successful? _____

Because these hosts are on different subnets and in different VLANs, they cannot communicate without a Layer 3 device to route between the separate subnetworks.

Ping from host PC2 to host PC5. Is the ping attempt successful? _____

Because PC2 is in the same VLAN and the same subnet as PC5, the ping is successful

Step 9: Move PC1 into the same VLAN as PC2.

The port connected to PC2 (S2 Fa0/18) is assigned to VLAN 20, and the port connected to PC1 (S2 Fa0/11) is assigned to VLAN 10. Reassign the S2 Fa0/11 port to VLAN 20. You do not need to first remove a port from a VLAN to change its VLAN membership. After you reassign a port to a new VLAN, that port is automatically removed from its previous VLAN.

```
S2#configure terminal

Enter configuration commands, one per line.  End with CNTL/Z.

S2(config)#interface fastethernet 0/11

S2(config-if)#switchport access vlan 20

S2(config-if)#end
```

Ping from host PC2 to host PC1. Is the ping attempt successful? _____

Even though the ports used by PC1 and PC2 are in the same VLAN, they are still in different subnetworks, so they cannot communicate directly.

Step 10: Change the IP address and network on PC1.

Change the IP address on PC1 to 172.17.20.22. The subnet mask and default gateway can remain the same. Once again, ping from host PC2 to host PC1, using the newly assigned IP address.

Is the ping attempt successful? _____

Why was this attempt successful?

_____
___

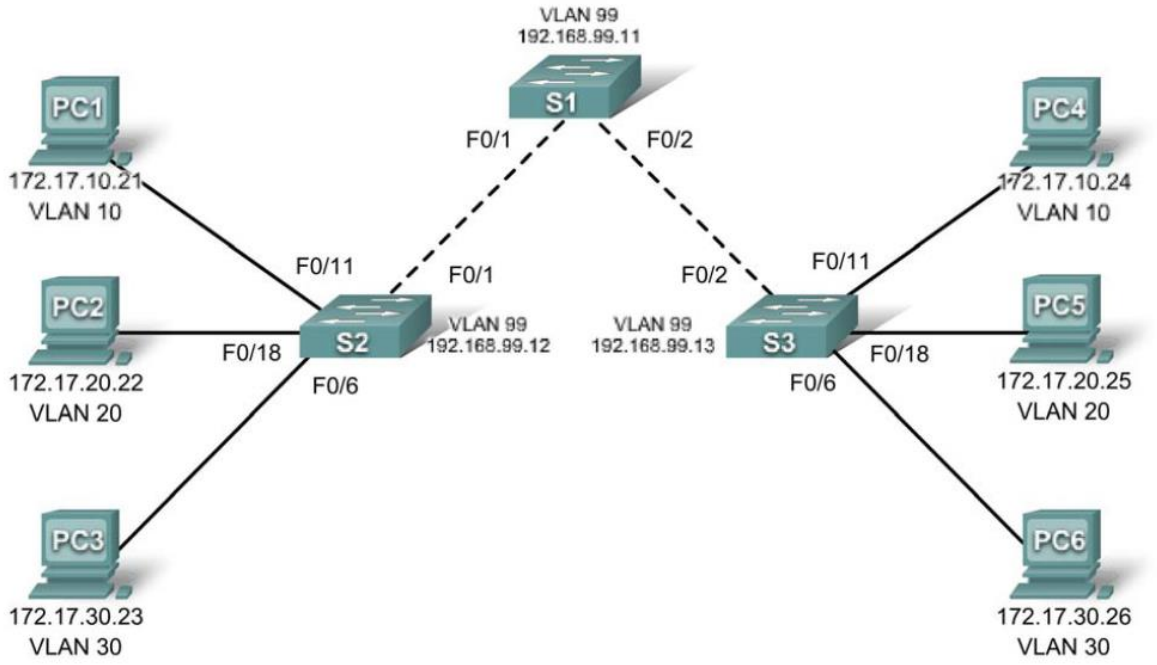
Task 7: Document the Switch Configurations

On each switch, capture the running configuration to a text file and save it for future reference.

LAB-11

Basic VTP Configuration

Topology Diagram



Learning Objective

Upon completion of this lab, you will be able to:
• Cable a network according to the topology diagram
• Erase the startup configuration and reload a switch to the default state
• Perform basic switch configuration
• Configure VLAN Trunking Protocol (VTP) on all switches
• Enable trunking on inter-switch connections
• Verify trunk configuration
• Modify VTP modes and observe the impact
• Create VLANs on the VTP server, and distribute this VLAN information to switches in the network
• Explain the differences in operation between VTP transparent mode, server mode, and client mode
• Assign switch ports to the VLANs
• Save the VLAN configuration
• Enable VTP pruning on the network
• Explain how pruning reduces unnecessary broadcast traffic on the LAN

Addressing Table

Device (Hostname)	Interface	IP Address	Subnet Mask	Default Gateway
Ports		Assignment		Network
Fa0/1 – 0/5		802.1q Trunks (Native VLAN 99)		172.17.99.0 /24
Fa0/6 – 0/10		VLAN 30 – Guest (Default)		172.17.30.0 /24
Fa0/11 – 0/17		VLAN 10 – Faculty/Staff		172.17.10.0 /24
Fa0/18 – 0/24		VLAN 20 – Students		172.17.20.0 /24
PC2	**NIC**	172.17.20.22	255.255.255.0	172.17.20.1
PC3	**NIC**	172.17.30.23	255.255.255.0	172.17.30.1
PC4	**NIC**	172.17.10.24	255.255.255.0	172.17.10.1
PC5	**NIC**	172.17.20.25	255.255.255.0	172.17.20.1
PC6	**NIC**	172.17.30.26	255.255.255.0	172.17.30.1

Port Assignment (Switches 2 and 3)

Task 1: Prepare the Network

Step 1: Cable a network that is similar to the one in the topology diagram.

You can use any current switch in your lab as long as it has the required interfaces shown in the topology.
The output shown in this lab is based on 2960 switches. Other switch types may produce different output.
If you are using older switches, then some commands may be different or unavailable.
Set up console connections to all three switches.

Step 2: Clear any existing configurations on the switches.

If necessary, refer to LAB-10 for the procedure to clear switch configurations and VLANs.
Use the **show vlan** command to confirm that only default VLANs exist and that all ports are assigned to VLAN 1.

```
S1#show vlan
VLAN Name Status Ports
---- -------------------------------- --------- -------------------------
--
1 default active Fa0/1, Fa0/2, Fa0/3, Fa0/4
Fa0/5, Fa0/6, Fa0/7, Fa0/8
Fa0/9, Fa0/10, Fa0/11, Fa0/12

Fa0/13, Fa0/14, Fa0/15,Fa0/16
Fa0/17, Fa0/18, Fa0/19,Fa0/20
Fa0/21, Fa0/22, Fa0/23,Fa0/24
Gig1/1, Gig1/2
1002 fddi-default active
1003 token-ring-default active
```

```
1004 fddinet-default active
1005 trnet-default active
```

Step 3: Disable all ports by using the shutdown command.

```
S1(config)#interface range fa0/1-24
S1(config-if-range)#shutdown
S1(config-if-range)#interface range gi0/1-2
S1(config-if-range)#shutdown
S2(config)#interface range fa0/1-24
S2(config-if-range)#shutdown
S2(config-if-range)#interface range gi0/1-2
S2(config-if-range)#shutdown
S3(config)#interface range fa0/1-24
S3(config-if-range)#shutdown
S3(config-if-range)#interface range gi0/1-2
S3(config-if-range)#shutdown
```

Step 4: Re-enable the user ports on S2 and S3.

Configure the user ports in access mode. Refer to the topology diagram to determine which ports are connected to end-user devices.

```
S2(config)#interface fa0/6
S2(config-if)#switchport mode access
S2(config-if)#no shutdown
S2(config-if)#interface fa0/11
S2(config-if)#switchport mode access
S2(config-if)#no shutdown
S2(config-if)#interface fa0/18
S2(config-if)#switchport mode access
S2(config-if)#no shutdown
S3(config)#interface fa0/6
S3(config-if)#switchport mode access
S3(config-if)#no shutdown
S3(config-if)#interface fa0/11
S3(config-if)#switchport mode access
S3(config-if)#no shutdown
S3(config-if)#interface fa0/18
S3(config-if)#switchport mode access
S3(config-if)#no shutdown
```

Task 2: Perform Basic Switch Configurations

Configure the S1, S2, and S3 switches according to the following guidelines and save all your Configurations:

- Configure the switch hostname as indicated on the topology.
- Disable DNS lookup.
- Configure an EXEC mode password of **class**.
- Configure a password of **cisco** for console connections.
- Configure a password of **cisco** for vty connections.

(Output for S1 shown)

```
Switch>enable
Switch#configure terminal
```

```
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname S1
S1(config)#enable secret class
S1(config)#no ip domain-lookup
S1(config)#line console 0
S1(config-line)#password cisco
S1(config-line)#login
S1(config-line)#line vty 0 15
S1(config-line)#password cisco
S1(config-line)#login
S1(config-line)#end
%SYS-5-CONFIG_I: Configured from console by console
S1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

Task 3: Configure the Ethernet Interfaces on the Host PCs

Configure the Ethernet interfaces of PC1, PC2, PC3, PC4, PC5, and PC6 with the IP addresses and default gateways indicated in the addressing table at the beginning of the lab.
Verify that PC1 can ping PC4, PC2 can ping PC5, and that PC3 can ping PC6.

Task 4: Configure VTP on the Switches

VTP allows the network administrator to control the instances of VLANs on the network by creating VTP domains. Within each VTP domain, one or more switches are configured as VTP servers. VLANs are then created on the VTP server and pushed to the other switches in the domain. Common VTP configuration tasks are setting the operating mode, domain, and password. In this lab, you will be using S1 as the VTP server, with S2 and S3 configured as VTP clients or in VTP transparent mode.

Step 1: Check the current VTP settings on the three switches.

```
S1#show vtp status
VTP Version : 2
Configuration Revision : 0
Maximum VLANs supported locally : 255
Number of existing VLANs : 5
VTP Operating Mode : Server
VTP Domain Name :
VTP Pruning Mode : Disabled
VTP V2 Mode : Disabled
VTP Traps Generation : Disabled
MD5 digest : 0x57 0xCD 0x40 0x65 0x63 0x59 0x47 0xBD
Configuration last modified by 0.0.0.0 at 0-0-00 00:00:00
Local updater ID is 0.0.0.0 (no valid interface found)

S2#show vtp status
VTP Version : 2
Configuration Revision : 0
Maximum VLANs supported locally : 255
Number of existing VLANs : 5
VTP Operating Mode : Server
VTP Domain Name :
VTP Pruning Mode : Disabled
VTP V2 Mode : Disabled
```

```
VTP Traps Generation : Disabled
MD5 digest : 0x57 0xCD 0x40 0x65 0x63 0x59 0x47 0xBD
Configuration last modified by 0.0.0.0 at 0-0-00 00:00:00
Local updater ID is 0.0.0.0 (no valid interface found)

S3#show vtp status
VTP Version : 2
Configuration Revision : 0
Maximum VLANs supported locally : 255
Number of existing VLANs : 5
VTP Operating Mode : Server
VTP Domain Name :
VTP Pruning Mode : Disabled
VTP V2 Mode : Disabled
VTP Traps Generation : Disabled
MD5 digest : 0x57 0xCD 0x40 0x65 0x63 0x59 0x47 0xBD
Configuration last modified by 0.0.0.0 at 0-0-00 00:00:00
```

Note that all three switches are in server mode. Server mode is the default VTP mode for most Catalyst switches.

Step 2: Configure the operating mode, domain name, and VTP password on all three switches.

Set the VTP domain name to **Lab4** and the VTP password to **cisco** on all three switches**.** Configure S1 in server mode, S2 in client mode, and S3 in transparent mode.

```
S1(config)#vtp mode server
Device mode already VTP SERVER.
S1(config)#vtp domain Lab4
Changing VTP domain name from NULL to Lab4
S1(config)#vtp password cisco
Setting device VLAN database password to cisco
S1(config)#end
S2(config)#vtp mode client
Setting device to VTP CLIENT mode
S2(config)#vtp domain Lab4
Changing VTP domain name from NULL to Lab4
S2(config)#vtp password cisco
Setting device VLAN database password to cisco
S2(config)#end
S3(config)#vtp mode transparent
Setting device to VTP TRANSPARENT mode.
S3(config)#vtp domain Lab4
Changing VTP domain name from NULL to Lab4
S3(config)#vtp password cisco
Setting device VLAN database password to cisco
S3(config)#end
```

Note: The VTP domain name can be learned by a client switch from a server switch, but only if the client switch domain is in the null state. It does not learn a new name if one has been previously set. For that reason, it is good practice to manually configure the domain name on all switches to ensure that the domain name is configured correctly. Switches in different VTP domains do not exchange VLAN information.

Step 3: Configure trunking and the native VLAN for the trunking ports on all three switches.

Use the **interface range** command in global configuration mode to simplify this task.

```
S1(config)#interface range fa0/1-5
S1(config-if-range)#switchport mode trunk
S1(config-if-range)#switchport trunk native vlan 99
S1(config-if-range)#no shutdown
S1(config-if-range)#end
S2(config)# interface range fa0/1-5
S2(config-if-range)#switchport mode trunk
S2(config-if-range)#switchport trunk native vlan 99
S2(config-if-range)#no shutdown
S2(config-if-range)#end
S3(config)# interface range fa0/1-5
S3(config-if-range)#switchport mode trunk
S3(config-if-range)#switchport trunk native vlan 99
S3(config-if-range)#no shutdown
S3(config-if-range)#end
```

Step 4: Configure port security on the S2 and S3 access layer switches.

Configure ports fa0/6, fa0/11, and fa0/18 so that they allow only a single host and learn the MAC address of the host dynamically.

```
S2(config)#interface fa0/6
S2(config-if)#switchport port-security
S2(config-if)#switchport port-security maximum 1
S2(config-if)#switchport port-security mac-address sticky
S2(config-if)#interface fa0/11
S2(config-if)#switchport port-security
S2(config-if)#switchport port-security maximum 1
S2(config-if)#switchport port-security mac-address sticky
S2(config-if)#interface fa0/18
S2(config-if)#switchport port-security
S2(config-if)#switchport port-security maximum 1
S2(config-if)#switchport port-security mac-address sticky
S2(config-if)#end
S3(config)#interface fa0/6
S3(config-if)#switchport port-security
S3(config-if)#switchport port-security maximum 1
S3(config-if)#switchport port-security mac-address sticky
S3(config-if)#interface fa0/11
S3(config-if)#switchport port-security
S3(config-if)#switchport port-security maximum 1
S3(config-if)#switchport port-security mac-address sticky

S3(config-if)#interface fa0/18
S3(config-if)#switchport port-security
S3(config-if)#switchport port-security maximum 1
S3(config-if)#switchport port-security mac-address sticky
S3(config-if)#end
```

Step 5: Configure VLANs on the VTP server.

There are four VLANS required in this lab:
• VLAN 99 (management)
• VLAN 10 (faculty/staff)

• VLAN 20 (students)
• VLAN 30 (guest)

Configure these on the VTP server.

```
S1(config)#vlan 99
S1(config-vlan)#name management
S1(config-vlan)#exit
S1(config)#vlan 10
S1(config-vlan)#name faculty/staff
S1(config-vlan)#exit
S1(config)#vlan 20
S1(config-vlan)#name students
S1(config-vlan)#exit
S1(config)#vlan 30
S1(config-vlan)#name guest
S1(config-vlan)#exit
```

Verify that the VLANs have been created on S1 with the **show vlan brief** command.

Step 6: Check if the VLANs created on S1 have been distributed to S2 and S3.

Use the **show vlan brief** command on S2 and S3 to determine if the VTP server has pushed its VLAN configuration to all the switches.

```
S2#show vlan brief
VLAN Name Status Ports
---- -------------------------------- --------- -------------------------
--
1 default active Fa0/1, Fa0/2, Fa0/4, Fa0/5
Fa0/6, Fa0/7, Fa0/8, Fa0/9
Fa0/10, Fa0/11, Fa0/12,Fa0/13
Fa0/14, Fa0/15, Fa0/16,Fa0/17
Fa0/18, Fa0/19, Fa0/20,Fa0/21
Fa0/22, Fa0/23, Fa0/24, Gi0/1
Gi0/2

10 faculty/staff active
20 students active
30 guest active
99 management active

S3#show vlan brief
VLAN Name                              Status      Ports
---- -------------------------------- --------- -------------------------
--
1 default                              active Fa0/1, Fa0/2, Fa0/4, Fa0/5
                                              Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                              Fa0/10, Fa0/11, Fa0/12,Fa0/13
                                              Fa0/14, Fa0/15, Fa0/16,Fa0/17
                                              Fa0/18, Fa0/19, Fa0/20,Fa0/21
                                              Fa0/22, Fa0/23, Fa0/24, Gi0/1
                                              Gi0/2
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                   act/unsup
1005 trnet-default                     act/unsup
```

Step 7: Create a new VLAN on switch 2 and 3.

```
S2(config)#vlan 88
%VTP VLAN configuration not allowed when device is in CLIENT mode.
S3(config)#vlan 88
S3(config-vlan)#name test
S3(config-vlan)#
```

Delete VLAN 88 from S3.

```
S3(config)#no vlan 88
```

Step 8: Manually configure VLANs.

Configure the four VLANs identified in Step 5 on switch S3.

```
S3(config)#vlan 99
S3(config-vlan)#name management
S3(config-vlan)#exit
S3(config)#vlan 10
S3(config-vlan)#name faculty/staff
S3(config-vlan)#exit
S3(config)#vlan 20
S3(config-vlan)#name students
S3(config-vlan)#exit
S3(config)#vlan 30
S3(config-vlan)#name guest
S3(config-vlan)#exit
```

Here you see one of the advantages of VTP. Manual configuration is tedious and error prone, and any error introduced here could prevent intra-VLAN communication. In addition, these types of errors can be difficult to troubleshoot.

Step 9: Configure the management interface address on all three switches.

```
S1(config)#interface vlan 99
S1(config-if)#ip address 172.17.99.11 255.255.255.0
S1(config-if)#no shutdown
S2(config)#interface vlan 99
S2(config-if)#ip address 172.17.99.12 255.255.255.0
S2(config-if)#no shutdown
S3(config)#interface vlan 99
S3(config-if)#ip address 172.17.99.13 255.255.255.0
S3(config-if)#no shutdown
```

Verify that the switches are correctly configured by pinging between them. From S1, ping the management interface on S2 and S3. From S2, ping the management interface on S3.

Step 10: Assign switch ports to VLANs.

Refer to the port assignment table at the beginning of the lab to assign ports to the VLANs. Use the **interface range** command to simplify this task. Port assignments are not configured through VTP. Port assignments must be configured on each switch manually or dynamically using a VMPS server. The commands are shown for S3 only, but both S2 and S1 switches should be similarly configured. Save the configuration when you are done.
32

```
S3(config)#interface range fa0/6-10
S3(config-if-range)#switchport access vlan 30
S3(config-if-range)#interface range fa0/11-17
```

```
S3(config-if-range)#switchport access vlan 10
S3(config-if-range)#interface range fa0/18-24
S3(config-if-range)#switchport access vlan 20
S3(config-if-range)#end
S3#copy running-config startup-config
Destination filename [startup-config]? [enter]
Building configuration...
[OK]
S3#
```

Task 5: Configure VTP Pruning on the Switches

VTP pruning allows a VTP server to suppress IP broadcast traffic for specific VLANs to switches that do not have any ports in that VLAN. By default, all unknown unicasts and broadcasts in a VLAN are flooded over the entire VLAN. All switches in the network receive all broadcasts, even in situations in which few users are connected in that VLAN. VTP pruning is used to eliminate or prune this unnecessary traffic. Pruning saves LAN bandwidth because broadcasts do not have to be sent to switches that do not need them. Pruning is configured on the server switch with the **vtp pruning** command in global configuration mode. The configuration is pushed to client switches. However, because S3 is in transparent mode, VTP pruning must be configured locally on that switch.

Confirm VTP pruning configuration on each switch using the **show vtp status** command. VTP pruning mode should be enabled on each switch.

```
S1#show vtp status
VTP Version : 2
Configuration Revision : 17
Maximum VLANs supported locally : 255
Number of existing VLANs : 9
VTP Operating Mode : Server
VTP Domain Name : Lab4
VTP Pruning Mode : Enabled
<output omitted>
```

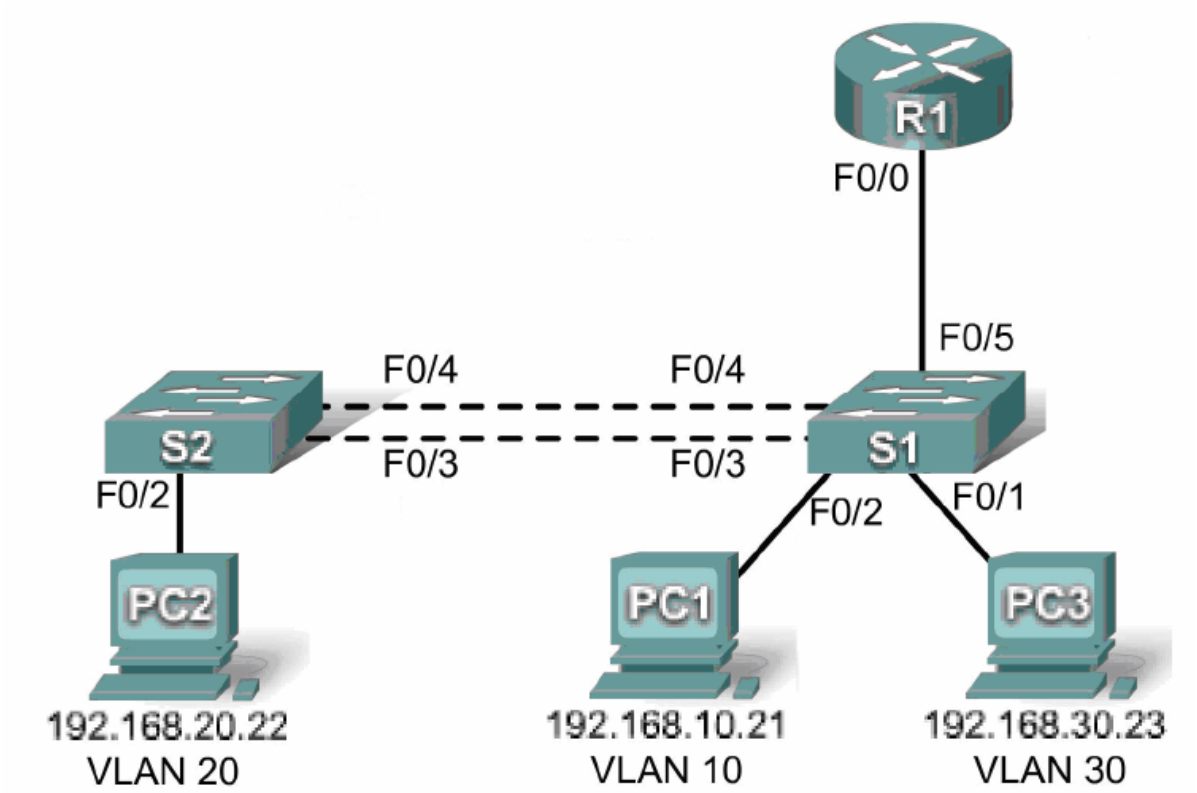
Task 6: Clean Up

Erase the configurations and reload the switches. Disconnect and store the cabling. For PC hosts that are normally connected to other networks (such as the school LAN or to the Internet), reconnect the appropriate cabling and restore the TCP/IP settings.

LAB-12

Inter VLAN-Routing

Topology Diagram



Addressing Table

Device (Hostname)	Interface	IP Address	Subnet Mask	Default Gateway
S1	VLAN 99	192.168.99.11	255.255.255.0	192.168.99.1
S2	VLAN 99	192.168.99.12	255.255.255.0	192.168.99.1
S3	VLAN 99	192.168.99.13	255.255.255.0	192.168.99.1
R1	Fa 0/0	See Subinterface Configuration Table		N/A
PC1	NIC	192.168.10.21	255.255.255.0	192.168.10.1
PC2	NIC	192.168.20.22	255.255.255.0	192.168.20.1
PC3	NIC	192.168.30.23	255.255.255.0	192.168.30.1

Subinterface Configuration Table – Router 1

Router Interface	Assignment	IP Address
Fa0/0.1	VLAN1	192.168.1.1
Fa0/0.10	VLAN 10	192.168.10.1
Fa0/0.20	VLAN 20	192.168.20.1
Fa0/0.30	VLAN 30	192.168.30.1
Fa0/0.99	VLAN 99	192.168.99.1

Task 1: Prepare the Network

Step 1: Cable a network that is similar to the one in the topology diagram.

Note that Ethernet (10Mb) LAN interfaces on routers do not support trunking, and Cisco IOS software earlier than version 12.3 may not support trunking on Fast Ethernet router interfaces. Set up console connections to all three switches and to the router.

Step 2: Clear any existing configurations on the switches if persistent.

Clear NVRAM, delete the vlan.dat file and reload the switches. Refer to VLAN configuration lab if necessary for the procedure. After the reload is complete, use the **show vlan** command to confirm that only default VLANs exist and that all ports are assigned to VLAN 1.

Step 3: Disable all ports using the shutdown command.

Ensure that the initial switch port states are inactive by disabling all ports. Use the **interface-range** command to simplify this task.

Step 4: Re-enable the active user ports on switches in access mode.

Use the **no shutdown** command, and configure them as access ports.

Task 2: Perform Basic Switch Configurations

Configure the S1and S2 switches according to the addressing table and the following guidelines:

- Configure the switch hostname.
- Disable DNS lookup.
- Configure an EXEC mode password of **class**.
- Configure a password of **cisco** for console connections.
- Configure a password of **cisco** for vty connections.
- Configure the default gateway on each switch.

Task 3: Configure the Ethernet Interfaces on host PCs

Configure the Ethernet interfaces of PC1, PC2, and PC3 with the IP addresses from the addressing table.

Connect these devices using the correct cables and interfaces.

Task 4: Configure VTP on the Switches

Step 1: Configure VTP on the switches.

Use the following table to configure the switches. Remember that VTP domain names and passwords are case-sensitive.

Switch Name	VTP Operating Mode	VTP Domain	VTP Password
S1	Client	Lab6	cisco
S2	Server	Lab6	cisco

Step 2: Configure trunking ports on switches and designate the native VLAN for the trunks.

Configure Fa0/3 through Fa0/4 as trunking ports, and designate VLAN 99 as the native VLAN for these trunks. Use the **interface range** command in global configuration mode to simplify this task.

Step 3: Configure VLANs on the VTP server.

Configure the following VLANS on the VTP server.

VLAN	VLAN Name
VLAN 99	Management
VLAN 10	R&D
VLAN 20	Engineering
VLAN 30	Sales

Verify that the VLANs have been created on S2 with the **show vlan brief** command.

Step 4: Verify that the VLANs created on S2 have been distributed to S1.

Use the **show vlan brief** command on S1 to verify that the four VLANs have been distributed to the client switch.

Step 5: Configure the Management interface address on all switches.

Refer to the addressing table at the beginning of the lab to assign the management IP address on all switches. Verify that the switches are correctly configured by pinging between them. From S1, ping the Management interface on S2. From S2, ping the Management interface on S1.

If not, troubleshoot the switch configurations and resolve.

Step 6: Assign switch ports to appropriate VLANs.

Refer to the topology diagram at the beginning of the lab to assign ports to VLANs.

Step 7: Check connectivity between VLANs.

Ping from PC1 (192.168.10.21) to PC2 (192.168.20.22). Ping from PC2 to PC3 (192.168.30.23).

Task 5: Configure the Router

Step 1: Clear the configuration on the router and reload.

Step 2: Create a basic configuration on the router.

- Configure the router with hostname R1.
- Disable DNS lookup.
- Configure an EXEC mode password of **class**.
- Configure a password of **cisco** for console connections.
- Configure a password of **cisco** for vty connections.

Step 3: Configure the trunking interface on R1.

You have demonstrated that connectivity between VLANs requires routing at the network layer, exactly like connectivity between any two remote networks. There are a couple of options for configuring routing between VLANs.

The first is something of a brute force approach. An L3 device, either a router or a Layer 3 capable switch, is connected to a LAN switch with multiple connections—a separate connection for each VLAN that requires inter-VLAN connectivity. Each of the switch ports used by the L3 device is configured in a different VLAN on the switch. After IP addresses are assigned to the interfaces on the L3 device, the routing table has directly connected routes for all VLANS, and inter-VLAN routing is enabled. The limitations to this approach are the lack of sufficient Fast Ethernet ports on routers, under-utilization of ports on L3 switches and routers, and excessive wiring and manual configuration. The topology used in this lab does not use this approach. An alternative approach is to create one or more Fast Ethernet connections between the L3 device (the router) and the distribution layer switch, and to configure these connections as dot1q trunks. This allows all inter-VLAN traffic to be carried to and from the routing device on a single trunk. However, it requires that the L3 interface be configured with multiple IP addresses. This can be done by creating "virtual" interfaces, called subinterfaces, on one of the router Fast Ethernet ports and configuring them to dot1q
aware.

Using the subinterface configuration approach requires these steps:

• Enter subinterface configuration mode
• Establish trunking encapsulation
• Associate a VLAN with the subinterface
• Assign an IP address from the VLAN to the subinterface

The commands are as follows:

R1(config)#**interface fastethernet 0/0**
R1(config-if)#**no shutdown**
R1(config-if)#**interface fastethernet 0/0.1**
R1(config-subif)#**encapsulation dot1q 1**
R1(config-subif)#**ip address 192.168.1.1 255.255.255.0**
R1(config-if)#**interface fastethernet 0/0.10**
R1(config-subif)#**encapsulation dot1q 10**
R1(config-subif)#**ip address 192.168.10.1 255.255.255.0**
R1(config-if)#**interface fastethernet 0/0.20**
R1(config-subif)#**encapsulation dot1q 20**
R1(config-subif)#**ip address 192.168.20.1 255.255.255.0**
R1(config-if)#**interface fastethernet 0/0.30**
R1(config-subif)#**encapsulation dot1q 30**
R1(config-subif)#**ip address 192.168.30.1 255.255.255.0**
R1(config-if)#**interface fastethernet 0/0.99**
R1(config-subif)#**encapsulation dot1q 99 native**
R1(config-subif)#**ip address 192.168.99.1 255.255.255.0**

Note the following points in this configuration:

- The physical interface is enabled using the **no shutdown** command, because router interfaces
- are down by default. The virtual interfaces are up by default.

- The subinterface can use any number that can be described with 32 bits, but it is good practice to assign the number of the VLAN as the interface number, as has been done here.
- The native VLAN is specified on the L3 device so that it is consistent with the switches.
- Otherwise, VLAN 1 would be the native VLAN by default, and there would be no communication between the router and the management VLAN on the switches.

Step 4: Verify the routing configuration.

At this point, there should be five networks configured on R1. Verify that you can route packets to all five by checking the routing table on R1.
If your routing table does not show all five networks, troubleshoot your configuration and resolve the problem before proceeding.

Step 5: Verify inter-VLAN routing

From PC1, verify that you can ping the other two hosts (192.168.20.22 and 192.168.30.23). It may take a couple of pings before the end-to-end path is established.
Are the pings successful?

_____

If not, troubleshoot your configuration. Check to make sure the default gateways have been set on all PCs and all switches.

At this point, you should be able to ping any node on any of the five networks configured on your LAN, including the switch management interfaces.

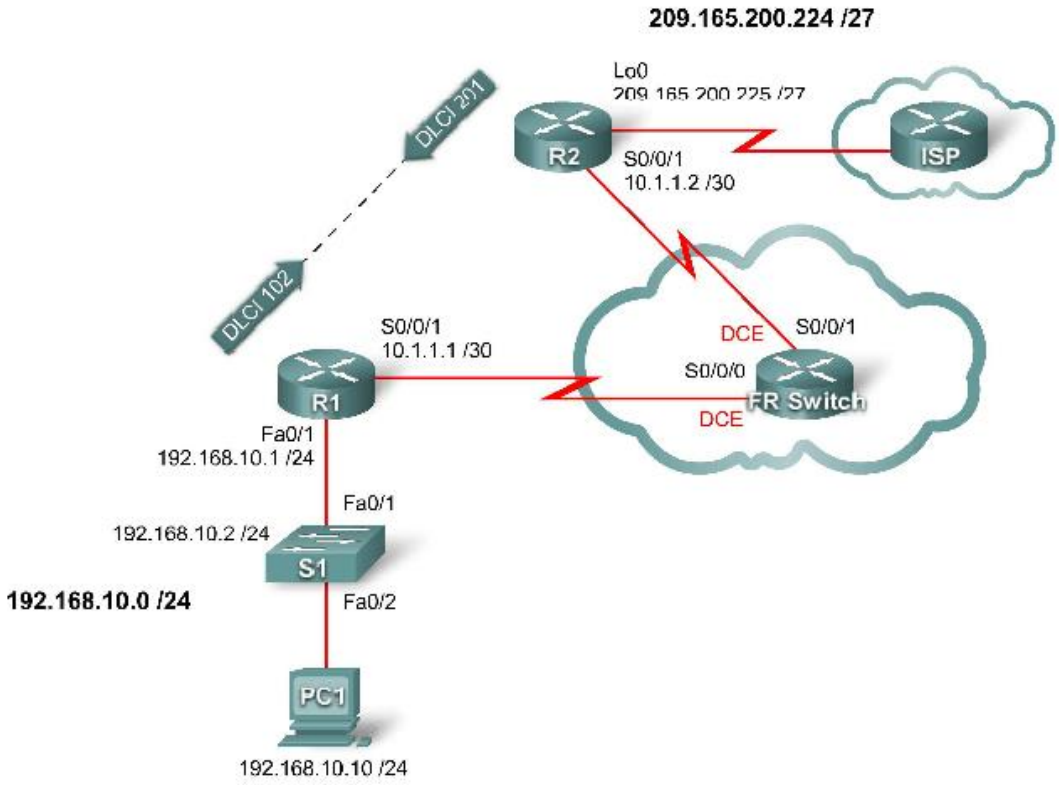
Task 6: Clean Up

Erase the configurations. Disconnect and store the cabling. For PC hosts that are normally connected to other network (such as the school LAN or to the Internet), reconnect the appropriate cabling and restore the TCP/IP settings.

LAB-13

BASIC FRAME RELAY

Topology Diagram



Learning Objectives

Upon completion of this lab, you will be able to:

- Cable a network according to the topology diagram
- Erase the startup configuration and reload a router to the default state
- Perform basic configuration tasks on a router
- Configure and activate interfaces
- Configure EIGRP routing on all routers
- Configure Frame Relay encapsulation on all serial interfaces
- Configure a router as a Frame Relay switch
- Understand the output of the **show frame-relay** commands
- Learn the effects of the **debug frame-relay lmi** command
- Intentionally break and restore a Frame Relay link
- Change the Frame Relay encapsulation type from the Cisco default to IETF
- Change the Frame Relay LMI type from Cisco to ANSI
- Configure a Frame Relay sub-interface

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	Fa0/0	192.168.10.1	255.255.255.0	N/A
	S0/0/1	10.1.1.1	255.255.255.252	N/A
R2	S0/0/1	10.1.1.2	255.255.255.252	N/A
	Lo 0	209.165.200.225	255.255.255.224	N/A
S1	VLAN1	192.168.10.2	255.255.255.0	192.168.10.1
PC1	NIC	192.168.10.10	255.255.255.0	192.168.10.1

Scenario

In this lab, you will learn how to configure Frame Relay encapsulation on serial links using the network shown in the topology diagram. You will also learn how to configure a router as a Frame Relay switch. There are both Cisco standards and Open standards that apply to Frame Relay. You will learn both. Pay special attention in the lab section in which you intentionally break the Frame Relay configurations.

Task 1: Prepare the Network

Step 1: Cable a network that is similar to the one in the topology diagram.

You can use any current router in your lab as long as it has the required interfaces shown in the topology. The Frame Relay labs, unlike any of the other labs in Exploration 4, have two DCE links on the same router. Be sure to change your cabling to reflect the topology diagram.

Note: If you use 1700, 2500, or 2600 routers, the router output and interface descriptions appear differently.

Step 2: Clear any existing configurations on the routers.

Task 2: Perform Basic Router Configuration

Configure the R1 and R2 routers and the S1 switch according to the following guidelines:

- Configure the router hostname.
- Disable DNS lookup.
- Configure an EXEC mode password.
- Configure a message-of-the-day banner.
- Configure a password for console connections.
- Configure a password for vty connections.
- Configure IP addresses on R1 and R2
- Important: Leave serial interfaces shut down.
- Enable EIGRP AS 1 on R1 and R2 for all networks.

```
enable
configure terminal
no ip domain-lookup
enable secret class
banner motd ^CUnauthorized access strictly prohibited, violators
will be prosecuted to the full extent of the law^C
!
!
!
line console 0
logging synchronous
password cisco
login
!
line vty 0 4
password cisco
login
end
copy running-config startup-config
```

!R1

```
interface serial 0/0/1
ip address 10.1.1.1 255.255.255.252
shutdown

!The serial interfaces should remain shutdown until the Frame Relay
!switch is configured

interface fastethernet 0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
router eigrp 1
no auto-summary
network 10.0.0.0
network 192.168.10.0
!
```

!R2

```
interface serial 0/0/1
ip address 10.1.1.2 255.255.255.252
shutdown
!The serial interfaces should remain shutdown until the Frame Relay
!switch is configured
interface loopback 0
ip address 209.165.200.225 255.255.255.224
router eigrp 1
no auto-summary
network 10.0.0.0
network 209.165.200.0
```

Task 3: Configure Frame Relay

You will now set up a basic point-to-point Frame Relay connection between routers 1 and 2. You
first need to configure FR Switch as a Frame Relay switch and create DLCIs.

What does DLCI stand for?

_____

What is a DLCI used for?

_____
__

What is a PVC and how is it used?

_____
__

Step 1: Configure FR Switch as a Frame Relay switch and create a PVC between R1 and R2.

This command enables Frame Relay switching globally on the router, allowing it to forward frames based on the incoming DLCI rather than on an IP address basis:

```
FR-Switch(config)#frame-relay switching
```

Change the interface encapsulation type to Frame Relay. Like HDLC or PPP, Frame Relay is a
data link layer protocol that specifies the framing of Layer 2 traffic.

```
FR-Switch(config)#interface serial 0/0/0
FR-Switch(config)#clock rate 64000
FR-Switch(config-if)#encapsulation frame-relay
```

Changing the interface type to DCE tells the router to send LMI keepalives and allows Frame Relay route statements to be applied. You cannot set up PVCs using the **frame-relay route** command between two Frame Relay DTE interfaces.

```
FR-Switch(config-if)#frame-relay intf-type dce
```

Note: Frame Relay interface types do not need to match the underlying physical interface type. A
physical DTE serial interface can act as a Frame Relay DCE interface, and a physical DCE interface can act as a logical Frame Relay DTE interface.
Configure the router to forward incoming traffic on interface serial 0/0/0 with DLCI 102 to serial
0/0/1 with an output DLCI of 201.

```
FR-Switch(config-if)#frame-relay route 102 interface serial 0/0/1 201
FR-Switch(config-if)#no shutdown
```

This configuration creates two PVCs: one from R1 to R2 (DLCI 102), and one from R2 to R1

(DLCI 201). You can verify the configuration using the **show frame-relay pvc** command.

```
FR-Switch(config-if)#interface serial 0/0/1
FR-Switch(config)#clock rate 64000
FR-Switch(config-if)#encapsulation frame-relay
FR-Switch(config-if)#frame-relay intf-type dce
FR-Switch(config-if)#frame-relay route 201 interface serial 0/0/0 102
FR-Switch(config-if)#no shutdown

FR-Switch#show frame-relay pvc

PVC Statistics for interface Serial0/0/0 (Frame Relay DCE)
                Active       Inactive        Deleted           Static
Local              0             0               0                0
Switched           0             1               0                0
Unused             0             0               0                0
DLCI = 102, DLCI USAGE = SWITCHED, PVC STATUS = INACTIVE, INTERFACE =
Serial0/0/0
input pkts 0                output pkts 0             in bytes 0
out bytes 0                 dropped pkts 0            in pkts dropped 0
out pkts dropped 0          out bytes dropped 0
in FECN pkts 0              in BECN pkts 0            out FECN pkts 0
out BECN pkts 0               in DE pkts 0              out DE pkts 0
out bcast pkts 0            out bcast bytes 0
30 second input rate 0 bits/sec, 0 packets/sec
30 second output rate 0 bits/sec, 0 packets/sec
switched pkts 0
Detailed packet drop counters:
no out intf 0                 out intf down 0           no out PVC 0
in PVC down 0                  out PVC down 0          pkt too big 0
shaping Q                     full 0 pkt above DE 0    policing drop 0

pvc create time 00:03:33, last time pvc status changed 00:00:19
PVC Statistics for interface Serial0/0/1 (Frame Relay DCE)

                  Active            Inactive       Deleted       Static
Local               0                 0              0             0
Switched            0                 1              0             0
Unused              0                 0              0             0

DLCI = 201, DLCI USAGE = SWITCHED, PVC STATUS = INACTIVE, INTERFACE =
Serial0/0/1

input pkts 0                   output pkts 0                in bytes 0
out bytes 0                   dropped pkts 0                in pkts dropped 0
out pkts dropped 0      out bytes dropped 0
in FECN pkts 0                in BECN pkts 0                  out FECN pkts 0
out BECN pkts 0                 in DE pkts 0                    out DE pkts 0
out bcast pkts 0                out bcast bytes 0

30 second input rate 0 bits/sec, 0 packets/sec
30 second output rate 0 bits/sec, 0 packets/sec
switched pkts 0
Detailed packet drop counters:
no out intf 0             out intf down 0               no out PVC 0

in PVC down 0                 out PVC down 0                pkt too big 0
shaping Q full 0              pkt above DE 0               policing drop 0
pvc create time 00:02:02,          last time pvc status changed 00:00:18
```

Notice the 1 in the Inactive column. The PVC you have created does not have any endpoints configured. The Frame Relay switch knows this and has marked the PVC as Inactive. Issue the **show frame-relay route** command. This command shows any existing Frame Relay routes, their interfaces, DLCIs, and status. This is the Layer 2 route that Frame Relay traffic takes through the network. Do not confuse this with Layer 3 IP routing.

```
FR-Switch#show frame-relay route

Input Intf       Input Dlci      Output Intf       Output Dlci        Status
Serial0/0/0          102         Serial0/0/1          201            inactive
Serial0/0/1         201         Serial0/0/0          102           inactive
```

Step 2: Configure R1 for Frame Relay.

Inverse ARP allows distant ends of a Frame Relay link to dynamically discover each other and provides a dynamic method of mapping IP addresses to DLCIs. Although Inverse ARP is useful, it is not always reliable. The best practice is to statically map IP addresses to DLCIs and to disable inverse-arp.

```
R1(config)#interface serial 0/0/1
R1(config-if)#encapsulation frame-relay
R1(config-if)#no frame-relay inverse-arp
```

The command **frame-relay map** statically maps an IP address to a DLCI. In addition to mapping IP to a DLCI, Cisco IOS software allows several other Layer 3 protocol addresses to mapped. The **broadcast** keyword in the following command sends any multicast or broadcast traffic destined for this link over the DLCI. Most routing protocols require the **broadcast** keyword to
properly function over Frame Relay. You can use the **broadcast** keyword on multiple DLCIs on
the same interface. The traffic is replicated to all PVCs.

```
R1(config-if)#frame-relay map ip 10.1.1.2 102 broadcast

R1(config-if)#no shutdown
```

Step 3: Configure R2 for Frame Relay.

```
R2(config)#interface serial 0/0/1
R2(config-if)#encapsulation frame-relay
R2(config-if)#no frame-relay inverse-arp
R2(config-if)#frame-relay map ip 10.1.1.1 201 broadcast
R2(config-if)#no shutdown
```

At this point, you receive messages indicating that the interfaces have come up and that EIGRP neighbor adjacency has been established.

```
R1#*Sep 9 17:05:08.771: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 1: Neighbor
10.1.1.2 (Serial0/0/1) is up: new adjacency
R2#*Sep 9 17:05:47.691: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 1: Neighbor
10.1.1.1 (Serial0/0/1) is up: new adjacency
```

The **show ip route** command shows complete routing tables.
R1:

```
R1#show ip route

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS

level-2
      ia - IS-IS inter area, * - candidate default, U - per-user
static route
      o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C        192.168.10.0/24 is directly connected, FastEthernet0/0
D        209.165.200.0/24 [90/20640000] via 10.1.1.2, 00:00:07, Serial0/0/1
         10.0.0.0/30 is subnetted, 1 subnets
C       10.1.1.0 is directly connected, Serial0/0/1
```

R2:

```
R2#show ip route

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static
route
o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

D         192.168.10.0/24 [90/20514560] via 10.1.1.1, 00:26:03, Serial0/0/1
          209.165.200.0/27 is subnetted, 1 subnets
C         209.165.200.224 is directly connected, Loopback0
          10.0.0.0/30 is subnetted, 1 subnets
C        10.1.1.0 is directly connected, Serial0/0/1
```

Task 4: Verify the Configuration

You should now be able to ping from R1 to R2. It may take several seconds after bringing up the
interfaces for the PVC to become active. You can also see EIGRP routes for each router.

Step 1: Ping R1 and R2.

Ensure that you can ping router R2 from router R1.

```
R1#ping 10.1.1.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 28/29/32ms

R2#ping 10.1.1.1

Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 10.1.1.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 28/29/32ms
```

Step 2: Get PVC information.

The **show frame-relay pvc** command displays information on all PVCs configured on the router. The output also includes the associated DLCI.

R1:

```
R1#show frame-relay pvc

PVC Statistics for interface Serial0/0/1 (Frame Relay DTE)
          Active         Inactive       Deleted        Static
Local         1              0              0             0
Switched      0              0              0             0
Unused        0              0              0             0
DLCI = 102, DLCI USAGE = LOCAL, PVC STATUS = ACTIVE, INTERFACE =
Serial0/0/1

input pkts 5               output pkts 5              in bytes 520
out bytes 520              dropped pkts 0             in pkts dropped 0
out pkts dropped 0         out bytes dropped 0
in FECN pkts 0             in BECN pkts 0             out FECN pkts 0
out BECN pkts 0            in DE pkts 0               out DE pkts 0
out bcast pkts 0           out bcast bytes 0
5 minute input rate 0      bits/sec, 0                packets/sec
5 minute output rate 0     bits/sec, 0                packets/sec

pvc create time 10:26:41, last time pvc status changed 00:01:04
```

R2:

```
R2#show frame-relay pvc

PVC Statistics for interface Serial0/0/1 (Frame Relay DTE)
          Active       Inactive          Deleted          Static
Local        1            0                0              0
Switched     0            0                0              0
Unused       0            0                0              0
DLCI = 201, DLCI USAGE = LOCAL, PVC STATUS = ACTIVE, INTERFACE =
Serial0/0/1

input pkts 5                output pkts 5                in bytes 520
out bytes 520               dropped pkts 0               in pkts dropped 0
out pkts dropped 0          out bytes dropped 0
in FECN pkts 0              in BECN pkts 0               out FECN pkts 0
out BECN pkts 0             in DE pkts 0                 out DE pkts 0
out bcast pkts 0            out bcast bytes 0
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
pvc create time 10:25:31, last time pvc status changed 00:00:00
```

FR Switch:

```
FR-Switch#show frame-relay pvc

PVC Statistics for interface Serial0/0/0 (Frame Relay DCE)

              Active     Inactive      Deleted       Static
  Local          0            0            0            0
  Switched       1            0            0            0
  Unused         0            0            0            0

DLCI = 102, DLCI USAGE = SWITCHED, PVC STATUS = ACTIVE, INTERFACE =
Serial0/0/0

  input pkts 0              output pkts 0             in bytes 0
  out bytes 0               dropped pkts 0            in pkts dropped 0
  out pkts dropped 0                 out bytes dropped 0
  in FECN pkts 0            in BECN pkts 0            out FECN pkts 0
  out BECN pkts 0           in DE pkts 0              out DE pkts 0
  out bcast pkts 0          out bcast bytes 0
  30 second input rate 0 bits/sec, 0 packets/sec
  30 second output rate 0 bits/sec, 0 packets/sec

  switched pkts 0
  Detailed packet drop counters:
  no out intf 0             out intf down 0           no out PVC 0
  in PVC down 0             out PVC down 0            pkt too big 0
  shaping Q full 0          pkt above DE 0            policing drop 0
  pvc create time 10:28:31, last time pvc status changed 00:03:57

PVC Statistics for interface Serial0/0/1 (Frame Relay DCE)

              Active     Inactive      Deleted       Static
  Local          0            0            0            0
  Switched       1            0            0            0
  Unused         0            0            0            0

DLCI = 201, DLCI USAGE = SWITCHED, PVC STATUS = ACTIVE, INTERFACE =
Serial0/0/1

  input pkts 0              output pkts 0             in bytes 0
  out bytes 0               dropped pkts 0            in pkts dropped 0
  out pkts dropped 0                 out bytes dropped 0
  in FECN pkts 0            in BECN pkts 0            out FECN pkts 0
  out BECN pkts 0           in DE pkts 0              out DE pkts 0
  out bcast pkts 0          out bcast bytes 0
  30 second input rate 0 bits/sec, 0 packets/sec
  30 second output rate 0 bits/sec, 0 packets/sec
  switched pkts 0
  Detailed packet drop counters:
  no out intf 0             out intf down 0           no out PVC 0
  in PVC down 0             out PVC down 0            pkt too big 0
  shaping Q full 0          pkt above DE 0            policing drop 0
  pvc create time 10:27:00, last time pvc status changed 00:04:03
```

Step 3: Verify Frame Relay mappings.

The **show frame-relay map** command displays information on the static and dynamic mappings
of Layer 3 addresses to DLCIs. Because Inverse ARP has been turned off, there are only static maps.

R1:

```
R1#show frame-relay map

Serial0/0/1 (up): ip 10.1.1.2 dlci 102(0x66,0x1860), static,
                  CISCO, status defined, active
```

R2:

```
R2#show frame-relay map

Serial0/0/1 (up): ip 10.1.1.1 dlci 201(0xC9,0x3090), static,
                  CISCO, status defined, active
```

FR Switch:

FR Switch acts as a Layer 2 device, so there is no need to map Layer 3 addresses to Layer 2 DLCIs.

Step 4: Debug the Frame Relay LMI.

What purpose does the LMI serve in a Frame Relay network?

_____
_____
_____
_____

What are the three different types of LMI?

_____

What DLCI does the LMI operate on?

_____

Issue the **debug frame-relay lmi** command. The output gives detailed information on all LMI data. Keepalives are sent every 10 seconds, so you may have to wait until you see any output.

The debug output shows two LMI packets: the first outgoing, the second incoming.

```
R1#debug frame-relay lmi
Frame Relay LMI debugging is on
Displaying all Frame Relay LMI data
R1#
*Aug 24 06:19:15.920: Serial0/0/1(out): StEnq, myseq 196, yourseen
195, DTE up
*Aug 24 06:19:15.920: datagramstart = 0xE73F24F4, datagramsize = 13
*Aug 24 06:19:15.920: FR encap = 0xFCF10309
*Aug 24 06:19:15.920: 00 75 01 01 00 03 02 C4 C3
*Aug 24 06:19:15.920:
*Aug 24 06:19:15.924: Serial0/0/1(in): Status, myseq 196, pak size 21
*Aug 24 06:19:15.924: RT IE 1, length 1, type 0
*Aug 24 06:19:15.924: KA IE 3, length 2, yourseq 196, myseq 196
*Aug 24 06:19:15.924: PVC IE 0x7 , length 0x6 , dlci 102, status 0x2
, bw 0
R1#undebug all
Port Statistics for unclassified packets is not turned on.

All possible debugging has been turned off
```

Notice that the output shows an outgoing LMI packet with a sequence number of 196. The last LMI message received from the FR Switch had sequence number 195.

```
*Aug 24 06:19:15.920: Serial0/0/1(out): StEnq, myseq 196, yourseen
195, DTE up
```

This line indicates an incoming LMI message from the FR Switch to R1 with sequence number 196.

```
*Aug 24 06:19:15.924: Serial0/0/1(in): Status, myseq 196, pak size 21
```

FR Switch sent this as sequence number 196 (myseq), and the last LMI message received by the
FR-Switch from R1 had sequence number 196 (yourseq).

```
*Aug 24 06:19:15.924: KA IE 3, length 2, yourseq 196, myseq 196
```

DLCI 102 is the only DLCI on this link, and it is currently active.

```
*Aug 24 06:19:15.924: PVC IE 0x7 , length 0x6 , dlci 102, status 0x2 , bw 0
```

Task 5: Troubleshooting Frame Relay.

A variety of tools are available for troubleshooting Frame Relay connectivity issues. To learn about troubleshooting, you will break the Frame Relay connection established earlier and then re-establish it.

Step 1: Remove the frame map from R1.

```
R1#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface serial0/0/1
```

```
R1(config-if)#encapsulation frame-relay
R1(config-if)#no frame-relay map ip 10.1.1.2 102 broadcast
```

Now that you have removed the frame map statement from R1, try to ping router R1 from router
R2. You will get no response.

```
R2#ping 10.1.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
```

Additionally, you should get console messages reporting the EIGRP adjacency going up and down.

```
R1(config-if)#*Sep 9 17:28:36.579: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 1:
Neighbor 10.1.1.2 (Serial0/0/1) is down: Interface Goodbye received

R1(config-if)#*Sep 9 17:29:32.583: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 1:
Neighbor 10.1.1.2 (Serial0/0/1) is up: new adjacency

R1(config-if)#*Sep 9 17:32:37.095: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 1:
Neighbor 10.1.1.2 (Serial0/0/1) is down: retry limit exceeded

R2#*Sep 9 17:29:15.359: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 1: Neighbor
10.1.1.1 (Serial0/0/1) is down: holding time expired
```

Issue the **debug ip icmp** command on R1:

```
R1#debug ip icmp

ICMP packet debugging is on
```

Now ping the serial interface of R1 again. The following debug message appears on R1:

```
R2#ping 10.1.1.1

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.1.1.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
R1#*Sep  9 17:42:13.415: ICMP: echo reply sent, src 10.1.1.1, dst
10.1.1.2
R1#*Sep  9 17:42:15.411: ICMP: echo reply sent, src 10.1.1.1, dst
10.1.1.2
R1#*Sep  9 17:42:17.411: ICMP: echo reply sent, src 10.1.1.1, dst
10.1.1.2
R1#*Sep  9 17:42:19.411: ICMP: echo reply sent, src 10.1.1.1, dst
10.1.1.2
R1#*Sep  9 17:42:21.411: ICMP: echo reply sent, src 10.1.1.1, dst
10.1.1.2
```

As is shown by this debug message, the ICMP packet from R2 is reaching R1.

Issuing the **show frame-relay map** command returns a blank line.

```
R1#show frame-relay map

R1#
```

Turn off all debugging with the **undebug all** command, and re-apply the **frame-relay map ip** command but without using the **broadcast** keyword.

```
R1#undebug all

Port Statistics for unclassified packets is not turned on.

All possible debugging has been turned off

R1#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R1(config)#interface serial0/0/1
R1(config-if)#encapsulation frame-relay
R1(config-if)#frame-relay map ip 10.1.1.2 102

R2#ping 10.1.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/41/44 ms
```

Notice that while pings are successful, the EIGRP adjacency continues to “flap” (go up and down).

```
R1(config-if)#*Sep 9 17:47:58.375: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 1:
Neighbor 10.1.1.2 (Serial0/0/1) is up: new adjacency

R1(config-if)#*Sep 9 17:51:02.887: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 1:
Neighbor 10.1.1.2 (Serial0/0/1) is down: retry limit exceeded

R1(config-if)#*Sep 9 17:51:33.175: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 1:
Neighbor 10.1.1.2 (Serial0/0/1) is up: new adjacency

R1(config-if)#*Sep 9 17:54:37.687: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 1:
Neighbor 10.1.1.2 (Serial0/0/1) is down: retry limit exceeded
```

Replace the Frame Relay map statement and include the **broadcast** keyword this time. Verify that the full routing table is restored and that you have full end-to-end connectivity.

```
R1#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface serial0/0/1
R1(config-if)#encapsulation frame-relay
R1(config-if)#frame-relay map ip 10.1.1.2 102 broadcast
```

```
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS
level-2
       ia - IS-IS inter area, * - candidate default, U - per-user
static  route  o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.10.0/24 is directly connected, FastEthernet0/0
     209.165.200.0/27 is subnetted, 1 subnets
D    209.165.200.224 [90/20640000] via 10.1.1.2, 00:00:05, Serial0/0/1
     10.0.0.0/30 is subnetted, 1 subnets
C    10.1.1.0 is directly connected, Serial0/0/1
```

Step 2: Change the Frame Relay encapsulation type.

Cisco IOS software supports two types of Frame Relay encapsulation: the default Cisco encapsulation and the standards-based IETF encapsulation. Change the Frame Relay encapsulation on serial0/0/1 on R2 to IETF.

```
R2(config-if)#encapsulation frame-relay ietf
```

Notice that the interface does not go down. You might be surprised by this. Cisco routers can correctly interpret Frame Relay frames that use either the default Cisco Frame Relay encapsulation or the IETF standard Frame Relay encapsulation. If your network is composed entirely of Cisco routers, then it does not make any difference whether you use the default Cisco
Frame Relay encapsulation or the IETF standard. Cisco routers understand both types of incoming frames. However, if you have routers from different vendors using Frame Relay, then the IETF standard must be used. The command **encapsulation frame-relay ietf** forces the Cisco router to encapsulate its outgoing frames using the IETF standard. This standard can be correctly understood by the router of another vendor.

```
R2#show interface serial 0/0/1

Serial0/0/1 is up, line protocol is up
Hardware is GT96K Serial
Internet address is 10.1.1.2/30
MTU 1500 bytes, BW 128 Kbit, DLY 20000 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation FRAME-RELAY IETF, loopback not set
<output omitted>

FR-Switch#show int s0/0/0

Serial0/0/0 is up, line protocol is up
Hardware is GT96K Serial
MTU 1500 bytes, BW 128 Kbit, DLY 20000 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation FRAME-RELAY, loopback not set
```

Note the difference in output between the two **show interface** commands. Also notice that the EIGRP adjacency is still up. Although FR Switch and R2 are using different encapsulation types,
they are still passing traffic.

Change the encapsulation type back to the default:

```
R2(config-if)#encapsulation frame-relay
```

Step 3: Change the LMI type.

On R2, change the LMI type to ANSI.

```
R2#configure terminal

Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#interface serial 0/0/1
R2(config-if)#encapsulation frame-relay
R2(config-if)#frame-relay lmi-type ansi
R2(config-if)#^Z
R2#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]

*Sep  9 18:41:08.351: %LINEPROTO-5-UPDOWN: Line protocol on Interface
Serial0/0/1, changed state to down

*Sep  9 18:41:08.351: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 1: Neighbor
10.1.1.1 (Serial0/0/1) is down: interface down

R2#show interface serial 0/0/1

Serial0/0/1 is up, line protocol is down

R2#show frame-relay lmi

LMI Statistics for interface Serial0/0/1 (Frame Relay DTE) LMI TYPE =
ANSI
   Invalid Unnumbered info 0               Invalid Prot Disc 0
   Invalid dummy Call Ref 0                Invalid Msg Type 0
   Invalid Status Message 0                Invalid Lock Shift 0
   Invalid Information ID 0                Invalid Report IE Len 0
   Invalid Report Request 0                Invalid Keep IE Len 0
   Num Status Enq. Sent 1391               Num Status msgs Rcvd 1382
   Num Update Status Rcvd 0                Num Status Timeouts 10
Last Full Status Req 00:00:27 Last Full Status Rcvd 00:00:27
```

If you continue issuing the **show frame-relay lmi** command, you will notice the highlighted times incrementing. When 60 seconds have passed, the interface changes its state to Up Down, because R2 and FR Switch are no longer exchanging Keepalives or any other link-state information.

Issue the **debug frame-relay lmi** command. Notice that LMI packets are no longer showing up in pairs. While all outgoing LMI messages are logged, no incoming messages are shown. This is because R2 is expecting ANSI LMI, and FR Switch is sending Cisco LMI.

```
R2#debug frame-relay lmi
*Aug 25 04:34:25.774: Serial0/0/1(out): StEnq, myseq 20, yourseen 0,
DTE down
*Aug 25 04:34:25.774: datagramstart = 0xE73F2634, datagramsize = 14
*Aug 25 04:34:25.774: FR encap = 0x00010308
*Aug 25 04:34:25.774: 00 75 95 01 01 00 03 02 14 00
*Aug 25 04:34:25.774:
```

Leave debugging on and restore the LMI type to Cisco on R2.

```
R2(config-if)#frame-relay lmi-type cisco
*Aug 25 04:42:45.774: Serial0/0/1(out): StEnq, myseq 2, yourseen 1, DTE
down
*Aug 25 04:42:45.774: datagramstart = 0xE7000D54, datagramsize = 13
*Aug 25 04:42:45.774: FR encap = 0xFCF10309
*Aug 25 04:42:45.774: 00 75 01 01 01 03 02 02 01
*Aug 25 04:42:45.774:
*Aug 25 04:42:45.778: Serial0/0/1(in): Status, myseq 2, pak size 21
*Aug 25 04:42:45.778: RT IE 1, length 1, type 0
*Aug 25 04:42:45.778: KA IE 3, length 2, yourseq 2 , myseq 2
*Aug 25 04:42:45.778: PVC IE 0x7 , length 0x6 , dlci 201, status 0x2 ,
bw 0
*Aug 25 04:42:55.774: Serial0/0/1(out): StEnq, myseq 3, yourseen 2, DTE
up
*Aug 25 04:42:55.774: datagramstart = 0xE7001614, datagramsize = 13
*Aug 25 04:42:55.774: FR encap = 0xFCF10309
*Aug 25 04:42:55.774: 00 75 01 01 01 03 02 03 02
*Aug 25 04:42:55.774:
*Aug 25 04:42:55.778: Serial0/0/1(in): Status, myseq 3, pak size 21
*Aug 25 04:42:55.778: RT IE 1, length 1, type 0
*Aug 25 04:42:55.778: KA IE 3, length 2, yourseq 1 , myseq 3
*Aug 25 04:42:55.778: PVC IE 0x7 , length 0x6 , dlci 201, status 0x2 ,
bw 0
*Aug 25 04:42:56.774: %LINEPROTO-5-UPDOWN: Line protocol on Interface
Serial0/0/1, changed state to up
```

As you can see, the LMI sequence number has been reset to 1, and R2 began to understand the LMI messages coming in from FR Switch. After FR Switch and R2 had successfully exchanged LMI messages, the interface changed state to Up.

Task 5: Configure a Frame Relay Sub-interface

Frame Relay supports two types of sub-interfaces: point-to-point and point-to-multipoint. Point-to-multipoint sub-interfaces support non-broadcast multi-access topologies. For example, a hub and spoke topology would use a point-to-multipoint sub-interface. In this lab, you will create a point-to-point sub-interface.

Step 1: On FR Switch, create a new PVC between R1 and R2.

```
FR-Switch(config)#interface serial 0/0/0
FR-Switch(config-if)#frame-relay route 112 interface serial 0/0/1 212
FR-Switch(config-if)#interface serial 0/0/1
FR-Switch(config-if)#frame-relay route 212 interface serial 0/0/0 112
```

Step 2: Create and configure a point-to-point sub-interface on R1.

Create subinterface 112 as a point-to-point interface. Frame Relay encapsulation must be specified on the physical interface before subinterfaces can be created.

```
R1(config)#interface serial 0/0/1.112 point-to-point
R1(config-subif)#ip address 10.1.1.5 255.255.255.252
R1(config-subif)#frame-relay interface-dlci 112
```

Step 3: Create and configure a point-to-point sub-interface on R2.

```
R2(config)#interface serial 0/0/1.212 point-to-point
R2(config-subif)#ip address 10.1.1.6 255.255.255.252
R2(config-subif)#frame-relay interface-dlci 212
```

Step 4: Verify connectivity.

You should be able to ping across the new PVC.

```
R1#ping 10.1.1.6
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.6, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 28/28/32 ms

R2#ping 10.1.1.5

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.5, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 28/28/32 ms
```

You can also verify the configuration using the **show frame-relay pvc** and **show frame-relay map** commands in Task 4.

R1:

```
R1#show frame-relay pvc

PVC Statistics for interface Serial0/0/1 (Frame Relay DTE)

                Active      Inactive      Deleted       Static
  Local            2             0            0            0
  Switched         0             0            0            0
  Unused           0             0            0            0

DLCI = 102, DLCI USAGE = LOCAL, PVC STATUS = ACTIVE, INTERFACE =
Serial0/0/1
```

```
  input pkts 319           output pkts 279          in bytes 20665
  out bytes 16665          dropped pkts 0           in pkts dropped 0
  out pkts dropped 0                 out bytes dropped 0
  in FECN pkts 0           in BECN pkts 0           out FECN pkts 0
  out BECN pkts 0          in DE pkts 0             out DE pkts 0
  out bcast pkts 193       out bcast bytes 12352
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
  pvc create time 04:43:35, last time pvc status changed 01:16:05

DLCI = 112, DLCI USAGE = LOCAL, PVC STATUS = ACTIVE, INTERFACE =
Serial0/0/1.112

  input pkts 15            output pkts 211          in bytes 2600
  out bytes 17624          dropped pkts 0           in pkts dropped 0
  out pkts dropped 0                 out bytes dropped 0
  in FECN pkts 0           in BECN pkts 0           out FECN pkts 0
  out BECN pkts 0          in DE pkts 0             out DE pkts 0
  out bcast pkts 200       out bcast bytes 16520
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
  pvc create time 00:19:16, last time pvc status changed 00:18:56
```

R2:

```
R2#show frame-relay pvc

PVC Statistics for interface Serial0/0/1 (Frame Relay DTE)

               Active     Inactive      Deleted       Static
  Local           2            0            0            0
  Switched        0            0            0            0
  Unused          0            0            0            0

DLCI = 201, DLCI USAGE = LOCAL, PVC STATUS = ACTIVE, INTERFACE =
Serial0/0/1

  input pkts 331           output pkts 374          in bytes 19928
  out bytes 24098          dropped pkts 0           in pkts dropped 0
  out pkts dropped 0                 out bytes dropped 0
  in FECN pkts 0           in BECN pkts 0           out FECN pkts 0
  out BECN pkts 0          in DE pkts 0             out DE pkts 0
  out bcast pkts 331       out bcast bytes 21184
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
  pvc create time 05:22:55, last time pvc status changed 01:16:36

DLCI = 212, DLCI USAGE = LOCAL, PVC STATUS = ACTIVE, INTERFACE =
Serial0/0/1.212

  input pkts 217           output pkts 16           in bytes 18008
  out bytes 2912           dropped pkts 0           in pkts dropped 0
  out pkts dropped 0                 out bytes dropped 0
  in FECN pkts 0           in BECN pkts 0           out FECN pkts 0
  out BECN pkts 0          in DE pkts 0             out DE pkts 0
  out bcast pkts 6         out bcast bytes 1872
```

```
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
pvc create time 00:19:37, last time pvc status changed 00:18:57
```

FR Switch:

```
FR-Switch#show frame-relay pvc

PVC Statistics for interface Serial0/0/0 (Frame Relay DCE)

                Active      Inactive      Deleted       Static
  Local            0             0            0            0
  Switched         2             0            0            0
  Unused           0             0            0            0

DLCI = 102, DLCI USAGE = SWITCHED, PVC STATUS = ACTIVE, INTERFACE =
Serial0/0/0

  input pkts 335            output pkts 376           in bytes 20184
  out bytes 24226           dropped pkts 2            in pkts dropped 2
  out pkts dropped 0                  out bytes dropped 0
  in FECN pkts 0            in BECN pkts 0            out FECN pkts 0
  out BECN pkts 0           in DE pkts 0              out DE pkts 0
  out bcast pkts 0          out bcast bytes 0
  30 second input rate 0 bits/sec, 0 packets/sec
  30 second output rate 0 bits/sec, 0 packets/sec
  switched pkts 333
  Detailed packet drop counters:
  no out intf 0             out intf down 0           no out PVC 0
  in PVC down 0             out PVC down 2            pkt too big 0
  shaping Q full 0          pkt above DE 0            policing drop 0
  pvc create time 05:23:43, last time pvc status changed 01:18:32

DLCI = 112, DLCI USAGE = SWITCHED, PVC STATUS = ACTIVE, INTERFACE =
Serial0/0/0

  input pkts 242            output pkts 18            in bytes 20104
  out bytes 3536            dropped pkts 0            in pkts dropped 0
  out pkts dropped 0                  out bytes dropped 0
  in FECN pkts 0            in BECN pkts 0            out FECN pkts 0
  out BECN pkts 0           in DE pkts 0              out DE pkts 0
  out bcast pkts 0          out bcast bytes 0
  30 second input rate 0 bits/sec, 0 packets/sec
  30 second output rate 0 bits/sec, 0 packets/sec
  switched pkts 242
  Detailed packet drop counters:
  no out intf 0             out intf down 0           no out PVC 0
  in PVC down 0             out PVC down 0            pkt too big 0
  shaping Q full 0          pkt above DE 0            policing drop 0
  pvc create time 00:21:41, last time pvc status changed 00:21:22

PVC Statistics for interface Serial0/0/1 (Frame Relay DCE)

                Active      Inactive      Deleted       Static
  Local            0             0            0            0
  Switched         2             0            0            0
```

```
  Unused          0              0              0              0

DLCI = 201, DLCI USAGE = SWITCHED, PVC STATUS = ACTIVE, INTERFACE =
Serial0/0/1

  input pkts 376            output pkts 333           in bytes 24226
  out bytes 20056           dropped pkts 0            in pkts dropped 0
  out pkts dropped 0                  out bytes dropped 0
  in FECN pkts 0            in BECN pkts 0            out FECN pkts 0
  out BECN pkts 0           in DE pkts 0              out DE pkts 0
  out bcast pkts 0          out bcast bytes 0
  30 second input rate 0 bits/sec, 0 packets/sec
  30 second output rate 0 bits/sec, 0 packets/sec
  switched pkts 376
  Detailed packet drop counters:
  no out intf 0             out intf down 0           no out PVC 0
  in PVC down 0             out PVC down 0            pkt too big 0
  shaping Q full 0          pkt above DE 0            policing drop 0
  pvc create time 05:23:14, last time pvc status changed 01:39:39

DLCI = 212, DLCI USAGE = SWITCHED, PVC STATUS = ACTIVE, INTERFACE =
Serial0/0/1

  input pkts 18             output pkts 243           in bytes 3536
  out bytes 20168           dropped pkts 0            in pkts dropped 0
  out pkts dropped 0                  out bytes dropped 0
  in FECN pkts 0            in BECN pkts 0            out FECN pkts 0
  out BECN pkts 0           in DE pkts 0              out DE pkts 0
  out bcast pkts 0          out bcast bytes 0
  30 second input rate 0 bits/sec, 0 packets/sec
  30 second output rate 0 bits/sec, 0 packets/sec
  switched pkts 18
  Detailed packet drop counters:
  no out intf 0             out intf down 0           no out PVC 0
```

```
  in PVC down 0             out PVC down 0            pkt too big 0
  shaping Q full 0          pkt above DE 0            policing drop 0
  pvc create time 00:21:36, last time pvc status changed 00:21:20
```

R1:

```
R1#show frame-relay map
Serial0/0/1 (up): ip 10.1.1.2 dlci 102(0x66,0x1860), static,
              broadcast,
              CISCO, status defined, active
Serial0/0/1.112 (up): point-to-point dlci, dlci 112(0x70,0x1C00),
broadcast
          status defined, active
```

R2:

```
R2#show frame-relay map
Serial0/0/1 (up): ip 10.1.1.1 dlci 201(0xC9,0x3090), static,
              broadcast,
              CISCO, status defined, active
```

```
Serial0/0/1.212 (up): point-to-point dlci, dlci 212(0xD4,0x3440),
broadcast
          status defined, active
```

FR Switch:

```
FR-Switch#show frame-relay route
```

Input Intf	Input Dlci	Output Intf	Output Dlci	Status
Serial0/0/0	102	Serial0/0/1	201	active
Serial0/0/0	112	Serial0/0/1	212	active
Serial0/0/1	201	Serial0/0/0	102	active
Serial0/0/1	212	Serial0/0/0	112	active

Now debug the Frame Relay LMI.

```
R1#debug frame-relay lmi
*Aug 25 05:58:50.902: Serial0/0/1(out): StEnq, myseq 136, yourseen 135,
DTE up
*Aug 25 05:58:50.902: datagramstart = 0xE7000354, datagramsize = 13
*Aug 25 05:58:50.902: FR encap = 0xFCF10309
*Aug 25 05:58:50.902: 00 75 01 01 00 03 02 88 87
*Aug 25 05:58:50.902:
*Aug 25 05:58:50.906: Serial0/0/1(in): Status, myseq 136, pak size 29
*Aug 25 05:58:50.906: RT IE 1, length 1, type 0
*Aug 25 05:58:50.906: KA IE 3, length 2, yourseq 136, myseq 136
*Aug 25 05:58:50.906: PVC IE 0x7 , length 0x6 , dlci 102, status 0x2 ,
bw 0
*Aug 25 05:58:50.906: PVC IE 0x7 , length 0x6 , dlci 112, status 0x2 ,
bw 0
```

Note that two DLCIs are listed in the LMI message from FR Switch to R1.

```
R2#debug frame-relay lmi
*Aug 25 06:08:35.774: Serial0/0/1(out):StEnq, myseq 7,yourseen 4,DTE up
*Aug 25 06:08:35.774: datagramstart = 0xE73F28B4, datagramsize = 13
*Aug 25 06:08:35.774: FR encap = 0xFCF10309
*Aug 25 06:08:35.774: 00 75 01 01 00 03 02 07 04
*Aug 25 06:08:35.774:
*Aug 25 06:08:35.778: Serial0/0/1(in): Status, myseq 7, pak size 29
*Aug 25 06:08:35.778: RT IE 1, length 1, type 0
*Aug 25 06:08:35.778: KA IE 3, length 2, yourseq 5 , myseq 7
*Aug 25 06:08:35.778: PVC IE 0x7,length 0x6, dlci 201, status 0x2, bw 0
*Aug 25 06:08:35.778: PVC IE 0x7,length 0x6, dlci 212, status 0x2, bw 0
```